IT Dashboard Manual

ZEPTO

Oct 24, 2024

FALCON AUTOTECH

NEO

IT Dashboard Manual

Dashboard User's Guide

Release 1.0

Oct 24, 2024

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1 Overview

The IT Dashboard Manual provides a comprehensive guide for users on how to effectively utilize this IT dashboard. The manual serves as a reference and instructional tool, enabling users to understand the purpose, features, and functionalities of the dashboard. It aims to facilitate user adoption, enhance data-driven decision-making, and maximize the value derived from the IT dashboard.

Key Components:

This manual covers the following key components related to the IT dashboard:

Introduction: The manual begins with an introduction to the IT dashboard, highlighting its purpose, benefits, and target audience. It provides an overview of the data visualizations, metrics, and insights that the dashboard offers, emphasizing the value it brings to users.

Accessing the Dashboard: This section explains how to access the IT dashboard, including login credentials, URL, or any specific access requirements. It provides step-by-step instructions on accessing the dashboard through different devices or platforms, ensuring users can easily connect to the system.

Dashboard Layout: The manual describes the layout and organization of the dashboard interface. It highlights the various sections, panels, and navigation options available to users. This section may include a visual representation or diagram showcasing the different components and their placement within the dashboard.

Interacting with the Dashboard: This section focuses on how users can interact with the dashboard to explore and analyze data. It covers functionalities such as filtering, sorting, searching, and drilling down into specific data points. Users are guided on how to customize the dashboard based on their preferences and requirements.

Data Visualizations: The manual provides an overview of the different types of data visualizations present in the dashboard, such as charts, graphs, tables, or maps. It explains the purpose and interpretation of each visualization, highlighting the key insights they convey. Users are guided on how to interpret and analyze the visualizations to gain actionable insights.

Troubleshooting and Support: In case users encounter issues or have questions while using the IT dashboard, this section offers troubleshooting tips and support information. It provides contact details for the IT support team or help desk, along with instructions on how to report and resolve any technical problems.

Best Practices and Guidelines: The manual outlines best practices for using the IT dashboard effectively. It provides guidelines on data hygiene, data visualization principles, and data interpretation. Users are encouraged to follow these practices to maximize the accuracy and relevance of the insights obtained from the dashboard.

Updates and Enhancements: The manual explains how updates and enhancements to the IT dashboard will be communicated to users. It provides information on versioning, release notes, and any scheduled maintenance activities. Users are informed about the process for requesting new features or suggesting improvements

2 User and Window Management

2.1 Supported Browsers

While the IT dashboard strives to support a wide range of browsers, some older versions or less popular browsers may not provide an optimal experience. It is advisable to use the latest stable versions of the supported browsers for the best performance, compatibility, and security.

Including information about supported browsers in your IT dashboard manual helps set clear expectations for users and provides guidance on the best browser choices for optimal performance, compatibility, and security. It can also reduce support requests related to browser-related issues and streamline the troubleshooting process by focusing on known and supported environments.

To begin working with the system, launch the dashboard link in a supported web browser. In case you don't have the Dashboard URL shared please contact the local IT team for the same.

Figure 2.1 Supported Browsers

NOTE: Using an unsupported browser will limit the functionality of the application.

2.2 Login

To access the dashboard, please open any of the suggested browsers mentioned earlier, such as Google Chrome, Mozilla Firefox, Microsoft Edge (Chromium-based version), or Safari. Once you have opened the browser, type the following URL into the address bar:

Dashboard URL: http://14.97.109.198:9091/dashboard/zepto/#/login

After entering the URL, press Enter, and you will be directed to a login window. This login window is where you will enter your credentials to access the dashboard.

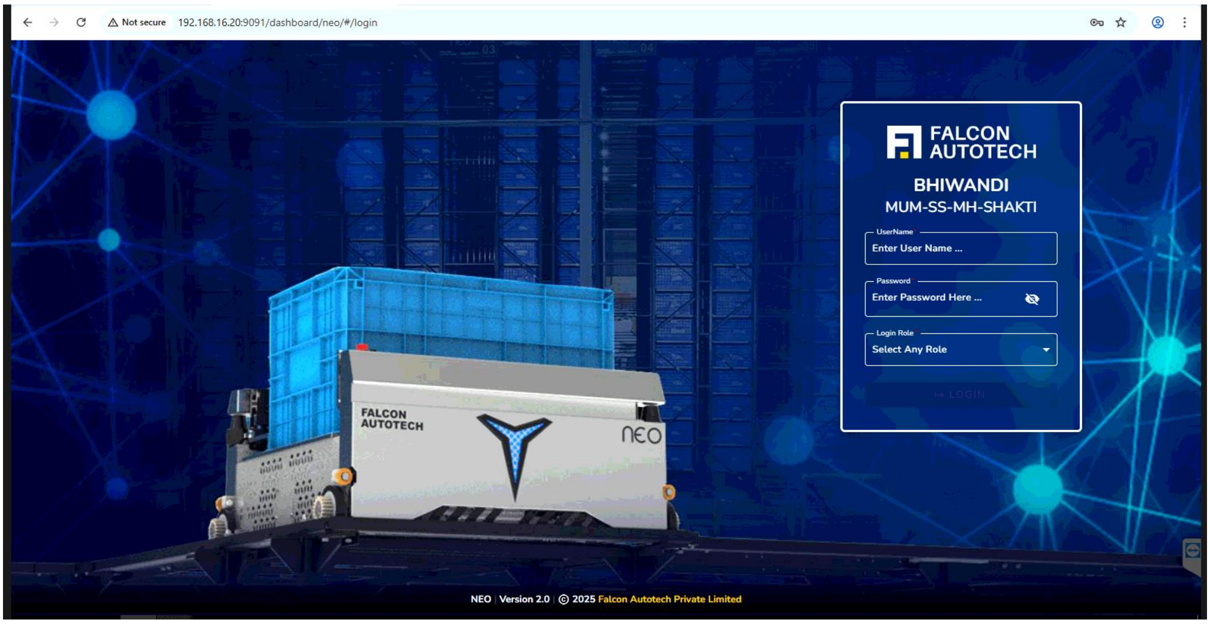


Figure 2.2 Login Window

To log in, please enter your assigned Username and Password into the respective fields on the login window. Once you have entered the correct credentials, click on the login button to proceed.

Upon successful login, you will be redirected to the homepage of the dashboard. This is where you will find various features, functions, and information relevant to your usage.

It's important to note that if this is your first time logging in and you do not have valid credentials, please reach out to Falcon support for assistance. You can contact them by sending an email to support@falconautoonline.com .

The support team will guide you through the process of creating new user credentials, enabling you to access the dashboard smoothly and securely.

2.3 Logout

Upon completing your task or work, it is important to properly log out from the IT dashboard workspace. To do so, follow the steps outlined below:

1. Locate the logout button in the upper right-hand corner of the navigation bar.
2. Click on the logout button to initiate the logout process.
3. To confirm your logout action, a popup window will appear in the middle of the screen.
4. In the popup window as shown in Figure 2.3, click on the "Logout" button to proceed with the logout.
5. After clicking "OK", you will be automatically redirected to the login screen.

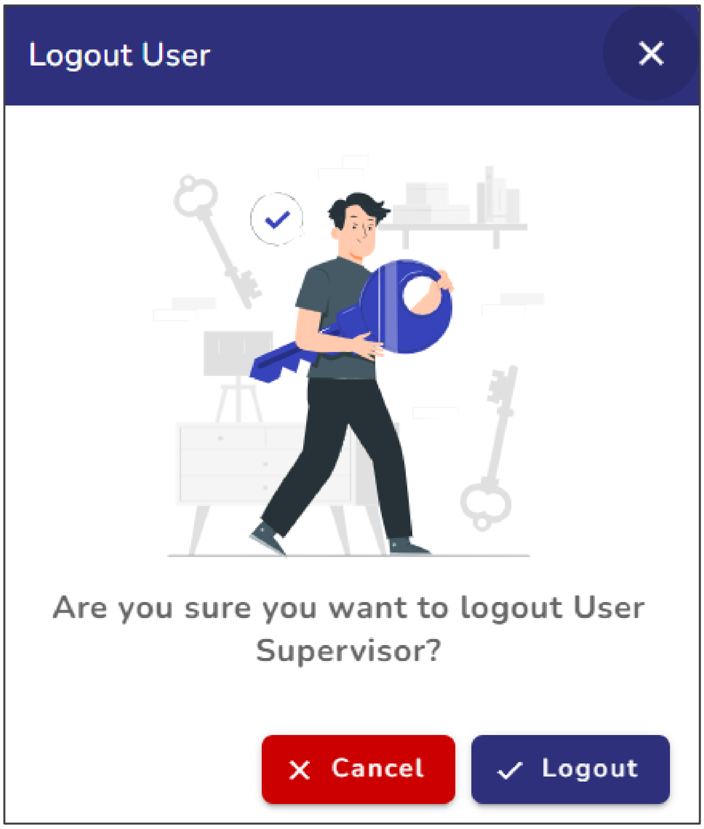


Figure 2.3 Popup Window-Logout

By following these steps, you ensure that your session is securely terminated, and you are logged out from the IT dashboard. This helps protect your account and sensitive information from unauthorized access.

Please note that logging out is a crucial step to maintaining the security and privacy of your data. It is recommended to always log out when you have finished using the IT dashboard, especially when accessing it from shared or public devices.

2.4 User Roles & Permissions

Roles within the dashboard are assigned to users based on their responsibilities. Key roles include:

- **Admin:** Full access to all features and configuration options.
- **Supervisor:** Access to view analytics, reports, and notifications.
- **Operator:** Access is limited to system monitoring and live status views.
- **Support:** Access to maintenance screens and other such things.

Supervisor

3 Navigation Bar

The navigation bar in the Falcon WCS Dashboard adapts dynamically based on the user's role. This section details the specific menus and tabs available to different roles to ensure clarity and usability.

A side navigation bar, also known as a menu bar, is a prominent section of a dashboard that serves as a navigational aid for visitors. Its primary purpose is to assist users in selecting topics, links, or sub-topics of interest within the dashboard. Users manually enter specific webpage URLs by utilising the navigation bar, as the navigation bar automatically takes care of this functionality.

The design and usability of a navigation bar are crucial factors that can significantly impact the overall user experience of a webpage. It plays a pivotal role in guiding users and allowing them to access various interactive screens and functionalities within the dashboard easily.

4 Supervisor Role Navigation

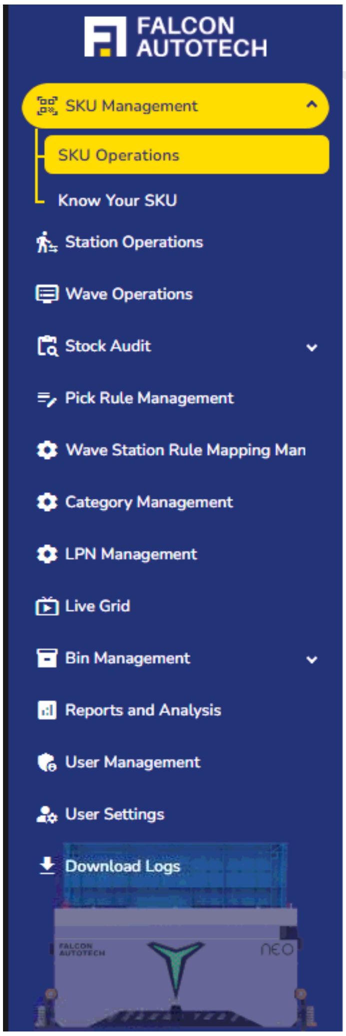


Figure 4.1 Navigation Bar

In the context of this IT dashboard, the navigation bar enables users to navigate to the following user interactive screens:

Accessible Features:
Supervisors have access to features primarily focused on operational oversight, reporting, and user management.

Menu Structure:

- **SKU Management**
  - Add, edit, or delete SKU information.
  - Assign SKUs to specific stations for operational purposes.
- **Station Operations**
  - Manage and monitor station-level activities.
  - Configure station settings to align with wave operations.
- **Wave Operations**
  - Oversee wave creation, execution, and closure.
  - Assign SKUs to waves and manage wave-level reporting.
- **Stock Audit**
  - Perform audits to ensure inventory accuracy.
  - Access audit logs and reports for reconciliation.
- **Pick Rule Management**
  - Create pick rules and manage them.
- **Wave Station Rule Mapping**
  - Update Rule Mapping to Stations
- **Category Management**
  - Maintains the category matrix for the warehouse to organize SKUs efficiently.
  - Enables mapping of SKUs to categories, facilitating rule-based picking, storage optimization, and reporting.
- **LPN Management**
  - Handles the tracking and maintenance of License Plate Numbers (LPNs) used for inventory movement and identification.
  - Enables supervisors to view, search, and update LPN details, such as contents, status, origin, and destination.
- **Live Grid**
  - Visualizes the entire layout or grid, showcasing the exact and real-time location of robots to simplify operational processes.
- **Bin Management**
  - View and update bin statuses.
  - Reallocate or reset bins as necessary.
- **Reports and Analysis**
  - Generate and analyze system performance reports.
  - Access shipment, throughput, and error reports.
- **User Management**
  - Add, edit, or deactivate users.
  - Assign roles and permissions.

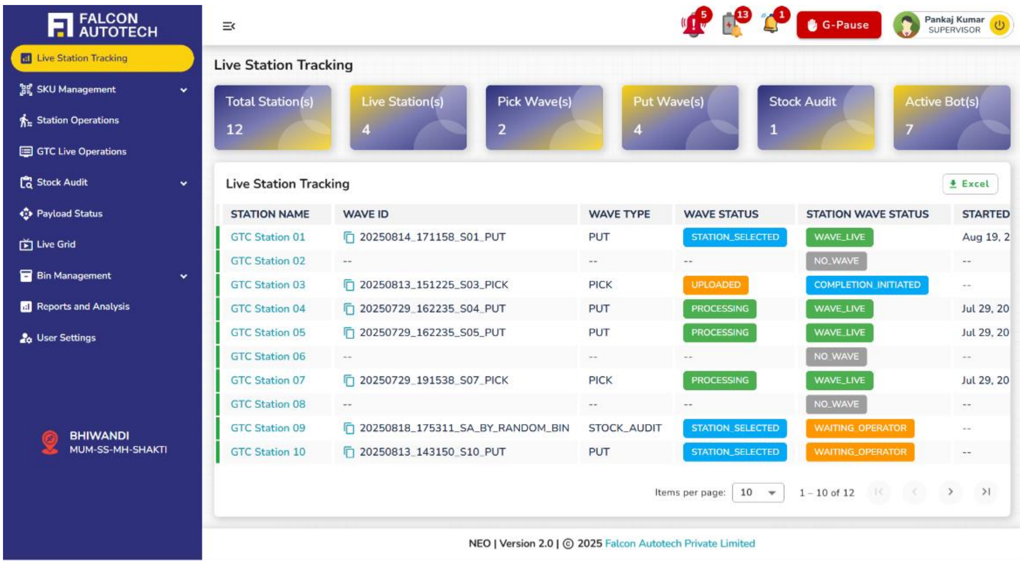
By utilizing the navigation bar, users can conveniently navigate between these user interactive screens, enabling them to access specific information, perform actions, and make informed decisions based on the data and functionalities offered by the IT dashboard.

4.1 Live Station Tracking

The **Live Station Tracking** page provides supervisors with real-time visibility into ongoing operations at each station. This feature enables quick monitoring of wave progress, PTL activity, and pallet handling, ensuring efficient tracking and timely decision-making.

Overview: Displays a summary view of all stations, including the waves currently running on each station. Provides drill-down options to monitor PTL activity during picking operations and pallet details during putaway operations.

- **Station Summary view:** A **station-level list** shows all active stations along with their associated running waves. Clicking on a station opens the detailed live view for that station.
- **Picking Operations (PTL View):** Clicking on a specific PTL reveals detailed information, including:
    - PTL Details: Identification and state of the PTL.
    - LPN Details: LPN(s) currently associated with the PTL.
    - Order Mapping: Orders mapped to the PTL.
    - Timestamps: Open and close time of the PTL.
    - Picking Details: SKU picked, picked quantity, and corresponding LPN.
- **Putaway Operations (Pallet View):** When the selected station is running a Putaway Operation.
- When the selected station is running a Putaway Operation. A live pallet data view is displayed. Users can view:
    - **Completed Pallets** – Pallets already processed.
    - **Active Pallets** – Pallets currently in putaway.
    - **Mapped Storage ID** – The destination storage location assigned to each pallet.




G-Pause
Pankaj Kumar
SUPERVISOR
LIVE STATION TRACKING > LIVE STATION-WISE TRACKING
Visual
Table
LEFT SIDE
RIGHT SIDE
722
719
716
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723
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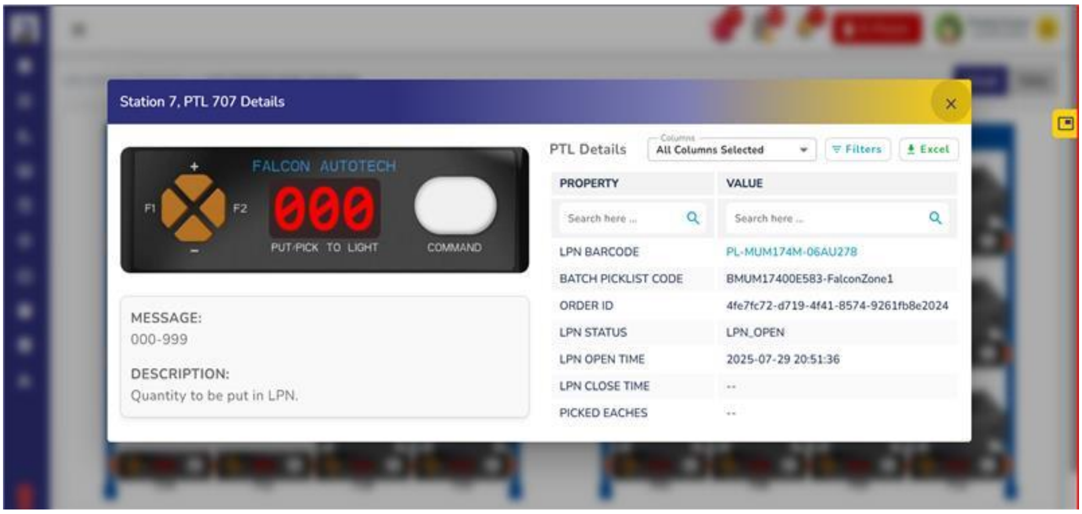
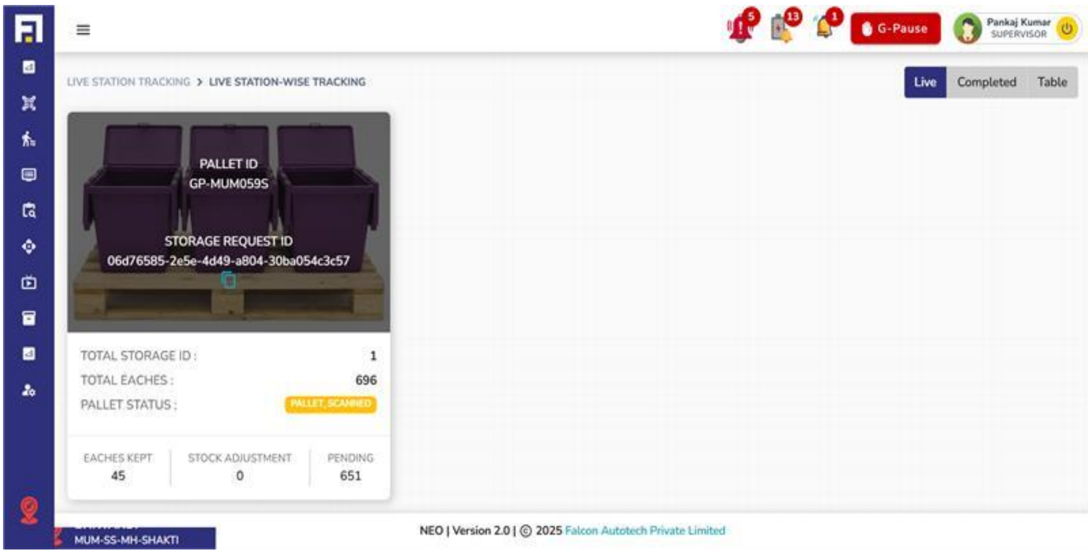

Station 7, PTL 707 Details
FALCON AUTOTECH
F1
F2
000
PUT/PICK TO LIGHT
COMMAND
MESSAGE:
000-999
DESCRIPTION:
Quantity to be put in LPN.
PTL Details
All Columns Selected
Filters
Excel
PROPERTY
VALUE
Search here ...
LPN BARCODE
PL-MUM174M-06AU278
BATCH PICKLIST CODE
BMUM17400E583-FalconZone1
ORDER ID
4fe7fc72-d719-4f41-8574-9261fb8e2024
LPN STATUS
LPN_OPEN
LPN OPEN TIME
2025-07-29 20:51:36
LPN CLOSE TIME
--
PICKED EACHES
--


G-Pause
Pankaj Kumar
SUPERVISOR
LIVE STATION TRACKING > LIVE STATION-WISE TRACKING
Live
Completed
Table
PALLET ID
GP-MUM059S
STORAGE REQUEST ID
06d76585-2e5e-4d49-a804-30ba054c3c57
TOTAL STORAGE ID : 1
TOTAL EACHES : 696
PALLET STATUS : PALLET_SCANNED
EACHES KEPT
45
STOCK ADJUSTMENT
0
PENDING
651
MUM-SS-MH-SHAKTI
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4.2 SKU Management

The **SKU Management** page is specifically designed to streamline the management of SKUs (Stock Keeping Units) in warehouse operations. It is located on the Navigation bar on the left side of the dashboard. This feature provides users with tools to manage SKU-related data effectively. Upon clicking on the SKU Management option, a drop-down menu appears with two sub-features: SKU Operations and Know Your SKU. These sub-features enable supervisors to perform specific tasks related to SKU handling.

4.2.1 SKU Operations

The SKU Operations section is one of the two sub-features under SKU Management. It provides access to two primary functionalities: SKU Master and SKU EAN Mapping, both accessible through respective buttons. This section is designed to help users upload, update, and manage SKU master data and SKU-EAN mappings seamlessly. When the user clicks on the SKU Operations option, a new screen opens displaying these two options, empowering users to choose based on the task they wish to perform.

Figure 4.2 SKU Operations

- **SKU Master**

The SKU Master functionality allows supervisors to upload or update the SKU master data for warehouse operations. The top section of this screen is dedicated to file uploads, with two convenient methods provided to the user. The first method involves clicking on the Select a CSV File to Upload button, which redirects the user to their file explorer to select a file from its saved location. Once selected, the user can click the Upload CSV button to complete the upload process. Alternatively, users can use the drag-and-drop functionality by simply dragging the file from its location and dropping it into the upload section, followed by clicking the Upload CSV button. After a successful upload, a confirmation message is displayed, and the uploaded data appears in a tabular format below the upload section. In the event of an error, an appropriate error message is shown, helping users address any issues, such as incorrect file formats or missing data.

Figure 4.3 SKU Master - CSV Upload

Additionally, the SKU Master screen offers a Sample CSV button, which allows users to download a sample file. This sample file serves as a reference to ensure the uploaded data adheres to the required format, thereby avoiding errors during uploads. Once the data is successfully uploaded, it is displayed in a table format containing columns like Max Bin Segment Storage, Max Quantity Storage, Volume of Each SKU, and Weight of Each SKU. Users can search for specific data within the table using the search boxes available for each column, enabling quick and efficient access to specific information.

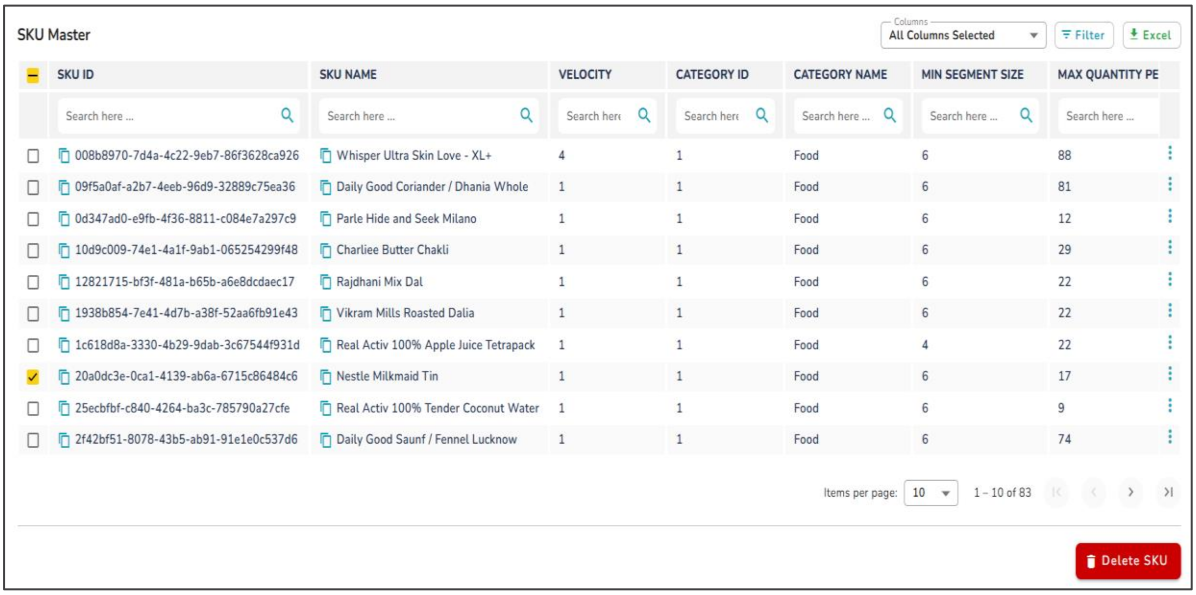


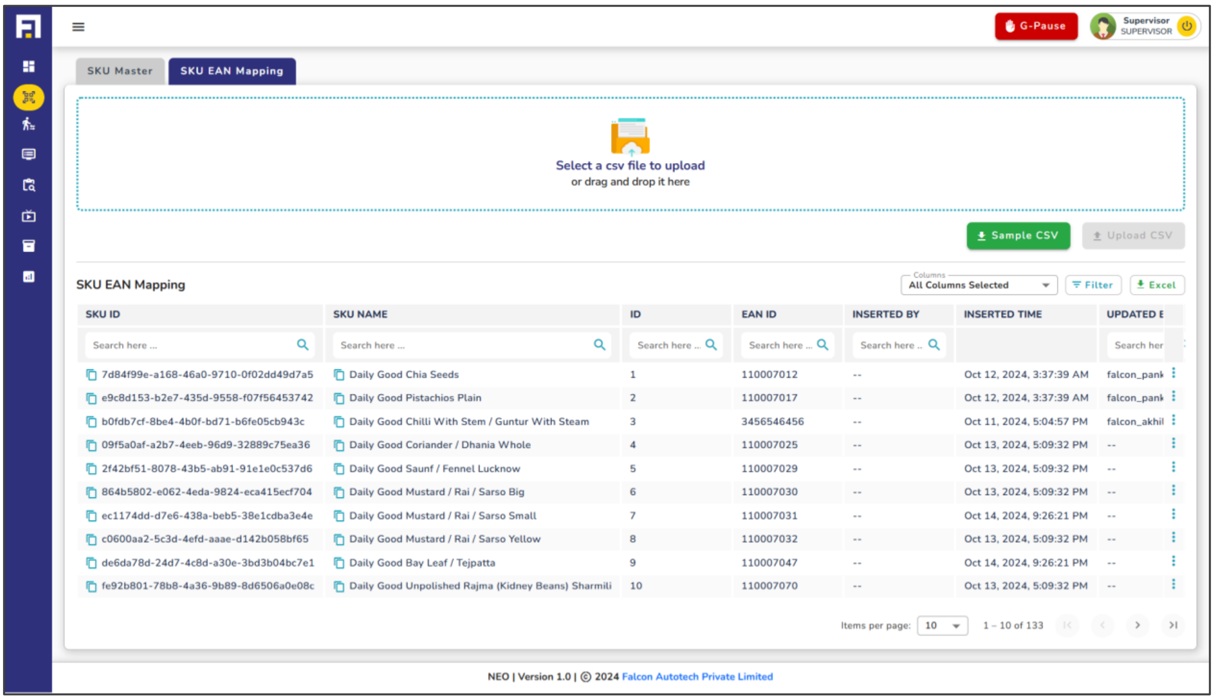
Figure 4.4 SKU - Master Tabular View

Description of Table Columns: -

- **SKU ID:** Unique alphanumeric identifier for each SKU.
- **SKU NAME:** Name of the product corresponding to the SKU.
- **VELOCITY**: Classification of the SKU based on movement speed (e.g., slow mover, fast mover, or medium mover).
- **CATEGORY NAME:** Specifies whether the SKU belongs to categories such as Food or Non-Food.
- **MIN SEGMENT SIZE:** Minimum storage segment size required for the SKU.
- **MAX QUANTITY PER SEGMENT:** Maximum number of SKUs allowed in a single storage segment.
- **MIN BIN SEGMENT STORAGE:** Minimum number of bins needed to store this SKU.
- **MAX BIN SEGMENT STORAGE:** Maximum number of bins permitted for storing this SKU.
- **MAX QUANTITY STORAGE (In Inventory):** Maximum allowable quantity of the SKU that can be stored in inventory.
- **VOLUME OF EACH SKU (cubic cm):** Physical volume of a single SKU.
- **WEIGHT OF EACH SKU (grams):** Weight of an individual SKU.
- **IMAGE URL:** Path to the SKU image for display on the dashboard.

- **EAN SKU Mapping: -**

The EAN SKU Mapping functionality allows users to upload or update the mapping between SKUs and their respective European Article Numbers (EANs) efficiently. Users can access this feature by clicking on the "EAN SKU Mapping" button available on the SKU Operations screen. This functionality supports two upload methods: selecting a file using the Select a CSV File to Upload button or utilizing the drag-and-drop feature for added convenience.



SKU ID	SKU NAME	ID	EAN ID	INSERTED BY	INSERTED TIME	UPDATED E
7d84f99e-a168-46a0-9710-0f02dd49d7a5	Daily Good Chia Seeds	1	110007012	--	Oct 12, 2024, 3:37:39 AM	falcon_pank
e9c8d153-b2e7-435d-9558-f07f56453742	Daily Good Pistachios Plain	2	110007017	--	Oct 12, 2024, 3:37:39 AM	falcon_pank
b0fdb7cf-8be4-4b0f-bd71-b6fe05cb943c	Daily Good Chilli With Stem / Guntur With Steam	3	3456546456	--	Oct 11, 2024, 5:04:57 PM	falcon_akhil
09f5a0af-a2b7-4eeb-96d9-32889c75ea36	Daily Good Coriander / Dhania Whole	4	110007025	--	Oct 13, 2024, 5:09:32 PM	--
2f42bf51-8078-43b5-ab91-91e1e0c537d6	Daily Good Saunf / Fennel Lucknow	5	110007029	--	Oct 13, 2024, 5:09:32 PM	--
864b5802-e062-4eda-9824-eca415ecf704	Daily Good Mustard / Rai / Sarso Big	6	110007030	--	Oct 13, 2024, 5:09:32 PM	--
ec1174dd-d7e6-438a-beb5-38e1cdba3e4e	Daily Good Mustard / Rai / Sarso Small	7	110007031	--	Oct 14, 2024, 9:26:21 PM	--
c0600aa2-5c3d-4efd-aaae-d142b058bf65	Daily Good Mustard / Rai / Sarso Yellow	8	110007032	--	Oct 13, 2024, 5:09:32 PM	--
de6da78d-24d7-4c8d-a30e-3bd3b04bc7e1	Daily Good Bay Leaf / Tejpatta	9	110007047	--	Oct 14, 2024, 9:26:21 PM	--
fe92b801-78b8-4a36-9b89-8d6506a0e08c	Daily Good Unpolished Rajma (Kidney Beans) Sharmili	10	110007070	--	Oct 13, 2024, 5:09:32 PM	--

Items per page: 10 1 – 10 of 133

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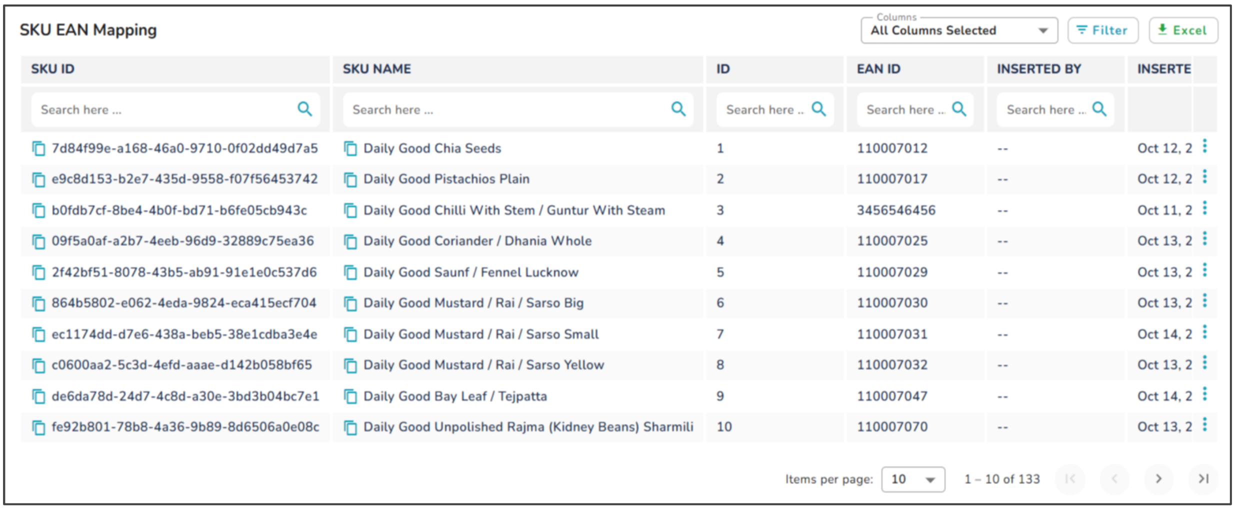
Once the file is selected or dragged, users must click the Upload CSV button to complete the upload process. After successful upload, the system displays a confirmation message, and the uploaded data is presented in a structured tabular format. This ensures the mapping data is accurate, up-to-date, and readily accessible for further operations.



SKU ID	SKU NAME	ID	EAN ID	INSERTED BY	INSERTE
7d84f99e-a168-46a0-9710-0f02dd49d7a5	Daily Good Chia Seeds	1	110007012	--	Oct 12, 2
e9c8d153-b2e7-435d-9558-f07f56453742	Daily Good Pistachios Plain	2	110007017	--	Oct 12, 2
b0fdb7cf-8be4-4b0f-bd71-b6fe05cb943c	Daily Good Chilli With Stem / Guntur With Steam	3	3456546456	--	Oct 11, 2
09f5a0af-a2b7-4eeb-96d9-32889c75ea36	Daily Good Coriander / Dhania Whole	4	110007025	--	Oct 13, 2
2f42bf51-8078-43b5-ab91-91e1e0c537d6	Daily Good Saunf / Fennel Lucknow	5	110007029	--	Oct 13, 2
864b5802-e062-4eda-9824-eca415ecf704	Daily Good Mustard / Rai / Sarso Big	6	110007030	--	Oct 13, 2
ec1174dd-d7e6-438a-beb5-38e1cdba3e4e	Daily Good Mustard / Rai / Sarso Small	7	110007031	--	Oct 14, 2
c0600aa2-5c3d-4efd-aaae-d142b058bf65	Daily Good Mustard / Rai / Sarso Yellow	8	110007032	--	Oct 13, 2
de6da78d-24d7-4c8d-a30e-3bd3b04bc7e1	Daily Good Bay Leaf / Tejpatta	9	110007047	--	Oct 14, 2
fe92b801-78b8-4a36-9b89-8d6506a0e08c	Daily Good Unpolished Rajma (Kidney Beans) Sharmili	10	110007070	--	Oct 13, 2

Items per page: 10 1 – 10 of 133

Description of Table Columns: -

SKU ID: This column represents the unique identifier for each Stock Keeping Unit (SKU) in the system. It ensures precise mapping and tracking of inventory items.

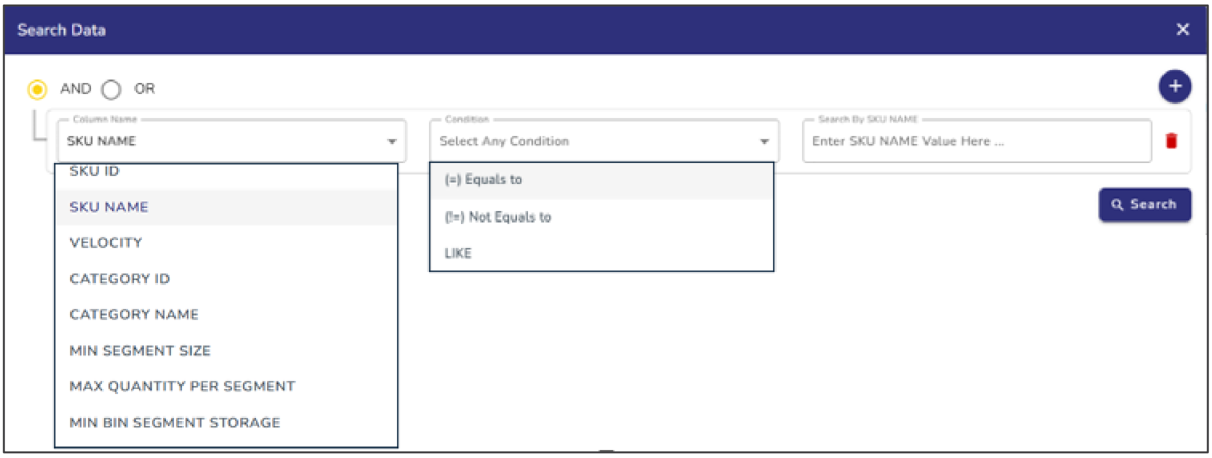
- SKU Name: This column contains the descriptive name of the SKU, providing easy identification and readability for the associated product.
- EAN ID: This field displays the European Article Number (EAN) associated with the SKU, which is critical for global product identification and standardization.
- Inserted By: This column records the name or ID of the user who initially uploaded or created the SKU-EAN mapping. It ensures the traceability of data entry activities.
- Inserted Time: This field captures the exact timestamp of when the SKU-EAN mapping was created, enabling effective tracking of data changes.
- Updated By: This column identifies the user who last modified the SKU-EAN mapping. It ensures accountability for any updates made to the data.
- Updated Time: This field logs the most recent timestamp when the SKU-EAN mapping was updated, allowing users to track the latest changes.

Table Functionalities: -

The **EAN-SKU Mapping** table provides advanced functionalities for enhanced data handling and user experience.

On the upper right side of the table, three options are available Columns, Filter, and Excel.

- **Columns Dropdown**:
  On the upper right side of the table, a **Columns** button is available. When clicked, a dropdown menu opens, allowing users to show or hide specific columns in the table. Users can select or deselect columns from this menu based on their preference, and the table will adjust accordingly to display only the selected columns.
- **Filter Button**:
  The **Filter** button allows users to perform detailed searches on the table data. When clicked, a filter window opens where users can apply search conditions to find the required data. This window includes a **+** button, enabling users to add multiple conditions for their search. This feature is especially useful for complex searches across the dataset.



- **Excel Download**:
  The **Excel** button is available to download all the tabular data in Excel format. Users can utilize this feature whenever they need to export the data for reporting or offline analysis.

4.2.2 Know Your SKU

The **Know Your SKU** page is a dedicated feature that allows users to easily search and retrieve detailed SKU-level information. This feature can be accessed via the **Know Your SKU** option available on the navigation bar.

The following key features are available on this page.

1. **Search Options:**

- A dropdown menu allows users to search for SKUs using the following parameters:
    - **SKU Name**: Search based on the exact or partial SKU name.
    - **SKU ID**: Search using the unique SKU identifier.
    - **EAN ID**: Search based on the European Article Number (EAN).
- **Input Box**: After selecting the desired search parameter, users can enter the appropriate value (e.g., SKU name, ID, or EAN) into the search box.
- **Auto-Suggest**: While typing the SKU Name, relevant matching SKUs are displayed in a dropdown list, providing quick selection.

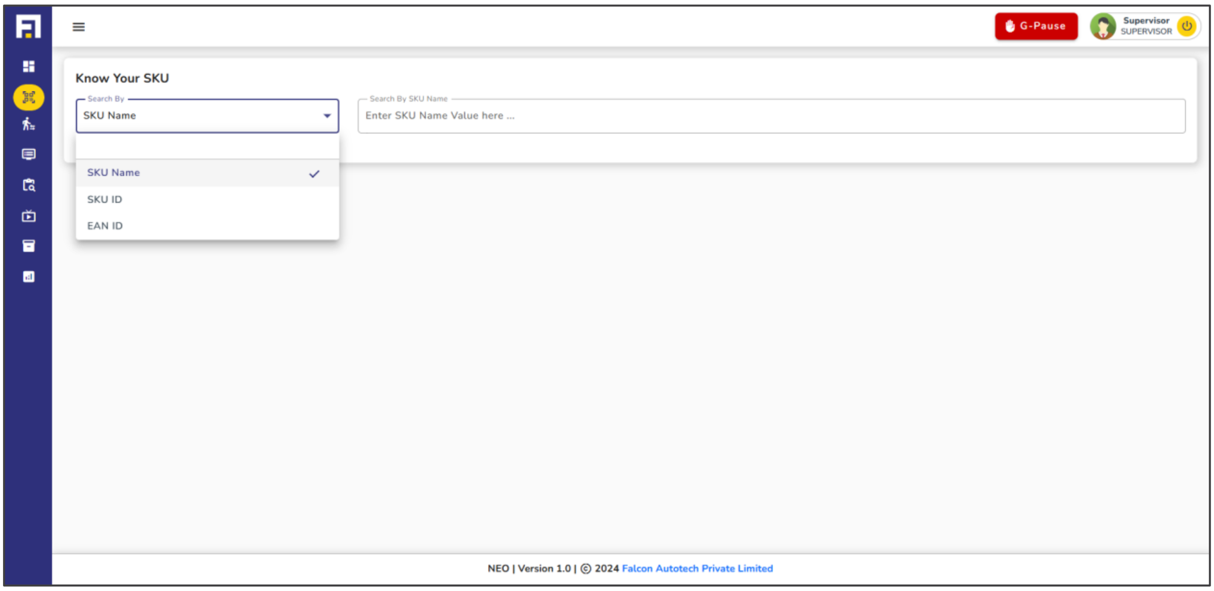


Figure 4.5 Know Your SKU – Search SKU

2. **SKU Information Display**:

- Upon selecting a specific SKU, a detailed information window appears below the search bar.
- This window contains **tabbed sections** that provide comprehensive details about the selected SKU. Tabs include:
  - **SKU Info**: Shows basic details such as SKU ID, SKU Name, and Quantity.
  - **SKU Master**: Provides master-level SKU information (not shown in the screenshot).
  - **EAN Details**: Displays mapping details between SKU and EAN IDs.
  - **Inventory Details**: Lists inventory availability and stock-related data.
  - **Batch Details**: Displays batch-level information for SKUs.

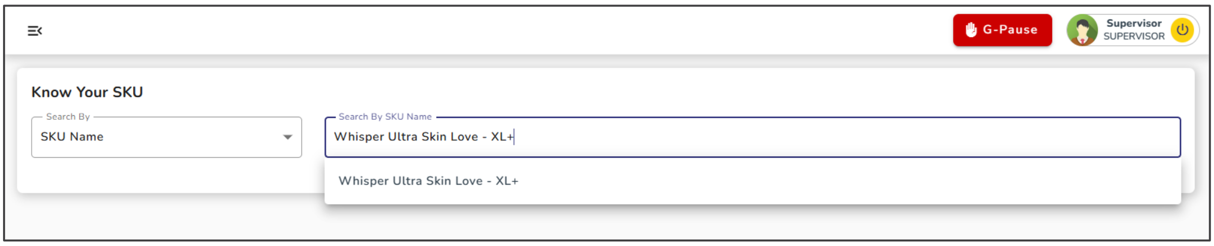


Figure 4.6 Know Your SKU - SKU Information Display

3. **Data Presentation:**

- SKU data is displayed in a tabular format with key columns such as:
  - **SKU ID**: A unique identifier for the SKU.
  - **SKU Name:** The product name for the SKU.
  - **SUM (Quantity):** Total quantity available for the SKU.
- Users can filter data or export it to Excel for further analysis.

Figure 4.7 Know Your SKU - Data Presentation

4.3 Station Operations

The Station Operations module on the dashboard is designed to provide users with tools for monitoring and configuring stations for various operations. It is accessible via the Navigation Bar and is divided into two sub-sections:

- Summary
- Operations

4.3.1 Summary

The Summary page provides a high-level overview of station statuses in a tabular format. This section is meant for monitoring purposes only, and users cannot perform any operations here. The tabular view offers detailed insights into various parameters of the stations, enabling users to understand their current state at a glance.

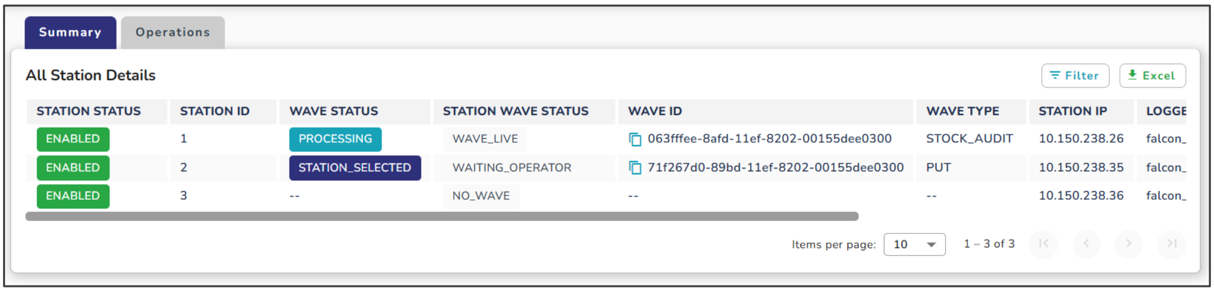


Figure 4.8 Station Operations - Summary

Table Columns in the Summary View: The table includes the following columns:

- **STATION STATUS**: Displays the current operational status of the station (e.g., Active, Inactive, Error).
- **STATION ID**: A unique identifier assigned to each station.
- **WAVE STATUS**: Indicates the status of the wave associated with the station.
- **STATION WAVE STATUS**: Provides a detailed status of the wave at the station level.
- **WAVE ID**: Displays the unique identifier of the active wave.
- **WAVE TYPE**: Specifies the type of wave assigned to the station (e.g., Pick Wave, Put Wave).
- **LOGGED IN USER ID**: Identifies the user currently logged into the station.
- **PICK RULE NAME(DEFAULT)**: The default rule applied to the pick process for that station.
- **PICK RULE NAME CURRENT**: The current rule applied to the pick process for that station.
- **BOT COUNT(DEFAULT)**: This represents the predefined or configured number of bots that are normally assigned to the station during standard operations.
- **BOT COUNT CURRENT**: This indicates the actual number of bots currently assigned to the station at the moment.

4.3.2 Operations

The Operations page allows supervisors to manage and configure the stations. This includes enabling or disabling stations and handling wave/process assignments.

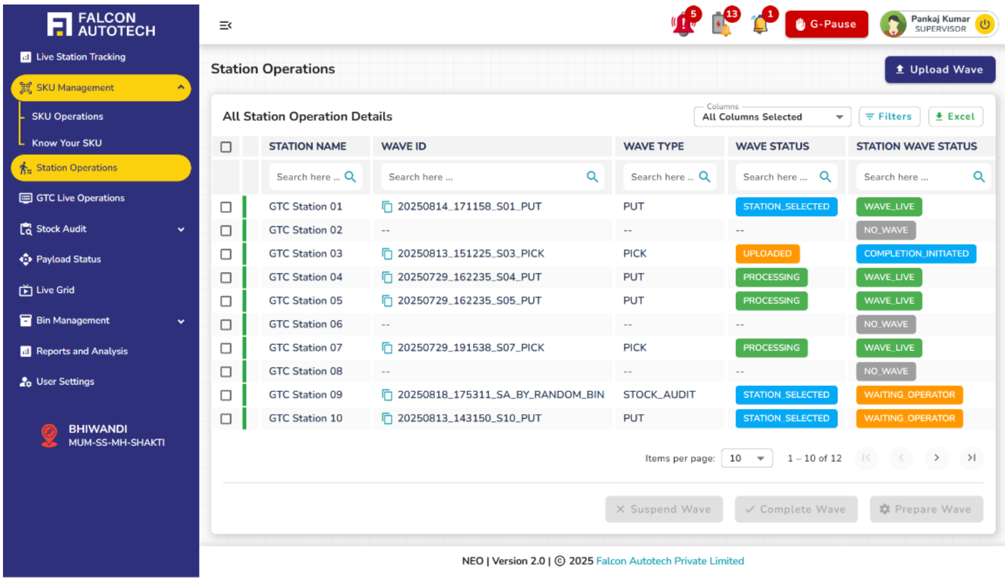


Figure 4.9 Station Operations – Operations

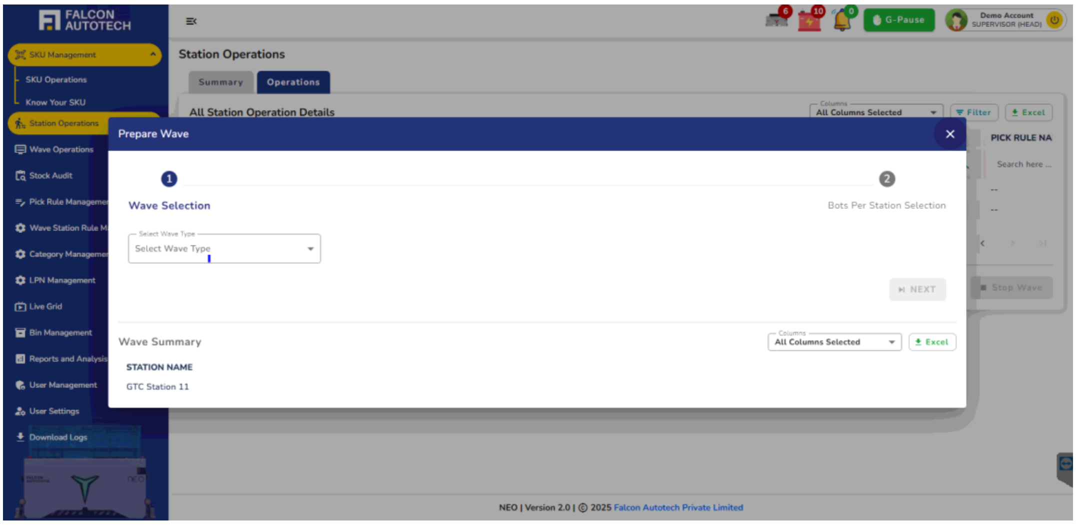
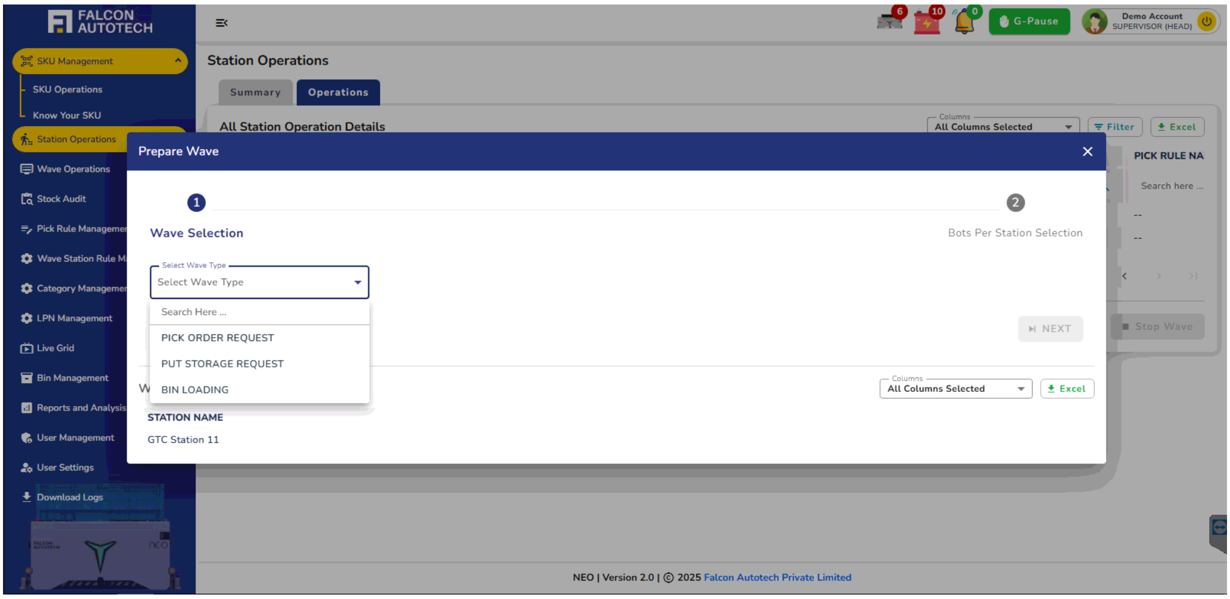


Figure 4.10 Station Operations – Wave Selection



FALCON AUTOTECH
SKU Management
SKU Operations
Know Your SKU
Station Operations
Wave Operations
Stock Audit
LPN Management
Live Grid
Bin Management
User Management
User Settings
Download Logs
G-Pause
Demo Account
SUPERVISOR (HEAD)
Station Operations
Summary
Operations
All Station Operation Details
All Columns Selected
Filter
Excel
Prepare Wave
Wave Selection
Bots Per Station Selection
Select Wave Type
Search Here ...
PICK ORDER REQUEST
PUT STORAGE REQUEST
BIN LOADING
NEXT
STATION NAME
GTC Station 11
PICK RULE NA
Stop Wave
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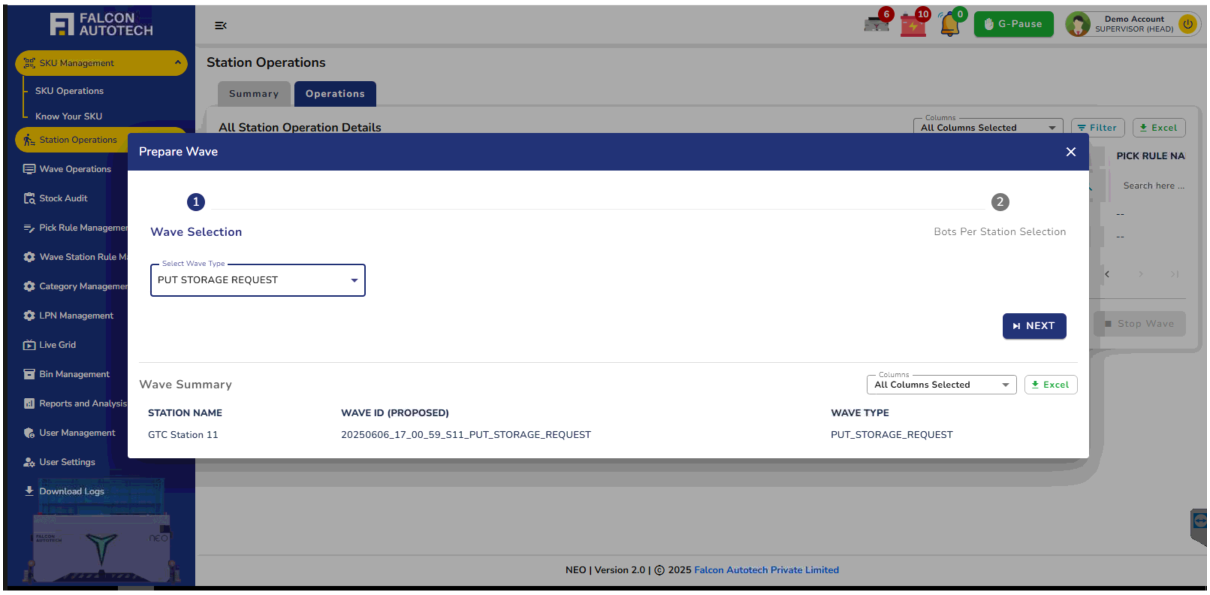

FALCON AUTOTECH
SKU Management
SKU Operations
Know Your SKU
Station Operations
Wave Operations
Stock Audit
LPN Management
Live Grid
Bin Management
User Management
User Settings
Download Logs
G-Pause
Demo Account
SUPERVISOR (HEAD)
Station Operations
Summary
Operations
All Station Operation Details
Prepare Wave
Wave Selection
Bots Per Station Selection
Select Wave Type
PUT STORAGE REQUEST
NEXT
Wave Summary
All Columns Selected
Excel
STATION NAME
WAVE ID (PROPOSED)
WAVE TYPE
GTC Station 11
20250606_17_00_59_S11_PUT_STORAGE_REQUEST
PUT_STORAGE_REQUEST
Stop Wave
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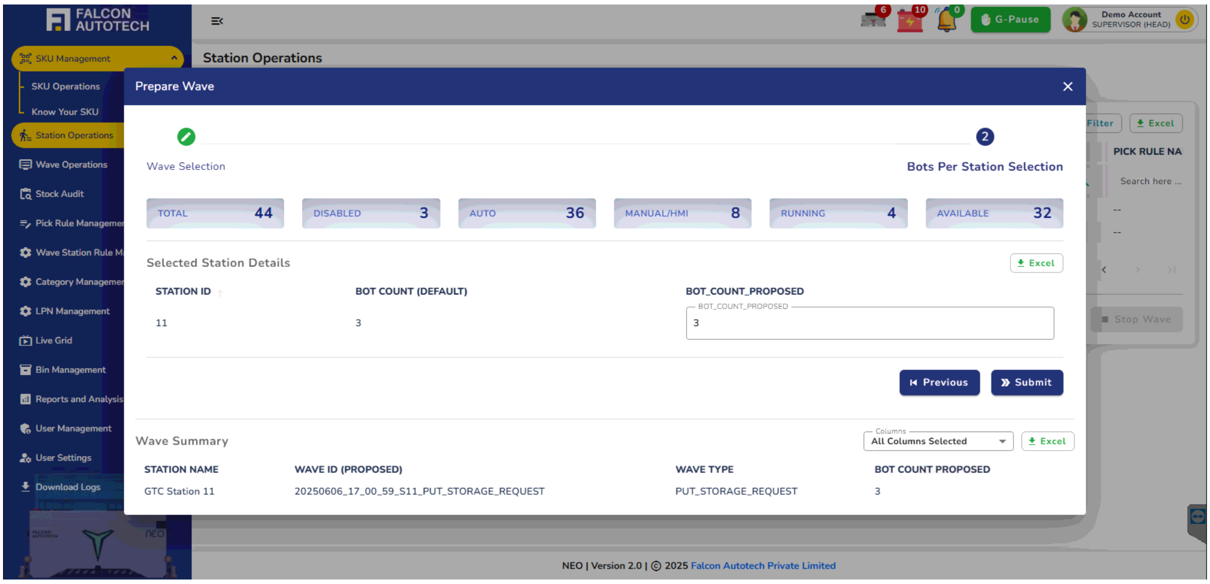

Station Operations
Prepare Wave
Wave Selection
Bots Per Station Selection
TOTAL 44
DISABLED 3
AUTO 36
MANUAL/HMI 8
RUNNING 4
AVAILABLE 32
Selected Station Details
STATION ID
BOT COUNT (DEFAULT)
BOT_COUNT_PROPOSED
11
3
3
Previous
Submit
Wave Summary
STATION NAME
WAVE ID (PROPOSED)
WAVE TYPE
BOT COUNT PROPOSED
GTC Station 11
20250606_17_00_59_S11_PUT_STORAGE_REQUEST
PUT_STORAGE_REQUEST
3
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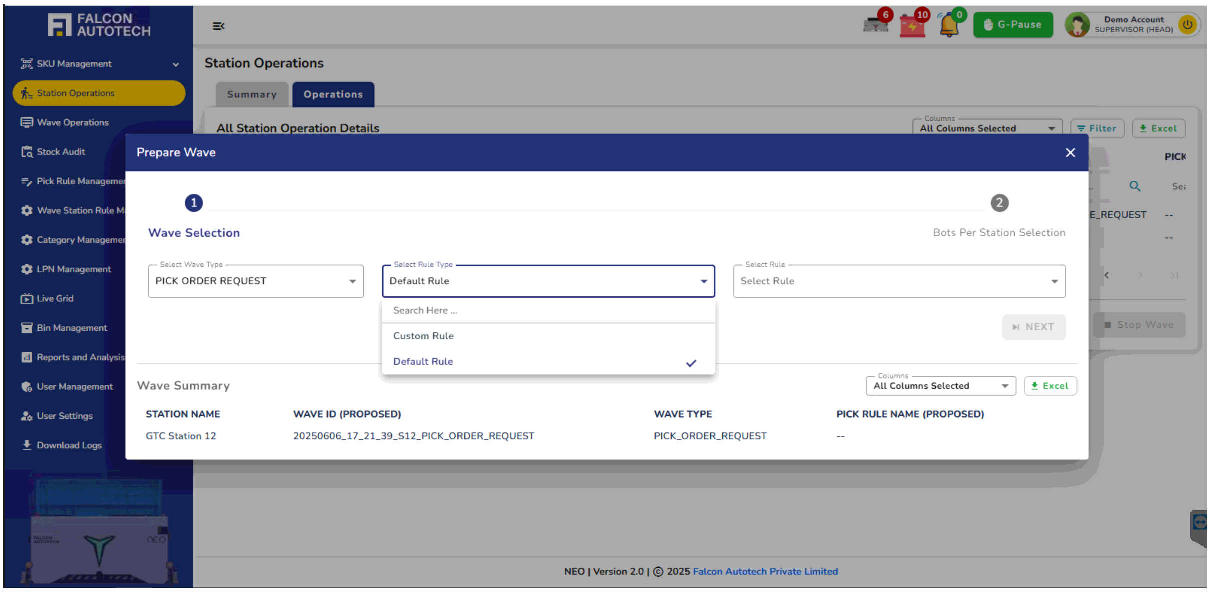

Station Operations
Summary
Operations
All Station Operation Details
Prepare Wave
Wave Selection
Bots Per Station Selection
PICK ORDER REQUEST
Default Rule
Select Rule
Search Here ...
Custom Rule
Default Rule
NEXT
Wave Summary
STATION NAME
WAVE ID (PROPOSED)
WAVE TYPE
PICK RULE NAME (PROPOSED)
GTC Station 12
20250606_17_21_39_S12_PICK_ORDER_REQUEST
PICK_ORDER_REQUEST
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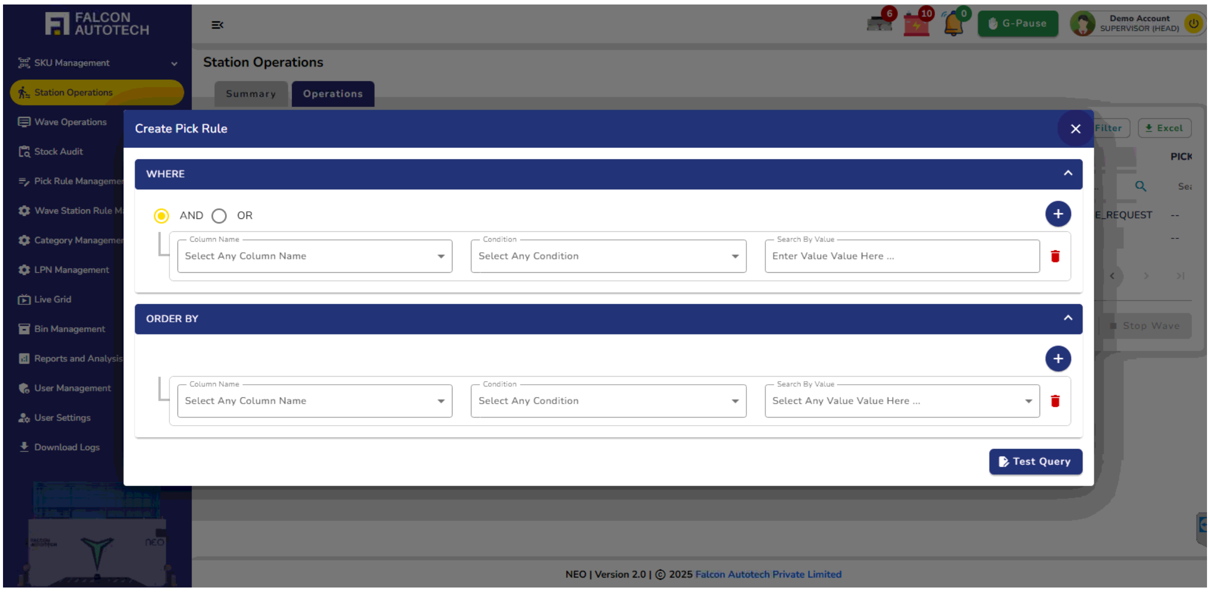


Purpose of the Operations Page:

- The Operations Page serves as a control panel for warehouse supervisors to manage and optimize station-level operations. It enables:
- Configuration of available stations for operational readiness.
- Dynamic enabling or disabling of stations based on real-time needs.
- Assignment and management of waves per station.
- Selection of wave type, bot allocation, and rule configuration tailored to operational requirements.

Key Functionalities of the Operations Page:

1. **Station Configuration**:
   - Supervisors can enable or disable individual stations.
   - If no wave is assigned to a station, its operational status can be toggled as required.
   - Disabled stations will not be considered for wave assignments or bot allocations.
2. **Wave Assignment**:
   - Supervisors can assign waves to available, enabled stations.
   - Stations without assigned waves display NO_WAVE in the STATION WAVE STATUS column.
   - This feature ensures clarity on which stations are idle and ready for allocation.
3. **Wave Type and Rule Management:**
   - Supervisors can select the wave type (Pick or Put) during assignment.
   - **For Order request waves:**

- Supervisors can choose an existing Pick Rule from a predefined list or create a new rule at the station level.
- Rule creation allows using logical conditions (AND/OR) and sorting (ORDER BY clause) to define order selection criteria.
- This provides flexibility in filtering and prioritizing orders for processing.

4. **Bot Allocation per station:**
   - Supervisors can configure the number of bots per station based on workload and wave type.
   - BOT COUNT (DEFAULT): The standard number of bots allocated to a station under normal conditions.
   - BOT COUNT (CURRENT): The actual number of bots currently allocated to that station, which may differ due to dynamic needs or manual overrides.

5. **Error Handling and Wave stop functionality**:
   - If a station has an active or in-progress wave, an error message will appear when attempting to disable it.
   - **Error Message**: "Stations with running wave can't be deactivated."
   - The pop-up provides details of the station, such as station_id, StationStatus, and WaveStatus, to help supervisors understand why the station cannot be disabled.
   - Instead of disabling the station, supervisors can use the **"Stop Wave"** button to manage active waves.
   - **Behavior when "Suspend Wave" is triggered**:
     1. All orders currently being processed (already picked or in progress) will be allowed to complete.
     2. Any pending or unstarted orders in the wave will be excluded from further processing.
     3. The wave status will transition to a terminal or interrupted state based on system design.

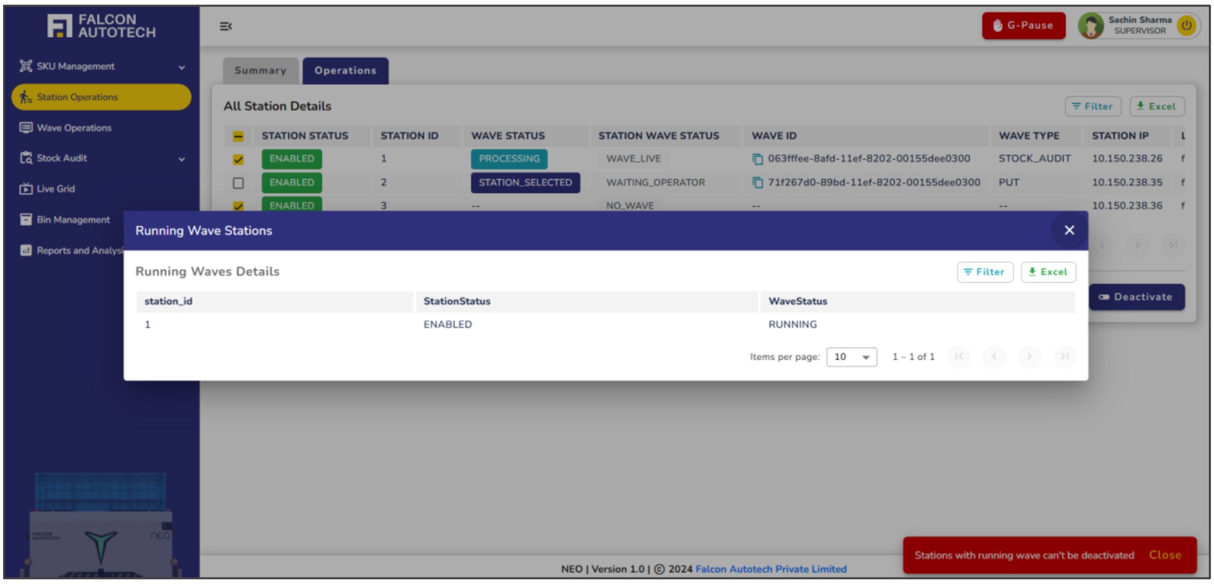


Figure 4.11 Station Operations - Station Enabling and Disabling

Table Information on the Operations Page

The table on the **Operations** page provides detailed information to help supervisors configure and manage station operations effectively. Below is the description of the columns:

- **STATION STATUS**: Displays whether the station is currently enabled or disabled.
- **STATION ID**: The unique identifier assigned to each station.
- **WAVE STATUS**: The status of the wave associated with the station (e.g., PROCESSING, STATION_SELECTED).
- **STATION WAVE STATUS**: Provides detailed information about the wave state at the station (e.g., WAVE_LIVE, WAITING_OPERATOR, NO_WAVE).
- **WAVE ID**: The unique identifier of the wave assigned to the station, if any.
- **WAVE TYPE**: Specifies the type of wave assigned to the station (e.g., PUT, STOCK_AUDIT).
- **STATION IP**: The IP address of the station used for network identification.
- **LOGGED IN USER ID**: The user ID of the operator currently logged in to the station.
- **LAST LOGIN TIME**: The timestamp of the most recent login by an operator on the station.
- **LAST LOGOUT TIME**: The timestamp of the most recent logout by an operator from the station.

4.4 Stock Audit

The **Stock Audit** feature in the NEO system allows supervisors and inspection engineers to audit inventory effectively. It supports various scenarios like damaged, missing, or expired items and ensures seamless inventory accuracy. The Stock Audit menu provides four options: **Audit Wave**, **Random Bin Recall**, **By Bin ID**, and **By SKU ID**.

Figure 4.12 Stock Audit Options

Each audit wave type follows a similar workflow involving actions like wave creation, leftover checks, approvals, and station selection. Below are the details for each wave type.

4.4.1 Audit Wave

The **Audit Wave** enables supervisors to generate an audit wave by selecting specific criteria like date ranges and short-put reasons (**Damaged Quantity**, **Missing Quantity**, or **Expired Quantity**).

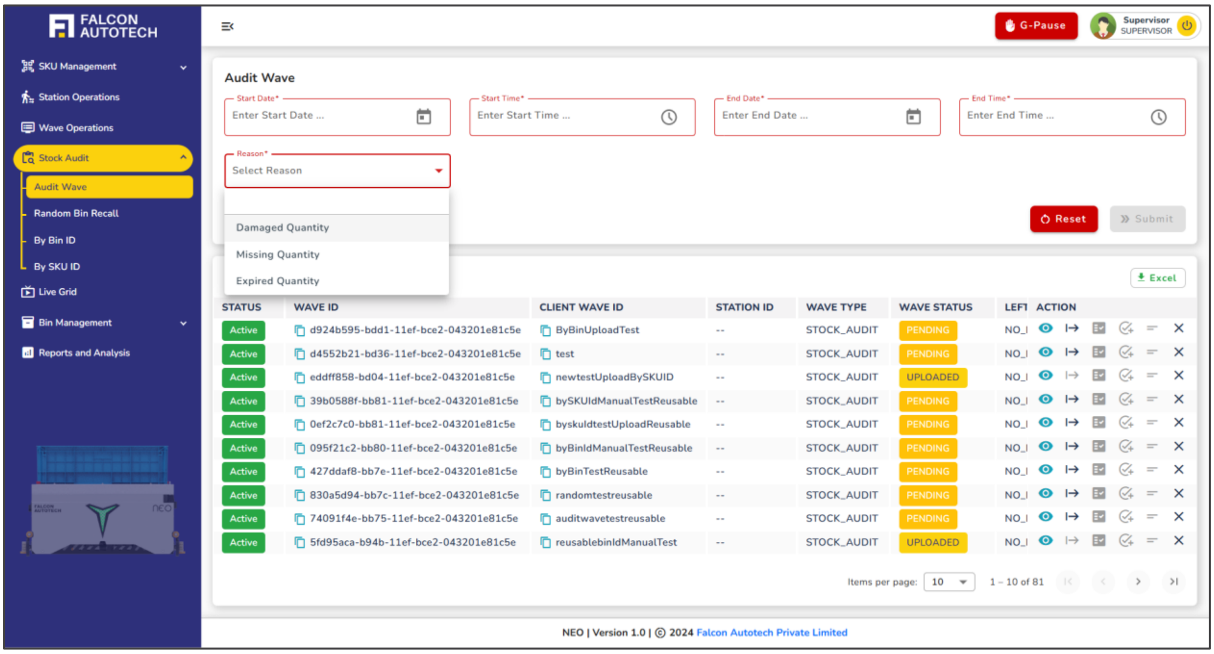


Figure 4.13 Audit Wave

Actions for Audit Wave

1. Upload or Create Wave:

   The system automatically creates an audit wave when the supervisor inputs the start date, end date, and selects a short-put reason. The user can view the list of bins included in the audit wave based on the given conditions, along with the corresponding SKUs/articles present in each bin.

2. Station Selection

   After the wave is created, the supervisor must click the Select Station button to assign one or more stations for processing the audit.

3. Wave Execution:

   Once stations are assigned, the audit tasks begin at the designated stations. The system guides operators through the bins and associated SKUs/articles that need to be verified. Supervisors can monitor the execution and progress of the audit in real-time through the Reports section.

4.4.2 Random Bin Recall

The **Random Bin Recall** option allows supervisors to verify system-indicated inventory accuracy by recalling a random number of bins from the system.

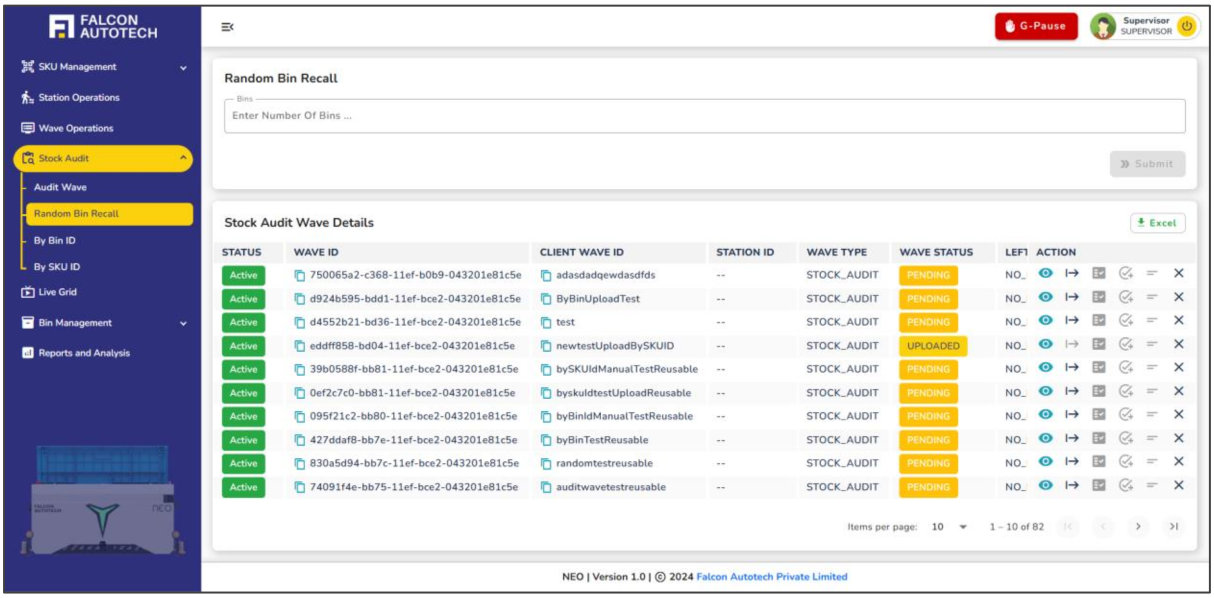


Figure 4.14 Random Bin Recall

Actions for Random Bin Recall Wave

1. Upload or Create Wave:

   The supervisor inputs the number of bins to recall, and the system generates a wave with all corresponding bins and their segments. The user can view the list of bins included in the audit wave based on the given conditions, along with the corresponding SKUs/articles present in each bin.

2. Station Selection

   After the wave is created, the supervisor must click the Select Station button to assign one or more stations for processing the audit.

3. Wave Execution:

   Once stations are assigned, the audit tasks begin at the designated stations. The system guides operators through the bins and associated SKUs/articles that need to be verified. Supervisors can monitor the execution and progress of the audit in real-time through the Reports section.

4.4.3 Bin Recall By Bin ID

The **By Bin ID** option allows the supervisor to create an audit wave for specific bins.

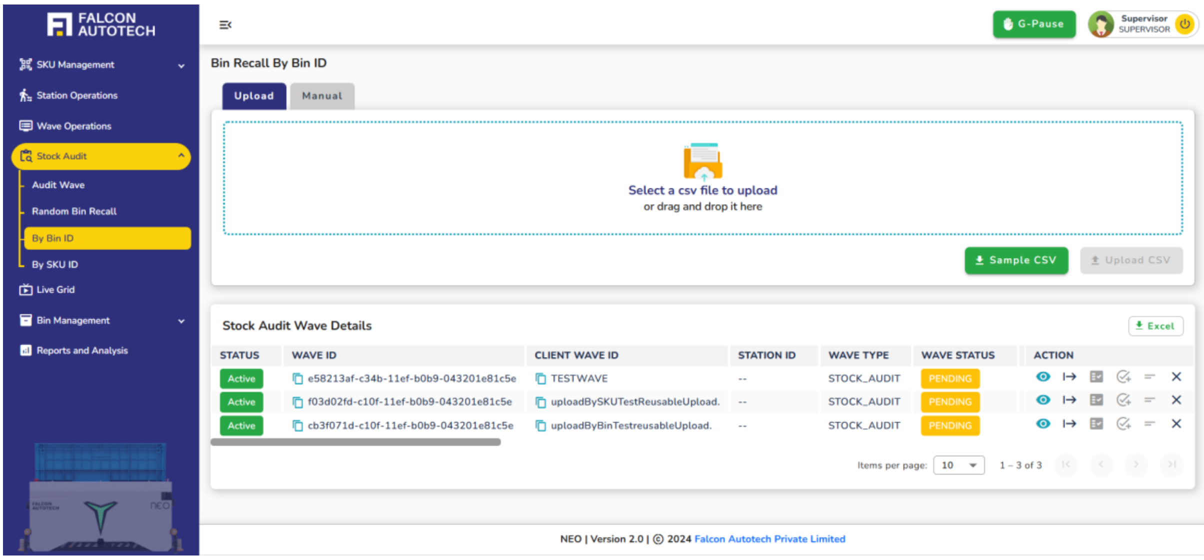


Figure 4.15 Bin Recall By Bin ID

Actions for Bin Recall By Bin ID

1. **Upload or Create Wave:**

   The supervisor uploads a CSV file with the Bin IDs. Manually enter the Bin IDs into the system. The user can view the list of bins included in the audit wave based on the given conditions, along with the corresponding SKUs/articles present in each bin.

2. **Station Selection**

   After the wave is created, the supervisor must click the Select Station button to assign one or more stations for processing the audit.

3. **Wave Execution:**

   Once stations are assigned, the audit tasks begin at the designated stations. The system guides operators through the bins and associated SKUs/articles that need to be verified. Supervisors can monitor the execution and progress of the audit in real-time through the Reports section.

4.4.4 Bin Recall By SKU ID

The **By SKU ID** option enables the supervisor to create an audit wave for specific SKUs.

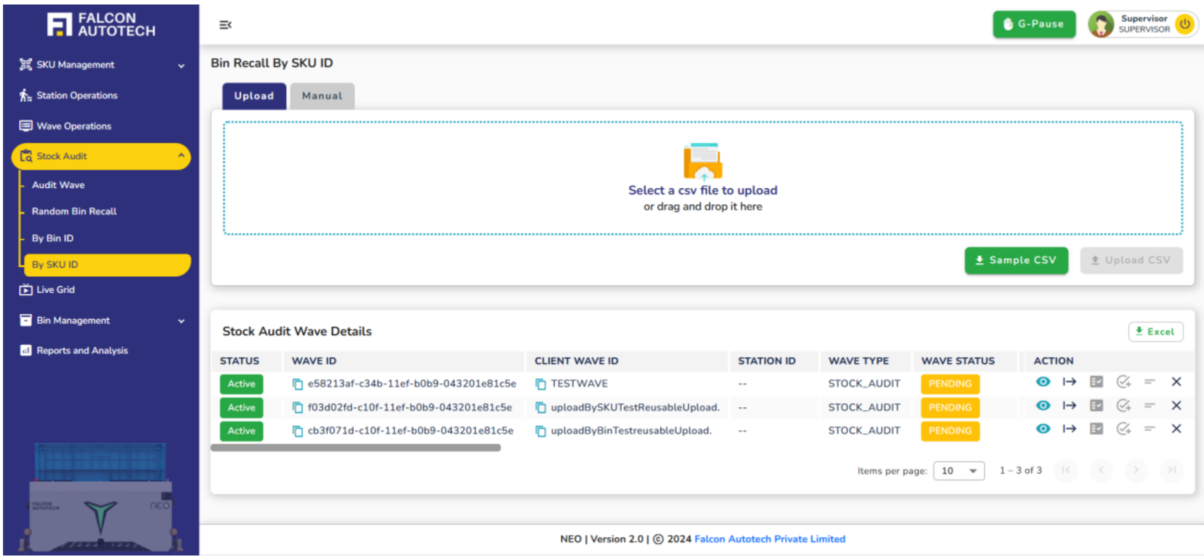


Figure 4.16 Bin Recall By SKU ID

Actions for Bin Recall By SKU ID

4. Upload or Create Wave:

   The supervisor selects one or more SKU IDs to create an audit wave. Manually enter the Bin IDs into the system. The user can view the list of bins included in the audit wave based on the given conditions, along with the corresponding SKUs/articles present in each bin.

5. Station Selection

   After the wave is created, the supervisor must click the Select Station button to assign one or more stations for processing the audit.

6. Wave Execution:

   Once stations are assigned, the audit tasks begin at the designated stations. The system guides operators through the bins and associated SKUs/articles that need to be verified. Supervisors can monitor the execution and progress of the audit in real-time through the Reports section.

4.5 Pick Rule Management

The Pick Rule Management module is used to define and manage rules that govern how items should be picked during the operation. This section displays a table listing all existing pick rules, along with their filter conditions, the supervisor who created them, and the timestamp of insertion.

7. Each row in the table represents a unique pick rule currently active in the system.

8. A dedicated "**+ Pick Rule**" button is available for supervisors to insert new rules into the system. On clicking this button, the user is presented with an interface like the Station Operations module, where they can configure the rule logic, define filters, and submit the rule for activation.

9. This module ensures traceability, consistency, and control over the picking logic used across the warehouse or station network.

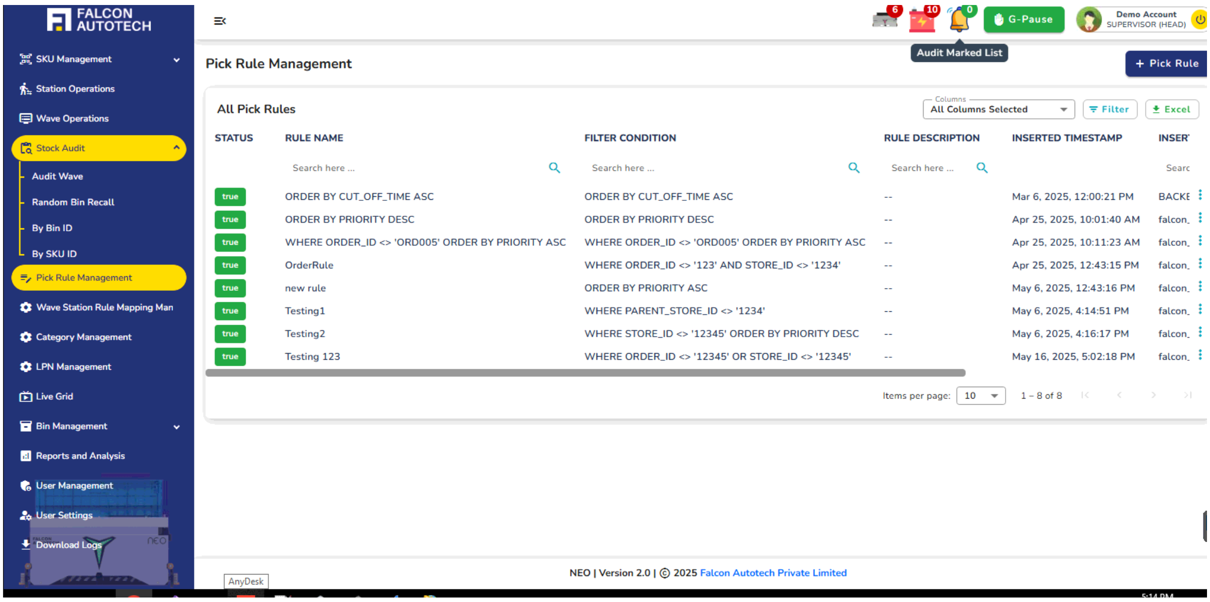
10. On clicking on **Filter condition** tab inside **Pick Rule interface,** the user is presented with a configurable section where they can:

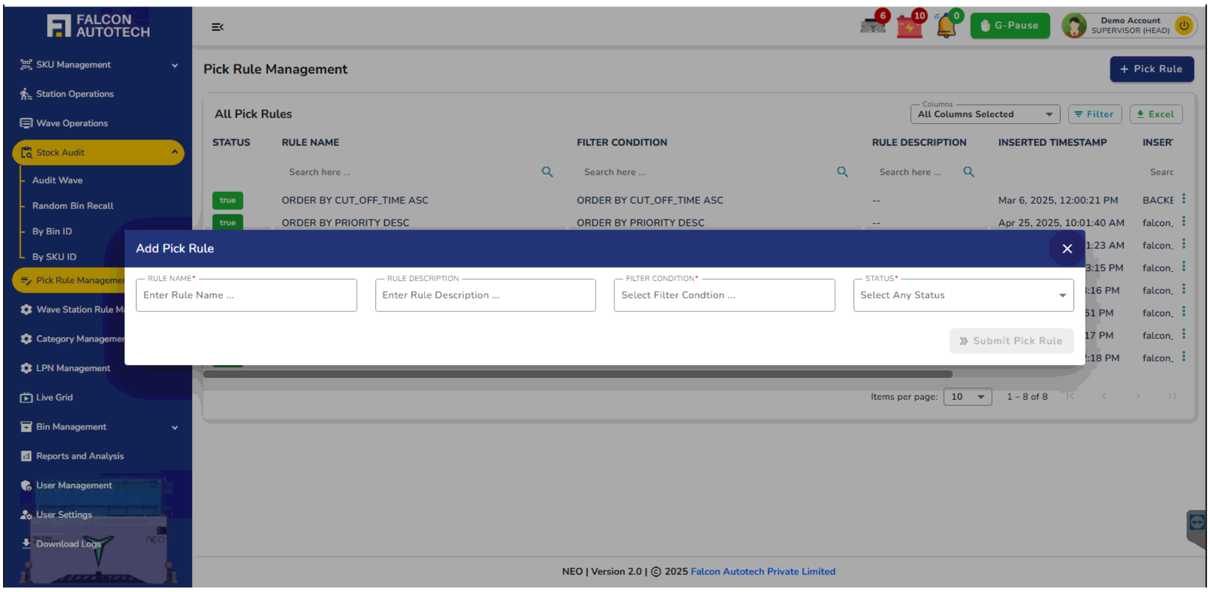
    - Select fields or parameters.
    - Set conditions using operators like equals (=), not equals (≠), contains, greater than, etc.
    - Add multiple filter conditions using "AND"/"OR" logic to define complex pick logic.
    - Preview the applied filter set dynamically before saving.

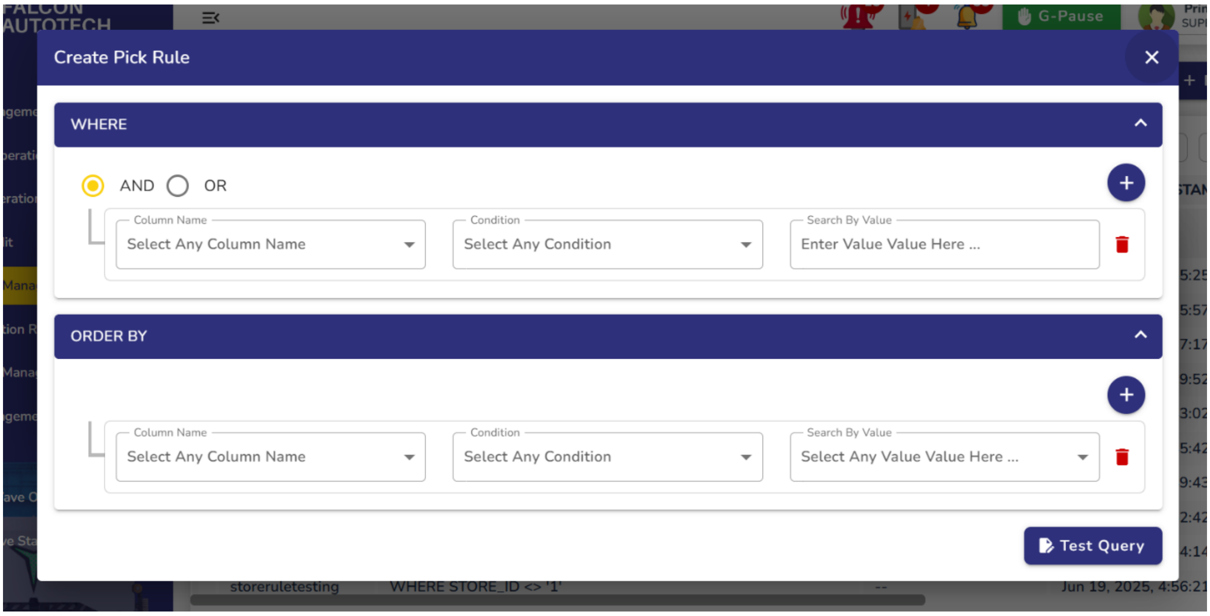
This tab essentially lets the user build the **logical foundation** for the pick rule. After defining the filter conditions. The user navigates back to the **main tab** (or rule metadata section) to enter:

- Rule Name
- Rule Description

- Once all information is entered, clicking the Submit button saves the rule to the system.
- Each rule in the table has an **Edit** option. On clicking **Edit**:
    - The rule configuration interface opens pre-filled with existing data.
    - The user can modify filter conditions, rule name, or description.
    - Updated changes are saved and reflected in the rule list.







4.6 Wave Station Rule Mapping Management

The Wave Station Rule Mapping Management module provides visibility and control over how different pick rules are mapped to stations in the system. This ensures that the correct rule logic is applied at each station during wave execution. The main table section displays a comprehensive list of:

- **Station IDs**
- The **current rule** mapped to each station
- The **default rule** (baseline configuration)
- **Filter conditions** defined in each rule

- **Current bot count** (bots actively allocated to that station based on current rule)
- **Default bot count** (initial or fallback allocation when no specific rule is applied)

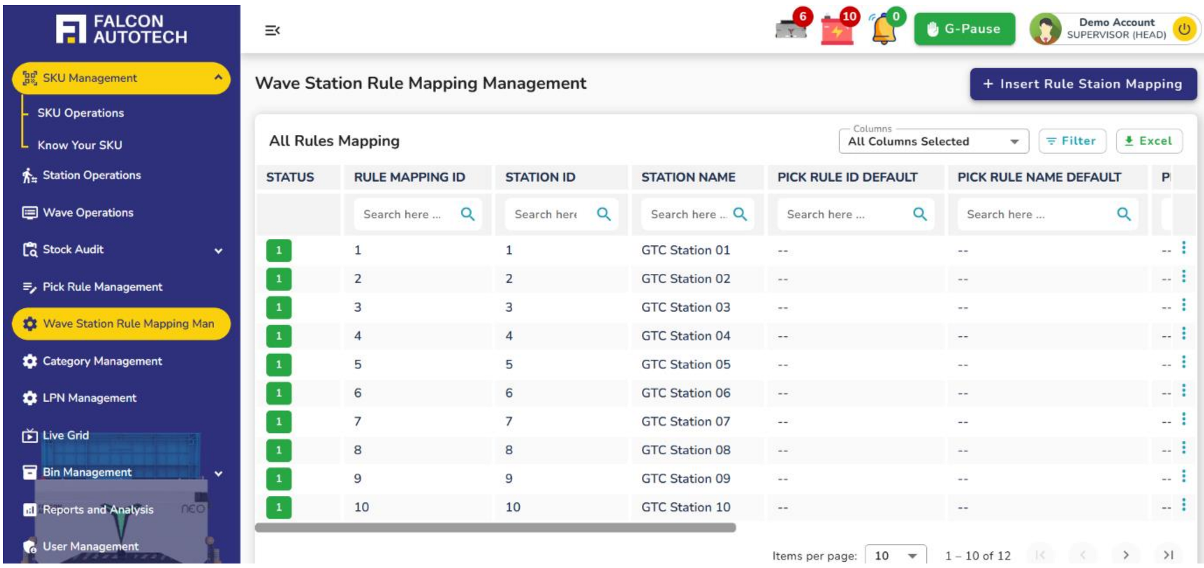
This table allows supervisors to quickly assess which rules are actively driving picking behaviour at each station and what deviations (if any) exist from the default setup.

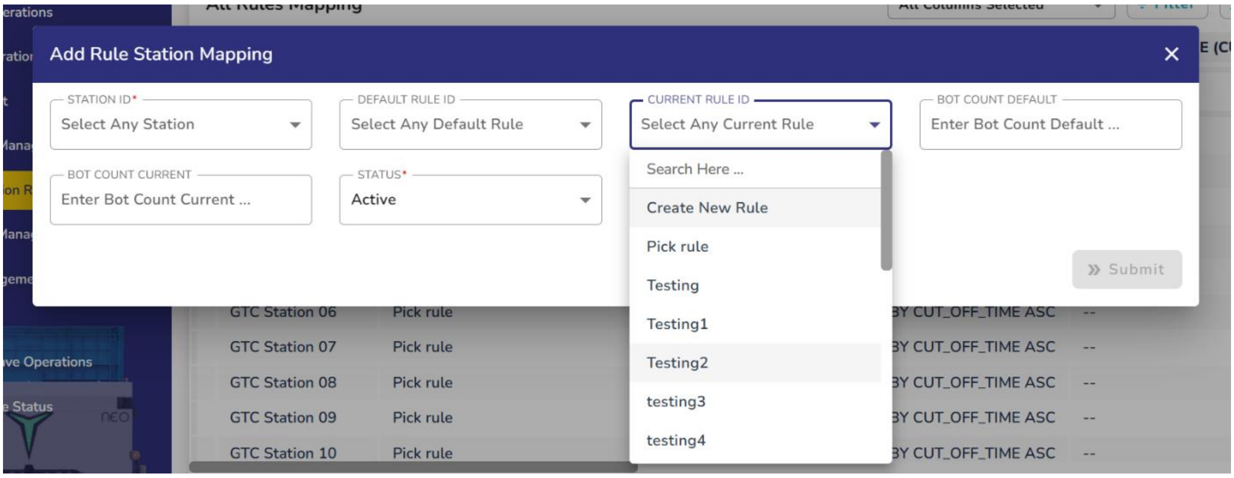
A **"+ Insert Wave Station Rule Mapping"** button is provided to allow supervisors to configure or update station-rule associations. On clicking this button:

1. A new interface opens where the supervisor can:
    - **Select a Station ID** from a dropdown or list.
    - **Choose an existing pick rule** to map to the selected station.
    - Or, if needed, **create a new pick rule** on the spot via an embedded rule creation flow (like Pick Rule Management), including access to the **Filter Condition tab**.
2. The supervisor can also specify:
    - The **current bot count** (number of bots currently assigned)
    - The **default bot count** (bots typically assigned under normal conditions)

Once submitted, the mapping is updated in the table and will reflect in wave operations logic. Each row in the mapping table includes an Edit option. On clicking **Edit**:

- The mapping form opens with existing data pre-filled.
- Users can update:
    - Mapped rule (switch to another existing rule or create a new one)
    - Bot counts (current/default)
- Saving the changes updates the live mapping for that station.




Add Rule Station Mapping
STATION ID*
Select Any Station
DEFAULT RULE ID
Select Any Default Rule
CURRENT RULE ID
Select Any Current Rule
BOT COUNT DEFAULT
Enter Bot Count Default ...
BOT COUNT CURRENT
Enter Bot Count Current ...
STATUS*
Active
Search Here ...
Create New Rule
Pick rule
Testing
Testing1
Testing2
testing3
testing4
Submit
GTC Station 06
GTC Station 07
GTC Station 08
GTC Station 09
GTC Station 10

4.7 Category Management

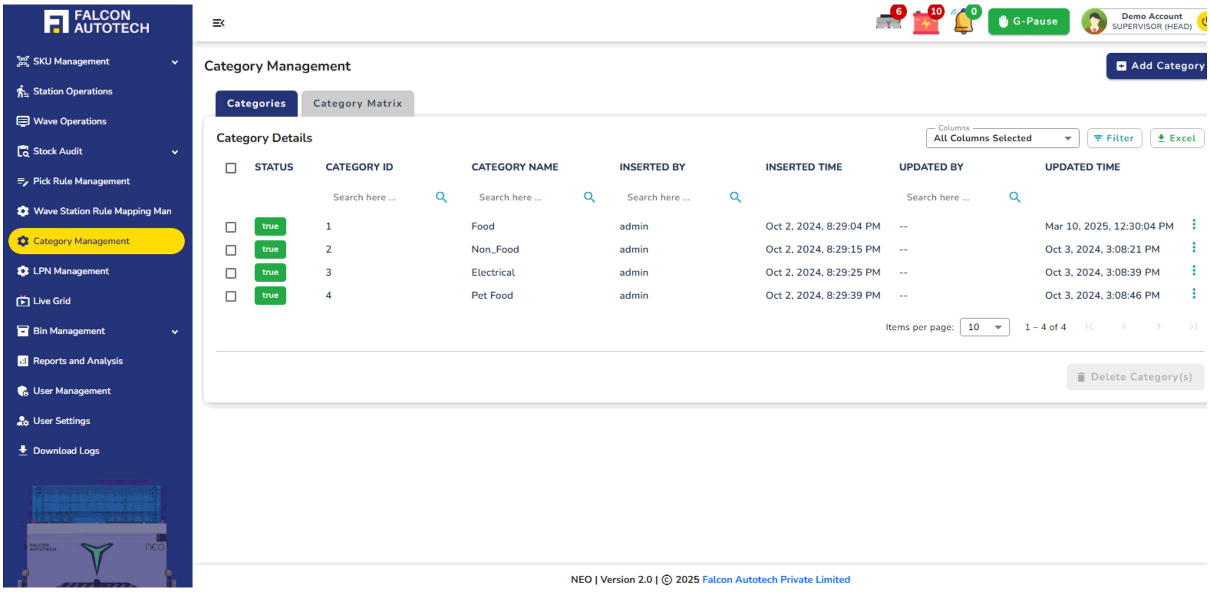
The **Category and Category Matrix Management** module is designed to define and maintain the hierarchical classification of SKUs (Stock Keeping Units) within the system. It plays a vital role in ensuring that all products are grouped logically into categories and subcategories, which helps in reporting, filtering, and applying rules based on product type.

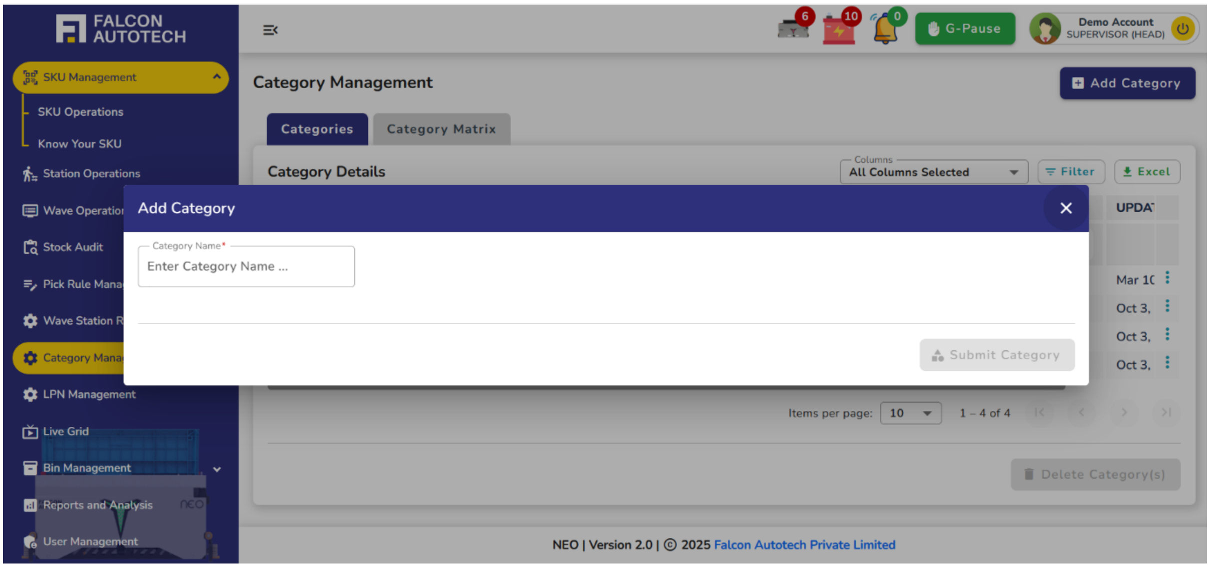
- This module is divided into two main tabs — **Categories** and **Category Matrix** — each serving a distinct purpose in the management of category structures.

The Categories tab provides a clear overview of all the categories present in the system. It displays a table that includes key details such as the category name, its status (active or inactive), the supervisor or user who inserted it, and the exact timestamp of when the category was created.

Each category in the list can be modified using the "Edit Category" option available for every row. When a user clicks this option, an edit form opens where the category name can be updated. Once the changes are submitted, the updated name is reflected in the main table, allowing supervisors to maintain consistency in naming conventions or correct input errors.

To create a new category, users can use the **"+ Add Category"** button. Clicking this opens a simple input form where the user can enter the category name and optionally set its status. This process ensures that new categories can be added on the fly, making the system flexible and adaptable to changing SKU classifications.

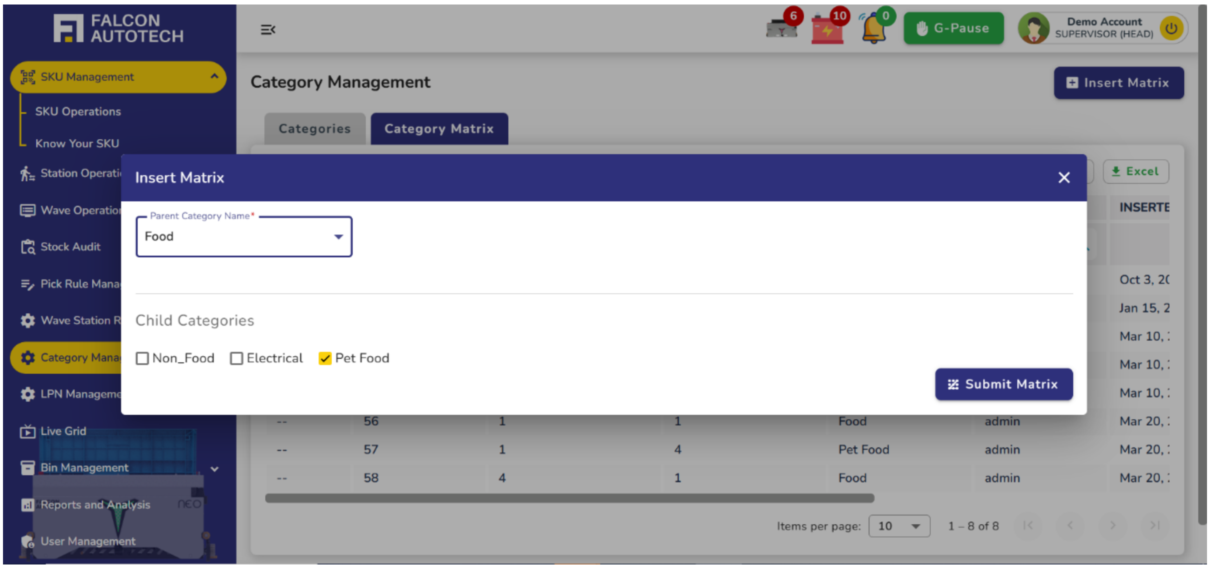




The Category Matrix tab focuses on defining relationships between parent categories and their respective subcategories. It displays a matrix-style table showing each parent category ID along with the list of subcategories that are mapped to it.

A key feature of this tab is the **"+ Add Matrix"** button, which allows users to create or modify these relationships. Upon clicking the button, a form opens where the user can:

- Select a parent category from a dropdown list
- View a list of all available subcategories
- Use checkboxes to select or deselect the subcategories that should be mapped to the chosen parent
- Importantly, any subcategories already mapped to the selected parent category are automatically pre-selected (ticked), helping the user easily see what's currently linked. This allows both adding new subcategories and removing existing ones in a single step. Once the desired configuration is set, the user submits the changes, and the updated mapping is reflected in the matrix table.



4.8 LPN Management

The LPN Management module is used to track and control the assignment of License Plate Numbers (LPNs) within the system. LPNs are unique identifiers used to label and trace containers during warehouse operations. This module provides visibility into how each LPN is being utilized — specifically, which station it is mapped to, when it was opened and closed, and the barcode associated with it.

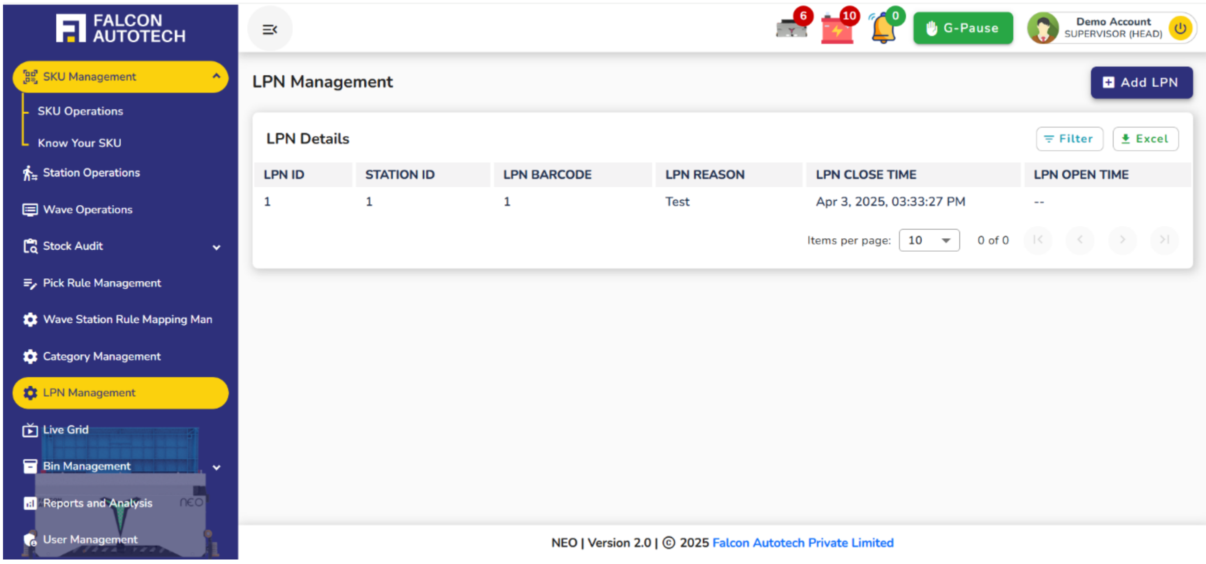
The central part of this module is a table that lists all active and historical LPN records. Each row in the table includes the following details:

- **LPN Barcode** – The unique identifier of the LPN
- **Mapped Station** – The station to which the LPN is currently or was previously mapped
- **Open Time** – The timestamp when the LPN was opened or activated for use
- **Close Time** – The timestamp when the LPN was closed or marked as complete
- **Status** – Whether the LPN is currently active or closed

This table gives supervisors or system users a clear view of how LPNs are being used across stations and helps monitor container flow during operations. To support dynamic operations, the module includes a **"+ Add LPN"** button that allows new LPNs to be added manually.

Upon clicking this button, a form appears where the user can:

1. **Enter the LPN Barcode** – The unique code that will identify the LPN.
2. **Select the Station** – Choose from a dropdown of available stations where this LPN will be used or was used.
3. **Provide a Reason for Mapping** – A brief explanation of why this LPN is being associated with the selected station. After submission, the LPN appears in the table with an automatically recorded **open time**, ready for tracking.


FALCON AUTOTECH
SKU Management
SKU Operations
Know Your SKU
Station Operations
Wave Operations
Stock Audit
Pick Rule Management
Wave Station Rule Mapping Man
Category Management
LPN Management
Live Grid
Bin Management
Reports and Analysis
User Management
6
10
0
G-Pause
Demo Account
SUPERVISOR (HEAD)
LPN Management
Add LPN
LPN Details
Filter
Excel
LPN ID
STATION ID
LPN BARCODE
LPN REASON
LPN CLOSE TIME
LPN OPEN TIME
1
1
1
Test
Apr 3, 2025, 03:33:27 PM
--
Items per page:
10
0 of 0
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4.9 Live Grid

The **Live Grid**, accessible from the navigation bar, offers a dynamic, read-only interface showcasing the real-time movements of all robots within the layout. This page is a visual representation of the operational flow, providing key insights into the bots' positions and movements without the need for any interaction or manual input.

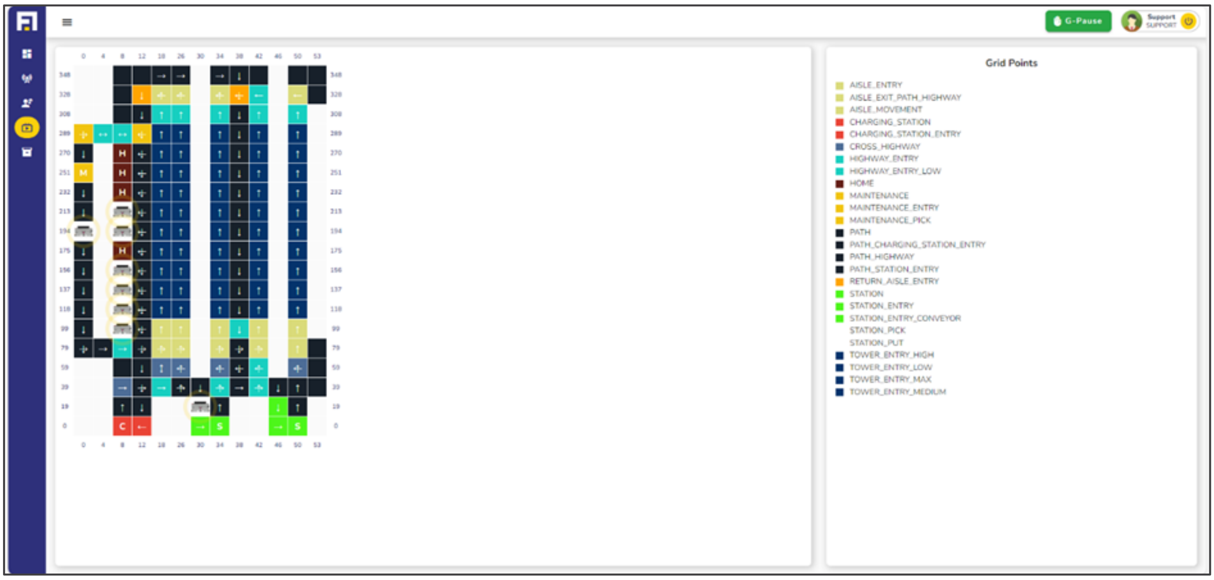


Figure 4.17 Live Grid

The key features of live grid page:

- **Real-Time Visualization:** The grid continuously updates to display the current locations and paths of all bots, offering a live snapshot of their activities on the layout.
- **Comprehensive Status Overview:** Users can instantly identify where each bot is located and observe how they navigate through the grid, making it easier to track their tasks and ensure smooth operations.

As a read-only page, Live Grid removes any potential for accidental changes or disruptions, ensuring the focus remains solely on monitoring. This tool is invaluable for understanding the spatial distribution and movement patterns of robots, helping users identify any congestion, inefficiencies, or delays in the system. With its clear, live updates and intuitive design, the **Live Grid** empowers users to stay informed about robot activity, offering a bird's-eye view of the layout in action.

4.10 Bin Management

The **Bin Management** page is an essential resource for users to understand and manage the inventory efficiently. It provides a comprehensive view of bin locations, contents, and associated details, making it an invaluable tool for inventory oversight and optimization. The page is divided into two intuitive submenus: **Bin Info** and **Know Your Bin**.

4.10.1 Bin Information

This section presents a dynamic table that lists all bins in the system, complete with their IDs, barcodes, locations within the tower, their placement (side and aisle), and other relevant attributes. Designed for flexibility, the table offers several user-friendly features:

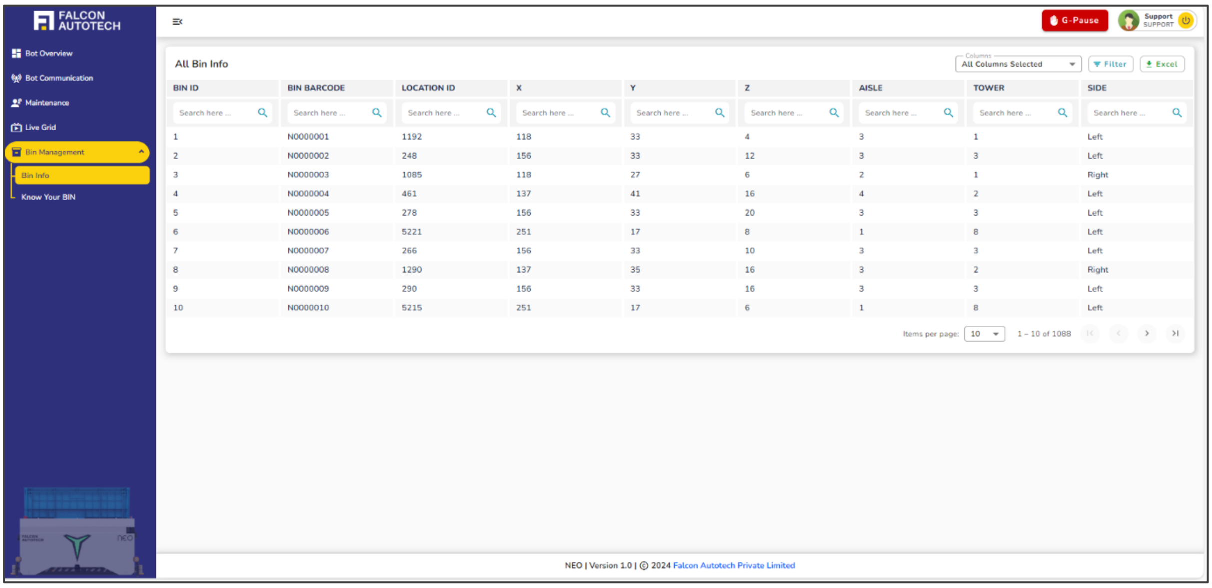


Figure 4.18 Bin Management - Bin Information

- **Customizable Columns**: Users can choose to display only specific columns by using the dropdown menu to select or deselect fields.

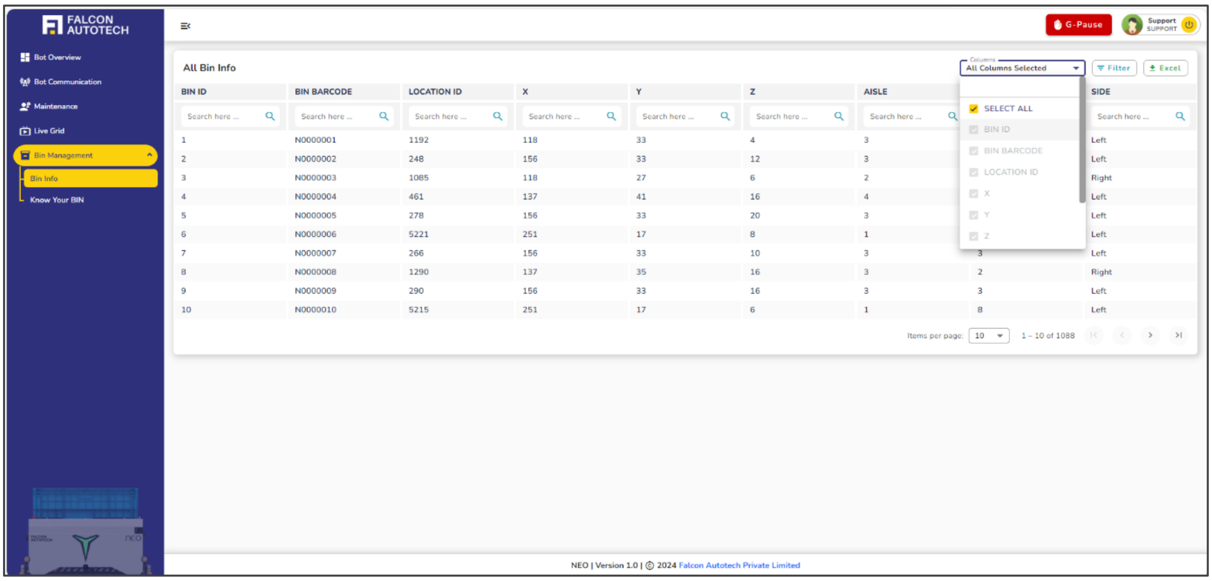


Figure 4.19 Bin Information - Customizable Columns

- **Excel Export**: The data can be downloaded as an Excel file with a single click on the export button, enabling offline analysis and reporting.

- **Advanced Filtering**: Multiple conditions can be applied to filter the data, allowing users to search columns with comparison operators or exact values, streamlining the process of finding specific information.

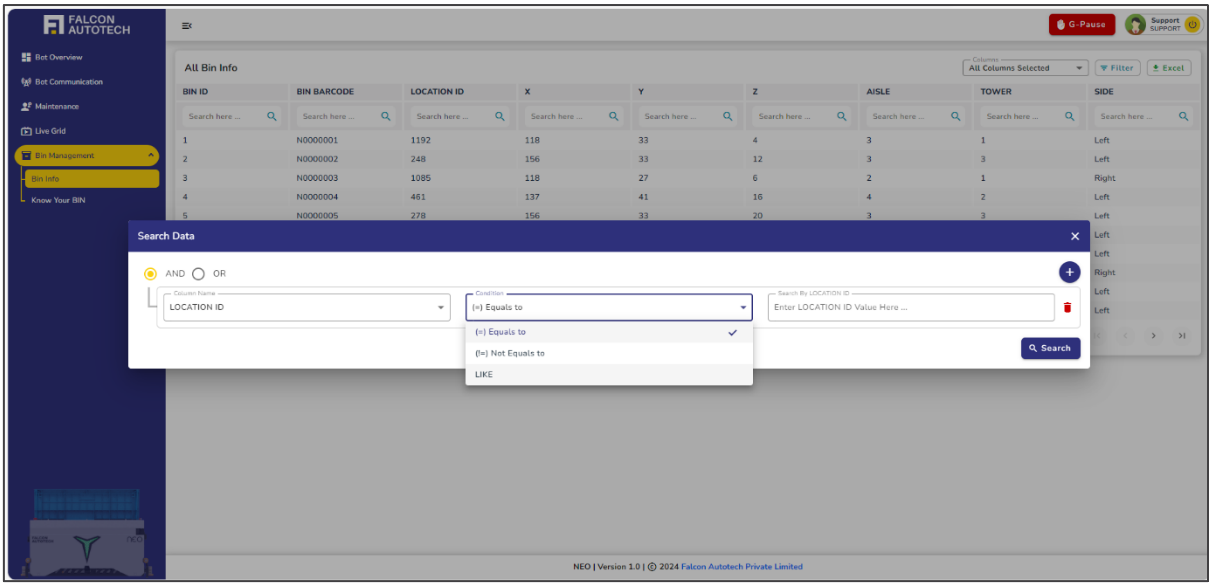


Figure 4.20 Bin Information - Advanced Filtering

4.10.2 Know Your Bin

This subsection provides a powerful search feature to fetch detailed information about a specific bin. Users can select a search method (Bin ID or Barcode) from a dropdown menu and input the corresponding value. Upon searching, a detailed breakdown of the selected bin is displayed, organized into the following categories:

Figure 4.21 Know Your Bin

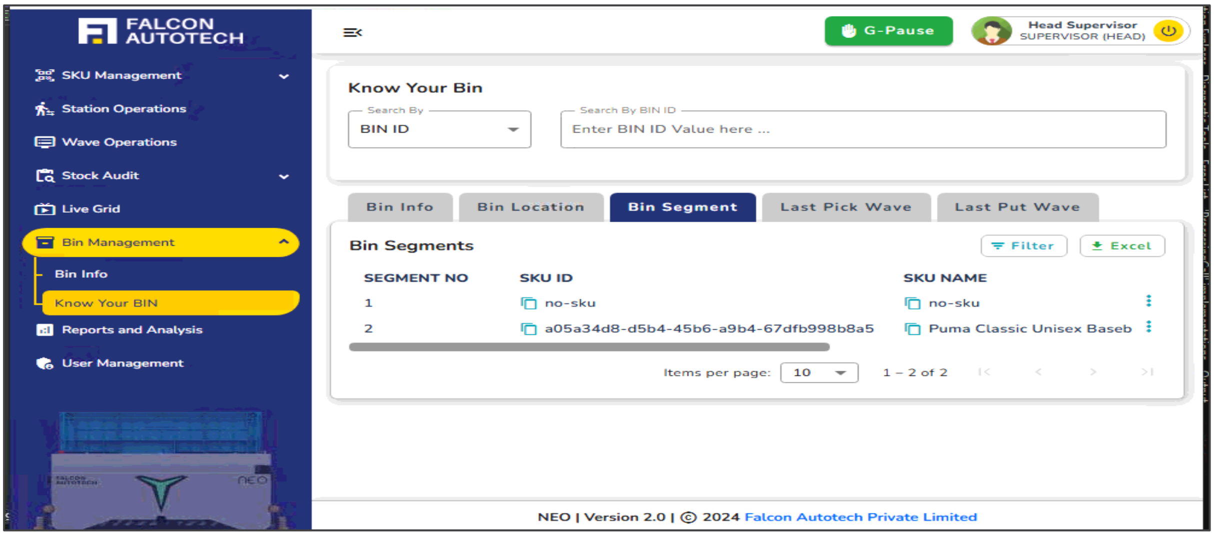


Figure 4.22 Know Your Bin - Tabular Information

- **Bin Info**: Highlights key details such as the bin's ID, barcode, and number of partitions/segments.
- **Bin Location**: Specifies where the bin is situated within the inventory.

4.11 Reports and Analysis

The system provides supervisors with comprehensive reporting capabilities through seven distinct report types, each designed to offer detailed insights into different aspects of operations.

4.11.1 Available Reports

Supervisors have access to the following reports:

1. **SKU Master**
    - Displays comprehensive product information for all SKUs
    - Shows product descriptions, dimensions, weights, and categories
    - Includes product status (active/inactive)
    - Lists storage requirements and handling instructions
    - Contains internal product codes and reference numbers
2. **SKU EAN Mapping**
    - Maps SKUs to their corresponding EAN (European Article Number) codes
    - Shows relationships between different product identifiers
    - Displays product names and descriptions
    - Includes barcode information for each SKU
    - Lists alternative product codes if applicable
3. **Live Inventory**
    - Provides real-time stock levels for all SKUs
    - Shows current bin locations of products
    - Displays available, reserved, and damaged quantities
    - Includes last updated timestamps
    - Shows inventory value and quantity thresholds
4. **Bin Loading Wave**
    - Details all bin loading operations
    - Shows timestamps of loading activities
    - Lists operators who performed the loading
    - Displays source and destination locations
    - Includes quantity and SKU information for loaded items

- Contains date and time filters for specific period analysis

5. **Pick Wave**

- Tracks all picking operations and their status
- Shows picker assignments and completion rates
- Lists picked quantities and locations
- Displays order references and priorities
- Includes picking timestamps and duration
- Features date and time filters for operation analysis

6. **Pick Overall Summary**

Pick Overall Summary offers a consolidated snapshot of picking operations, covering:

- Overall status of pick operations.
- Batch code associated with the wave.
- Total number of orders processed.
- Total order lines included.
- Count of distinct SKUs picked.
- Picked eaches vs. expected eaches.
- Stock adjustments recorded during picking.
- Total LPNS (License Plate Numbers) handled.
- Overall picked eaches and pending eaches.

7. **Pick Wave Bin Level Data**

The Pick Wave Bin Level Data module provides a detailed breakdown of picking activity at the bin level. It captures the following parameters:

- Wave ID – Identifier of the pick wave.
- Station ID – The station where the bin is processed.
- Batch Picklist Code – Batch code associated with the picklist.
- Order ID – Unique order reference.
- SKU ID – Identifier of the SKU being picked.
- Bin ID – The bin location from which items are picked.
- Bin Segment No. – Segment within the bin.
- MRP – Maximum Retail Price of the SKU.
- Expiry – Expiry date of the SKU (if applicable).
- Expected Quantity – Quantity system expected in the bin.

- Pick Quantity – Quantity picked.
- Short Pick Quantity – Difference between expected and picked.
- On Station Time – Timestamp when bin arrived at station.
- Pick Started Time – Timestamp when picking began.
- Pick End Time – Timestamp when picking was completed.
- Difference in Seconds – Duration of the pick activity.

8. Put Wave

- Records all put-away operations
- Shows operator assignments and completion status
- Lists source and destination locations
- Includes quantity and product information
- Displays put-away timestamps
- Contains date and time filters for specific period analysis

9. Put Overall Summary

Put Overall Summary module provides a consolidated view of putaway operations, including key performance indicators and operational details:

- **Status** – Current state of the putaway process.
- **Pallet ID** – Identifier of the pallet being processed.
- **Total Storage Requests** – Number of storage requests generated.
- **Total Storage** – Number of successful putaway completions.
- **Distinct SKUs** – Count of unique SKUs involved.
- **Expected Eaches** – System-expected quantity to be put away.
- **Put Eaches** – Actual quantity put away.
- **Total Stock Adjustment** – Adjustments made during putaway.
- **Overall Put Eaches** – Consolidated count of successfully put away eaches.
- **Pending Eaches** – Remaining items pending putaway.
- **Started Time & Completed Time** – Time window of the putaway operation.

10. Put Wave bin level data

The Put Wave Bin Level Data module provides detailed insights into putaway operations at the bin level. It captures the following parameters:

- Wave ID – Identifier of the put wave.
- Station ID – The station handling the putaway.
- Storage Request ID – Unique reference for the storage request.

- SKU ID – Identifier of the SKU being put away.
- Bin ID – Storage bin location where the SKU is placed.
- Bin Segment No. – Segment number within the bin.
- MRP – Maximum Retail Price of the SKU.
- Expiry – Expiry date of the SKU (if applicable).
- Expected Quantity – System-expected quantity to be put away.
- Put Quantity – Actual quantity put away.
- Short Put Quantity – Difference between expected and put quantity.
- On Station Time – Timestamp when the pallet/bin arrived at the station.
- Put Started Time – Timestamp when putaway started.
- Put Completed Time – Timestamp when putaway finished.
- Difference in Seconds – Duration of the putaway activity.

11. Stock Audit Wave

- Tracks all inventory audit activities
- Shows discrepancies found during audits
- Lists audit operators and timestamps
- Includes original and adjusted quantities
- Displays reason codes for adjustments
- Features date and time filters for audit period analysis

4.11.2 Standard Filtering and Export Features

Each report includes powerful filtering and customization options to enhance data analysis:

Customizable Column Display The supervisor can modify the visible columns through a dropdown menu, selecting only the relevant fields for their analysis. This feature allows for a cleaner, more focused view of the data.

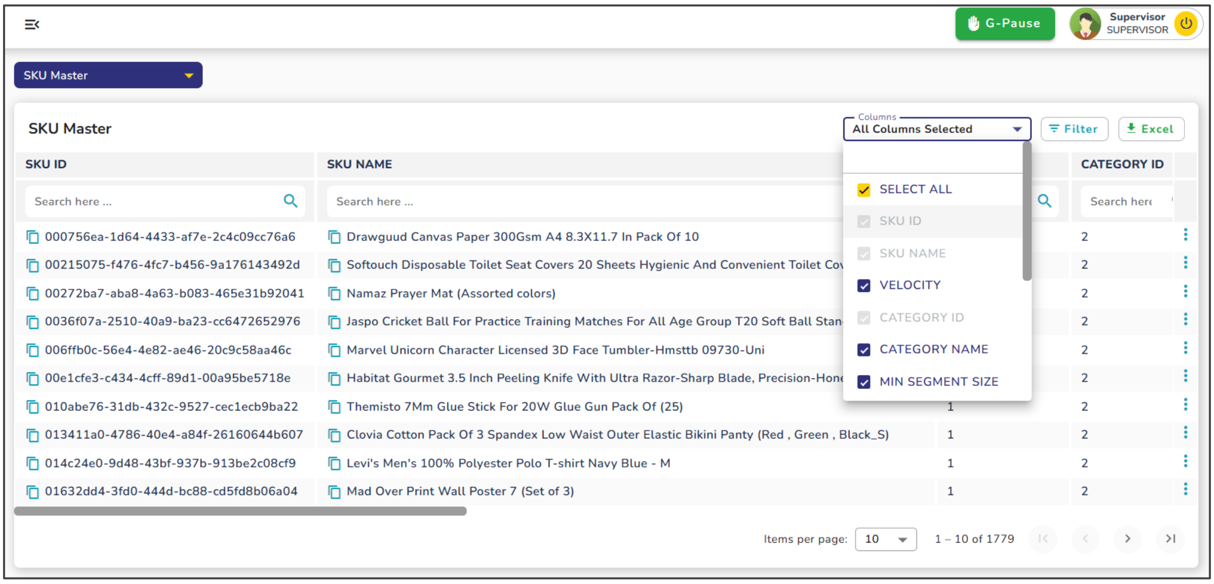


Figure 4.23 Custom Columns

Excel Export Functionality For offline analysis and record-keeping, supervisors can export the displayed data to Excel format with a single click of the export button. The exported file includes all filtered and customized data as shown in the current view.

Advanced Filtering Capabilities The system supports sophisticated filtering mechanisms where supervisors can:

- Apply multiple filter conditions simultaneously
- Search for exact values within specific columns
- Combine different filter criteria for precise data retrieval

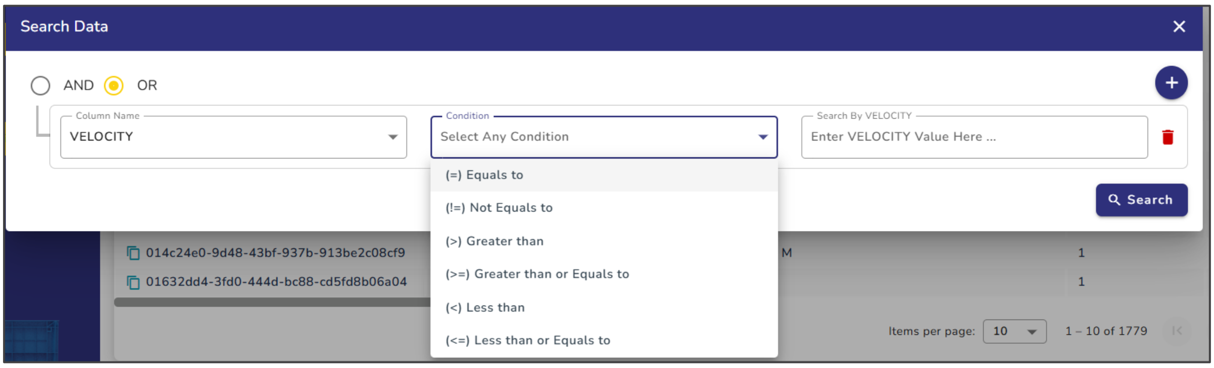


Figure 4.24 Advance Filter

Time-Based Filtering

Four specific reports include additional datetime filtering capabilities:

- Bin Loading Wave
- Pick Wave
- Put Wave
- Stock Audit Wave

For these reports, supervisors can specify date and time ranges to narrow down the data to specific operational periods, enabling more targeted analysis of temporal patterns and workflows.

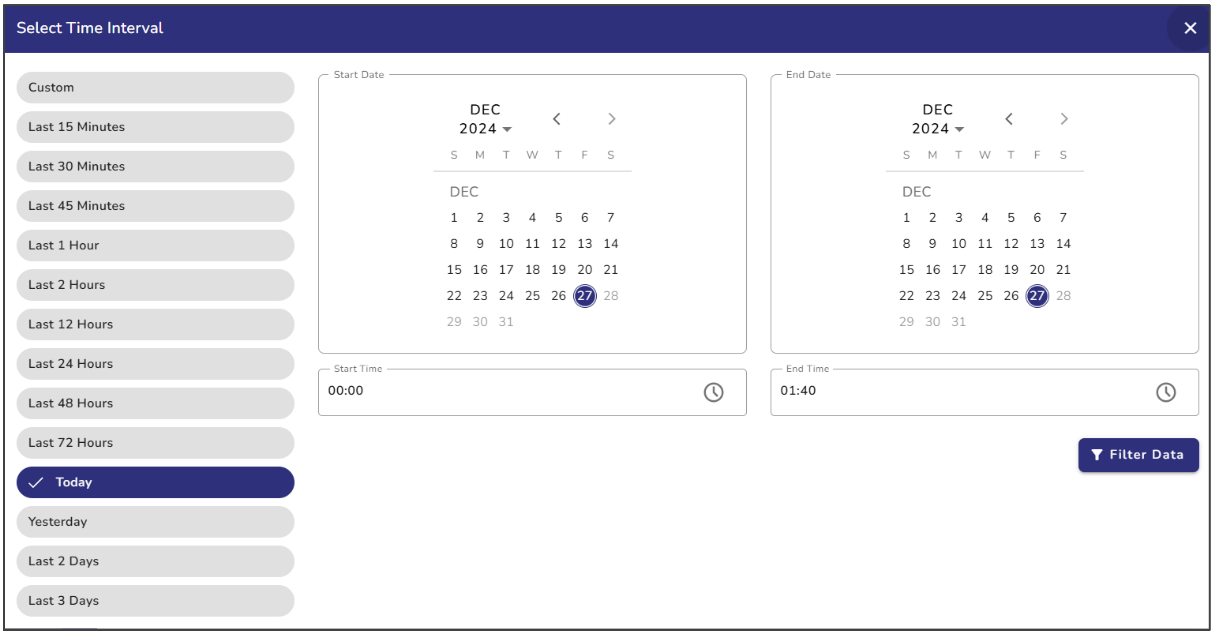


Figure 4.25 Time Filter

4.12 User Management

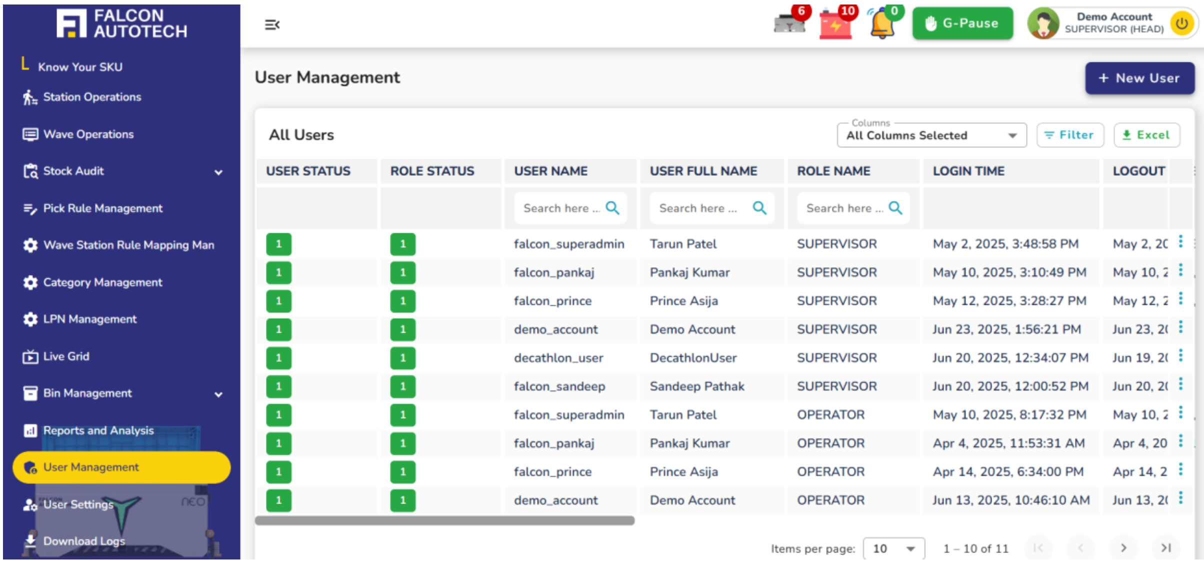
The User Management module is designed to maintain and monitor all user accounts within the system, including both supervisors and operators. It provides complete visibility into user roles, activity status, and login behavior, enabling administrators to manage access control effectively and ensure accountability across the system.

At the core of the module is a user list table that displays comprehensive details for all registered users. Each row represents a user and includes the following key information:

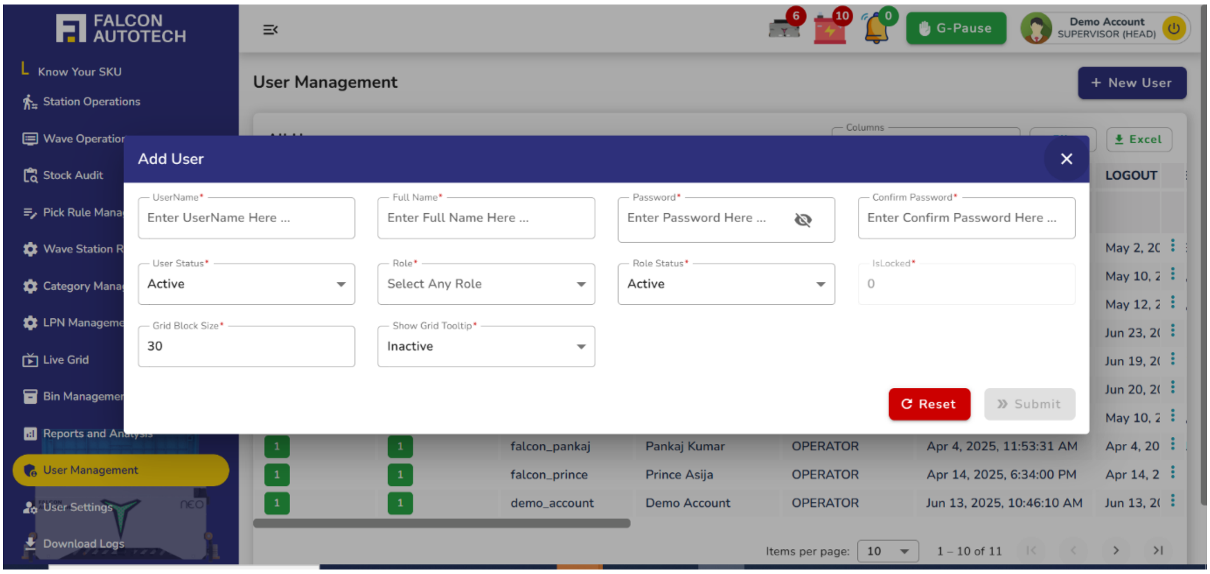
- **Username** – The unique ID used for logging in
- **Full Name** – The complete name of the user for identification

- **User Role** – Indicates whether the user is a **Supervisor**, **Operator**, or another defined role
- **User Status** – Active or Inactive, indicating if the user is currently enabled in the system
- **Login Time** – Timestamp of the user's last login
- **Logout Time** – Timestamp of the user's last logout or system timeout
- **Locked Status** – Indicates if the user account is currently **locked**, often due to multiple failed login attempts or manual administrative action.

This table allows administrators to monitor who is currently using the system, view recent activity, and manage user statuses with ease.



- A **"+ Add User"** button is available at the top of the page to facilitate the creation of new user accounts.
- On clicking this button, an input form appears where the administrator is prompted to enter:
    1. **Username** – Unique identifier for login
    2. **Full Name** – User's full name as it should appear in the system
    3. **Password** – Secure password for authentication
    4. **Confirm Password** – To ensure no entry errors
    5. **Role** – Selectable from predefined roles such as **Supervisor**, **Operator**, etc.
- Once the information is filled in and submitted, the new user is added to the system and becomes visible in the user table.


FALCON AUTOTECH
Know Your SKU
Station Operations
Stock Audit
User Management
User Settings
Download Logs
G-Pause
Demo Account
SUPERVISOR (HEAD)
User Management
+ New User
Excel
Add User
UserName*
Enter UserName Here ...
Full Name*
Enter Full Name Here ...
Password*
Enter Password Here ...
Confirm Password*
Enter Confirm Password Here ...
User Status*
Active
Role*
Select Any Role
Role Status*
Active
IsLocked*
0
Grid Block Size*
30
Show Grid Tooltip*
Inactive
Reset
Submit
LOGOUT
falcon_pankaj
Pankaj Kumar
OPERATOR
Apr 4, 2025, 11:53:31 AM
falcon_prince
Prince Asija
OPERATOR
Apr 14, 2025, 6:34:00 PM
demo_account
Demo Account
OPERATOR
Jun 13, 2025, 10:46:10 AM
Items per page:
10
1 – 10 of 11

Support

5 Navigation Bar

5.1 Overview

The navigation bar in the Falcon WCS Dashboard adapts dynamically based on the user's role. This section details the specific menus and tabs available to different roles to ensure clarity and usability.

A side navigation bar, also known as a menu bar, is a prominent section of a dashboard that serves as a navigational aid for visitors. Its primary purpose is to assist users in selecting topics, links, or sub-topics of interest within the dashboard. Users manually enter specific webpage URLs by utilizing the navigation bar, as the navigation bar automatically takes care of this functionality.

The design and usability of a navigation bar are crucial factors that can significantly impact the overall user experience of a webpage. It plays a pivotal role in guiding users and allowing them to easily access various interactive screens and functionalities within the dashboard.

5.2 Menu Structure:

Accessible Features:
Support users have access to robot management, enabling them to handle all manual and operational tasks. They can perform maintenance, manage HMI-related operations, and must be on-site to monitor robots in real-time, ensuring smooth functioning and taking corrective actions when necessary.

- **4.2.1 Bot Overview**
  - Displays a comprehensive summary of each bot, enabling users to switch between auto and manual operations seamlessly.
- **4.2.2 Bot Communication**
  - Shows real-time communication between robots and the server, monitored via heartbeat signals.
- **4.2.3 Maintenance**
  - Access a dedicated page for maintenance tasks when a bot requires servicing or cannot perform operations during live activities.
- **4.2.4 Live Grid**
  - Visualizes the entire layout or grid, showcasing the exact and real-time location of robots to simplify operational processes.
- **4.2.5 Bin Management**
  - Provides detailed information about the bins in the inventory for efficient management and tracking.

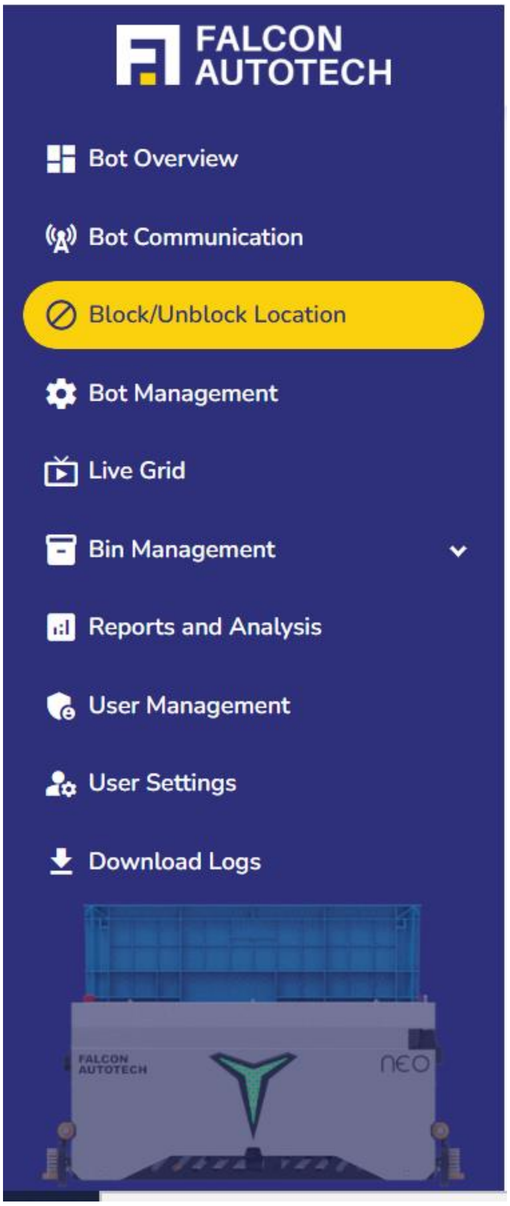


Figure 5.1 Navigation Bar

By utilizing the navigation bar, users can conveniently navigate between these user interactive screens, enabling them to access specific information, perform actions, and make informed decisions based on the data and functionalities offered by the IT dashboard.

6 Bot Overview

When the **Bot Overview** option is selected, the right side of the screen displays **multiple tabs** for accessing comprehensive bot-related information. These views are designed to cater to different operational roles and user preferences for **monitoring, managing, and troubleshooting bots**.

- **Visual Display**: This tab provides a graphical representation of all bots. Users can easily identify vital information and alarms at a glance, making it ideal for a quick visual overview.
- **Mode-Based Filters:** Users can filter the visual display by bot operating mode:
    - **Autostart** – Bots currently in autostart mode
    - **Autostop** – Bots that have stopped automatically
    - **Manual** – Bots operating in manual override
    - **HMI** – Bots being controlled via HMI panel
    - **Alarms** – Only bots currently showing active alarms
    - **Disabled** – Bots that are offline or deactivated
- **Table Display**: This tab offers a structured tabular view of all bot-related information. The table supports sorting and filtering options, enabling users to locate specific information quickly and efficiently.
- In addition to mode-based filters, a **dropdown menu** is available to allow **bot-wise selection**. This enables users to **focus on a single bot** to monitor its status and behaviour in detail. Useful for tracking individual bot performance or diagnosing issues.

These views complement each other, catering to different user preferences and operational needs for monitoring and managing bots effectively.

6.1 Visual Display

A dynamic, circular layout showcasing all bots. Hovering over each bot reveals essential information, including its current operational mode, bot ID, status (enabled and running), IP address, alarm code, and description. Users can also check the bin status on each bot, indicating whether it is loaded. The display is color-coded: red highlights any active alarms, while green indicates that everything is functioning properly.

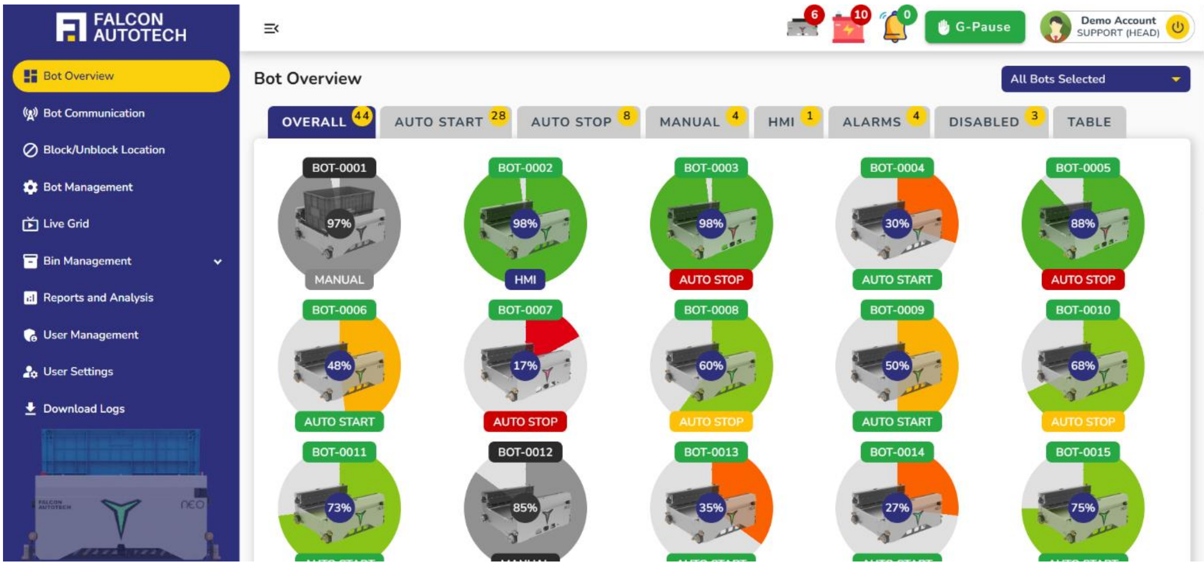


Figure 6.1 Visual Display

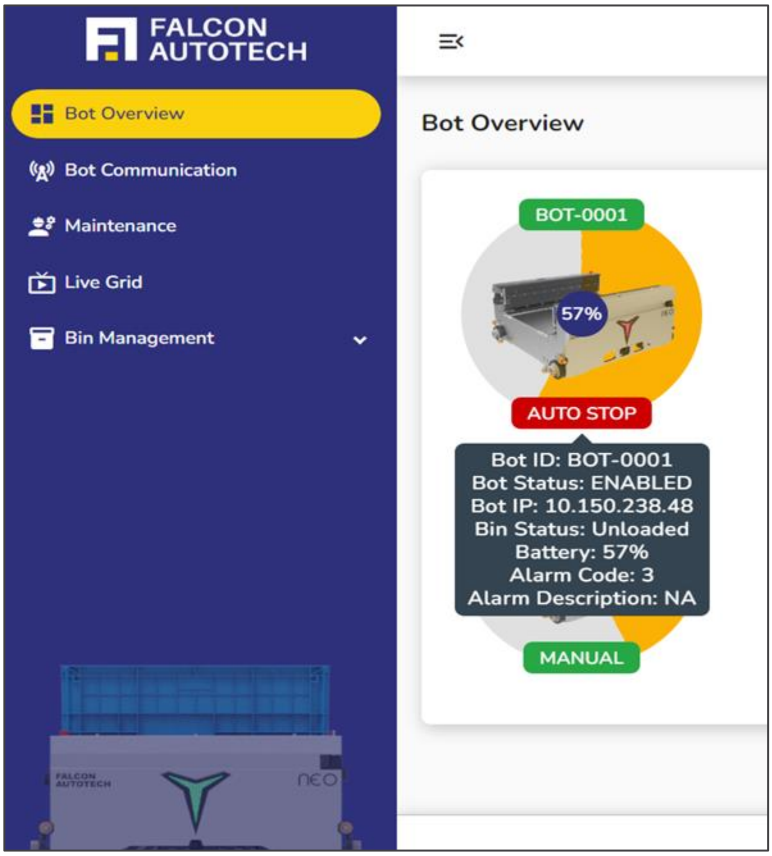


Figure 6.2 Bot Display on Hovering

6.2 Table Display

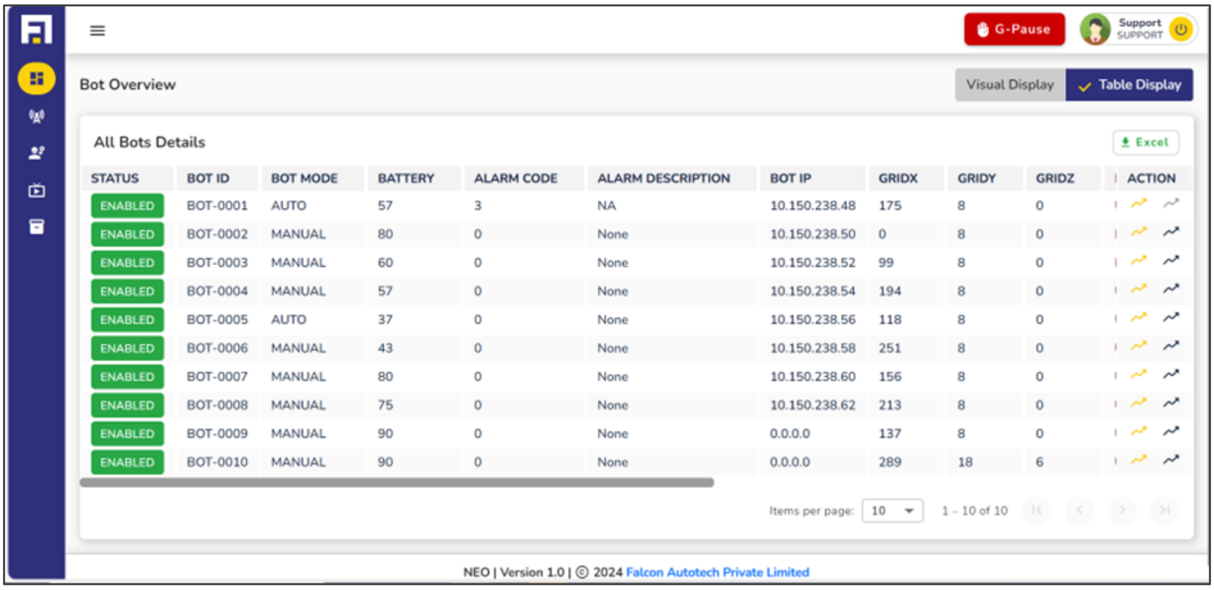
This tab provides a comprehensive tabular view of all bot-related information. It includes detailed data such as the bot ID, operating mode, current location, alarm code, description, bin load status, and other critical feedback. Users can perform action for each bot by redirecting to a detailed page (see Actions) for in-depth information about the specific bot or access additional functionalities directly from the tab.

Vital information of Bot displayed are:

- **Status**: Enabled/Disabled – *Enabled/Disabled* – The bot is marked as *Enabled* when it is operational, healthy, and live, and as *Disabled* when it is not in use (very rarely).
- **Bot ID**: Unique identifier for each bot.
- **Bot Mode**: Indicates the communication mode with the server, which can be *Auto*, *Manual*, or *HMI (Human-Machine Interface).*
- **Battery**: Displays the battery percentage of each bot, with a lower threshold set at 18%.
- **Alarm Code**: A unique code representing the specific alarm.
- **Alarm Description**: A detailed explanation of the alarm.
- **Bot IP**: The unique IP address assigned to each bot.
- **Grid X, Grid Y and Grid Z**: The current location coordinates of the bot within the grid.
- **Load Status**: Indicates whether the bot is *loaded* or *unloaded*, reflecting the bin status.
- **Counter**: Indicates the continuity of the bot's communication with the server,
- **Auto Start Feedback**: Confirms if the bot is operating in *Auto Mode*.
- **HMI Mode Feedback**: Confirms if the bot is operating in *Manual Mode.*

NOTE:

- **Bot Accessibility:** *Disabled* bots are completely non-operational and cannot be used. However, *enabled* bots may also be unavailable for use depending on specific conditions.
- The information displayed is read-only; users cannot edit the status or details of any bot.



STATUS	BOT ID	BOT MODE	BATTERY	ALARM CODE	ALARM DESCRIPTION	BOT IP	GRIDX	GRIDY	GRIDZ	ACTION
ENABLED	BOT-0001	AUTO	57	3	NA	10.150.238.48	175	8	0	
ENABLED	BOT-0002	MANUAL	80	0	None	10.150.238.50	0	8	0	
ENABLED	BOT-0003	MANUAL	60	0	None	10.150.238.52	99	8	0	
ENABLED	BOT-0004	MANUAL	57	0	None	10.150.238.54	194	8	0	
ENABLED	BOT-0005	AUTO	37	0	None	10.150.238.56	118	8	0	
ENABLED	BOT-0006	MANUAL	43	0	None	10.150.238.58	251	8	0	
ENABLED	BOT-0007	MANUAL	80	0	None	10.150.238.60	156	8	0	
ENABLED	BOT-0008	MANUAL	75	0	None	10.150.238.62	213	8	0	
ENABLED	BOT-0009	MANUAL	90	0	None	0.0.0.0	137	8	0	
ENABLED	BOT-0010	MANUAL	90	0	None	0.0.0.0	289	18	6	

Figure 6.3 Table Display

This table can be downloaded in excel, by clicking on **Excel** button shown just below Table Display.

6.3 Bot Details

The Bots page offers an in-depth view of each individual bot, including comprehensive real-time data, operational parameters, and a set of actionable controls. This page is primarily used by engineers, supervisors, or maintenance personnel who need to interact directly with a bot for monitoring or troubleshooting purposes.

Password prompt before access:

Before accessing a specific bot's detail page, the user is prompted to **enter their login password**. This ensures that only authenticated personnel can interact with the bot.

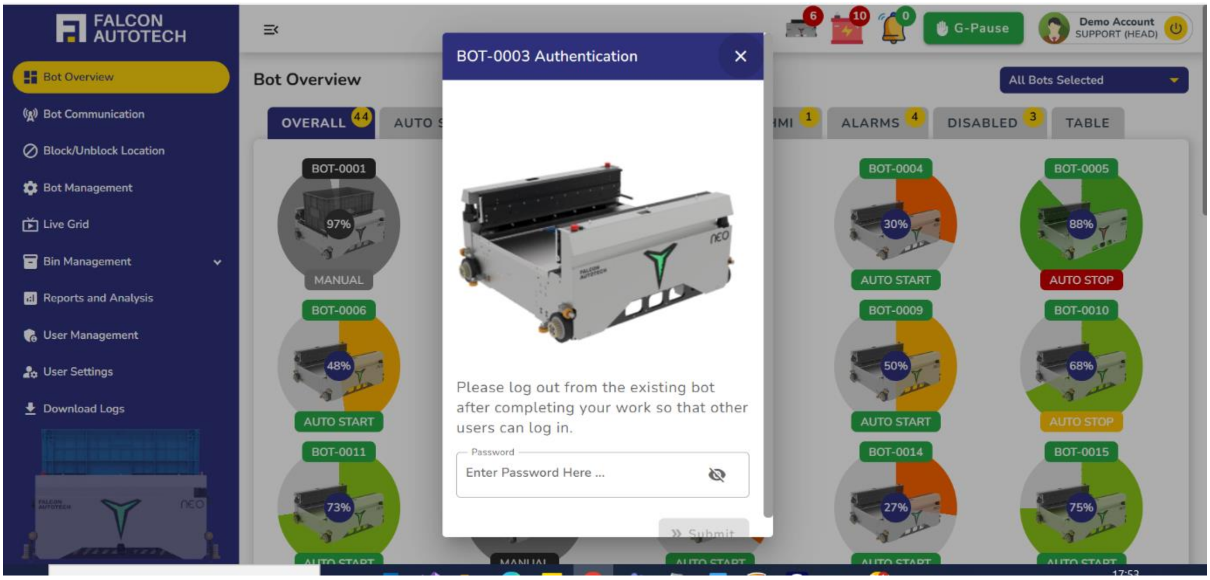


Figure 6.4 Bot Details Authentication

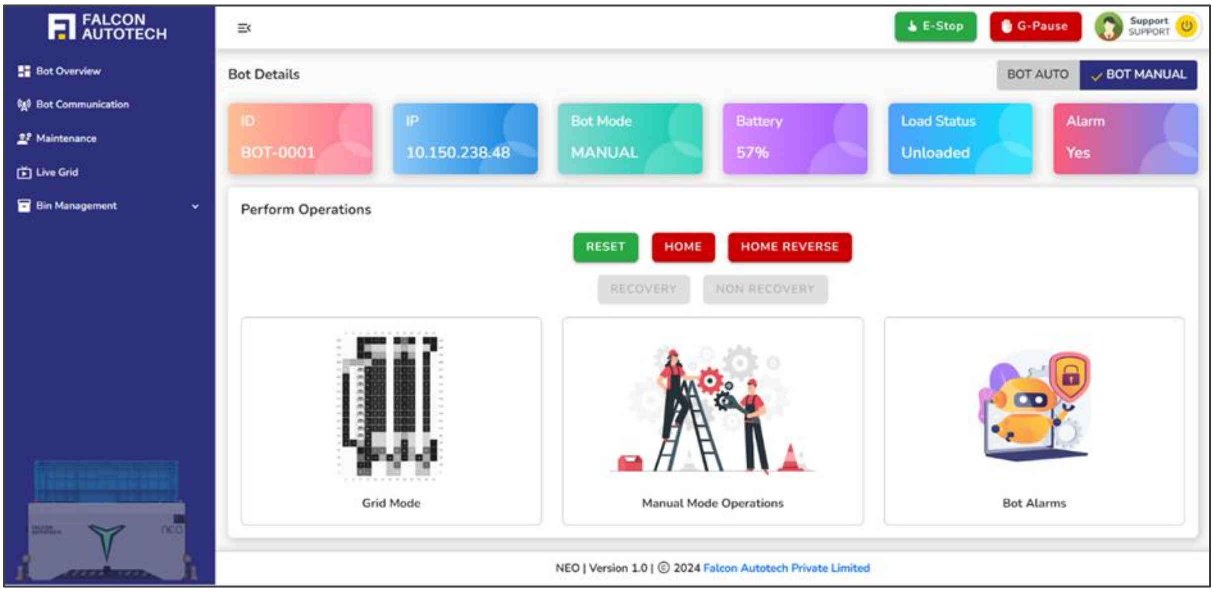


Figure 6.5 Bot Details Main Page

On each bot detail page, the first two options displayed are **Bot Auto** and **Bot Manual**.

6.3.1 Bot Manual:

NOTE:

- Before working with any controls, it is important to understand the layout, which is designed with towers, aisles, and a track system that guides the bot's movement. Barcodes are placed along the tracks to help locate the bot.
- When the bot is switched on for the first time or in case of an error, it must follow a set of sequential controls in manual mode to ensure it becomes healthy and starts communicating properly.

The **Bot Manual** mode indicates that there is no live wave or process running on the bot, allowing manual operations to be performed.

In the **Bot Manual** section on the bot details page, users will find operation buttons such as **Reset**, **Home**, **Home Reverse**, **Recovery**, and **Non-Recovery**. Below these options, the page is divided into three sections such as **Grid Mode**, **Manual Mode Operations**, and **Bot Alarms**. Each button is clickable and includes a feedback mechanism. The color coding, along with the enabling and disabling of the buttons, depends on the feedback status, which is also displayed on the dashboard.

Operation that is performed when clicked on the below buttons:

6.3.1.1 Reset:

Clear any active alarms on the bot. If the feedback is red, it indicates that the reset was unsuccessful, and a message will appear in the snack bar upon clicking the button. If the message displayed says "Reset Successful" and the feedback turns green, it confirms that the reset operation was successfully performed.

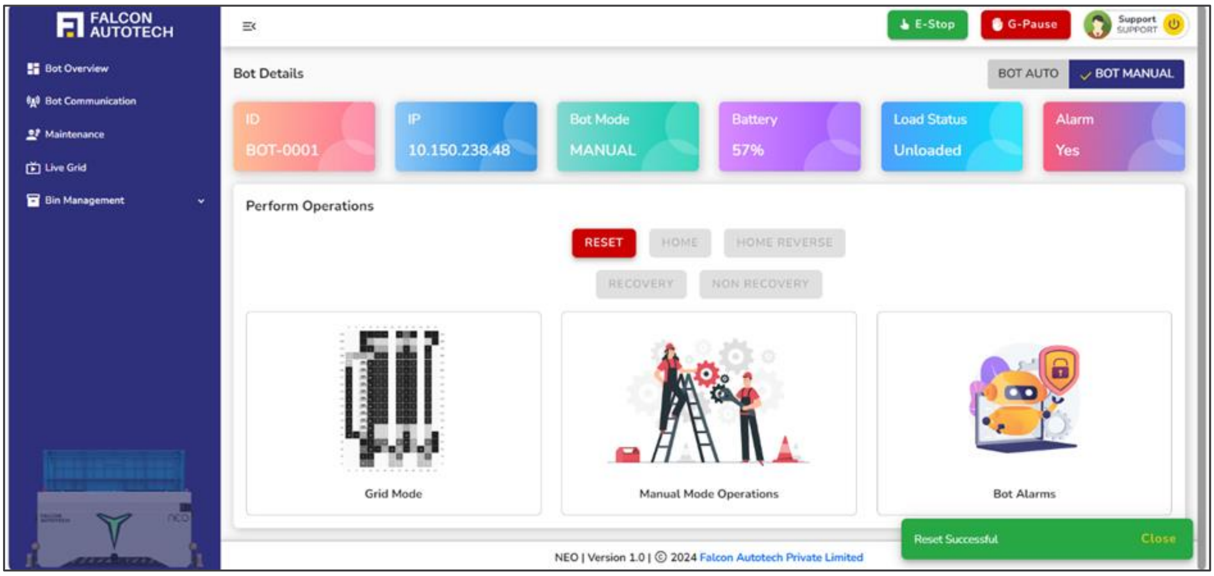


Figure 6.5 Reset Feedback True

When the reset feedback is false, the above figure 3.1 will be displayed, indicating that the user cannot perform any other operations until the current alarm is reset. To reset the alarm, the user clicks the reset button, after which the screen will update to show the feedback, confirming whether the reset operation was successful.

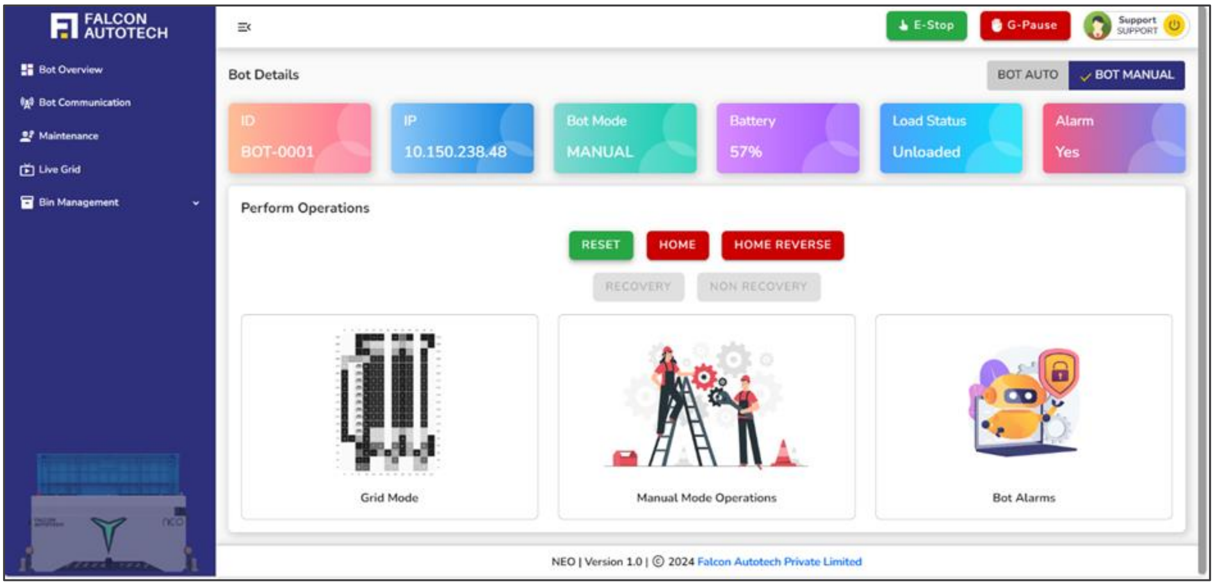


Figure 6.6 Reset Feedback True

6.3.1.2 Home and Home reverse:

When clicked, these buttons will localize the bot to the current barcode's position. The bot may move slightly forward (Home) or backward (Home Reverse), and the support user will need to observe the bot's movement to determine its location. Whether the user has to click on **Home** or **Home Reverse** buttons successfully to locate the barcodes depending on the bot's current position and the required direction.

These buttons are both feedback and clickable buttons. When the **Home** feedback is false, it indicates that the bot is not properly located at the barcode position. The support user will need to click on either the **Home** or **Home Reverse** button to perform the homing operation. Similar to the **Reset** operation, the snack bar on the dashboard will display the appropriate message based on the outcome.

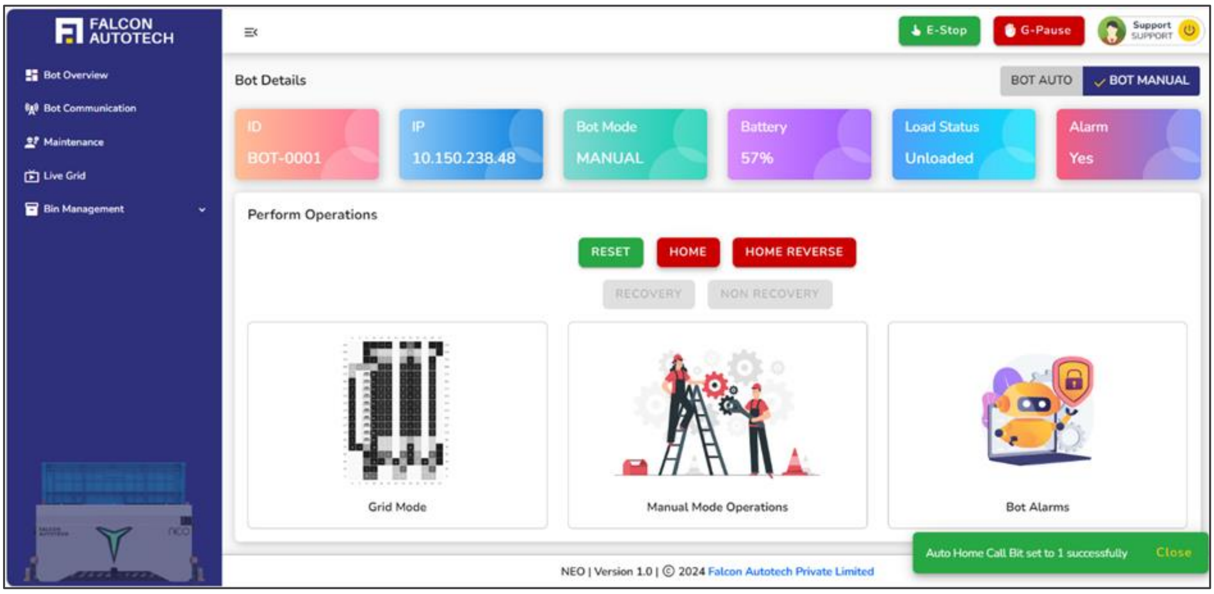


Figure 6.7 Home Feedback False

When the home feedback is false, the above figure will be displayed. To localize the bot, the user clicks the Home or Home Reverse button, after which the screen will update to show the feedback, confirming whether the `*home*` operation was successful.

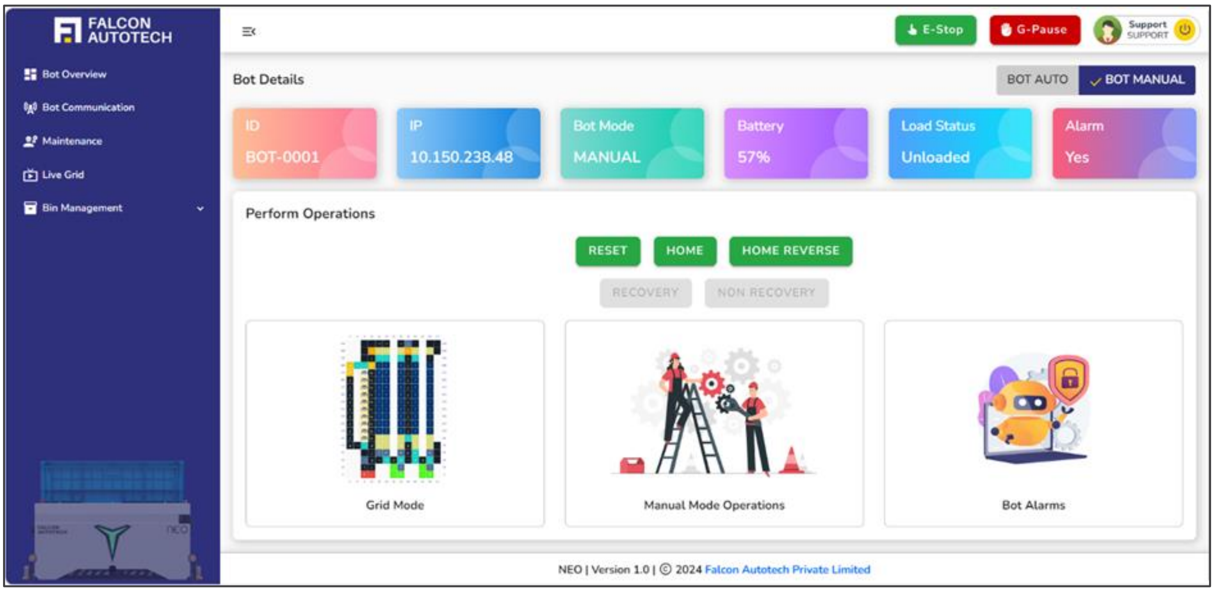


Figure 6.8 Home Feedback True

Once both the **Home** and **Reset** feedback are true, the following sections are enabled:

Grid Mode and **Manual Mode Operations**. This indicates that the bot is in a healthy state, allowing the support user to perform other controls and operations on the robot.

6.3.1.3 Recovery Button:

The **Recovery** button is intended for use only in specific scenarios and will be enabled only when the situation requires it. **Recovery** is used to resume a task that the robot was previously performing but could not complete due to an alarm or issue. When an operational robot encounters an alarm that interrupts its assigned task, clicking the **Recovery** button sends a trigger to the robot, instructing it to complete the pending/processing task.

NOTE:

- The Recovery operation is highly critical and cannot be used arbitrarily. It should only be executed following specific key guidelines to ensure proper usage and avoid potential issues.
- The Recovery button will only be enabled once both the Reset and Home feedback are true. It must be **clicked only after these conditions are met, following the sequential steps precisely.**

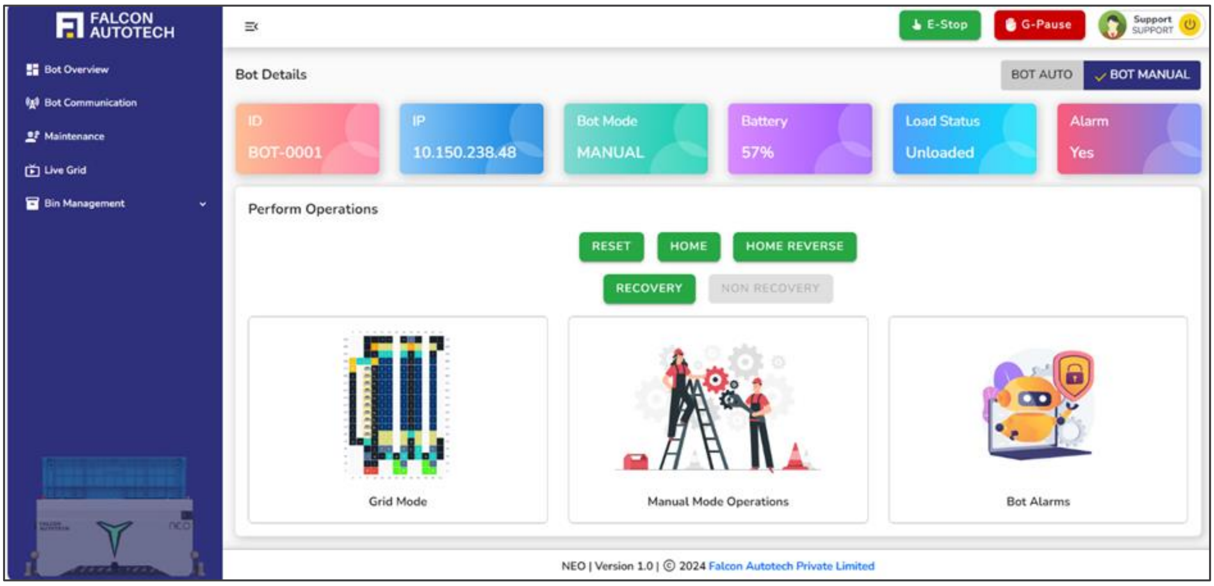


Figure 6.9 Recovery Button Enabled

6.3.1.4 Non-Recovery Button:

The **non-recovery** button is a specialized and situationally enabled feature designed to handle exceptional scenarios where a bot encounters an error and cannot complete its assigned task. This functionality is essential for maintaining workflow continuity by seamlessly transferring the incomplete task to another bot while ensuring the faulty bot is sent to maintenance for repair.

The **non-recovery** operation is initiated only under the following circumstances:

- **Bot Failure**: When a bot encounters an error during its task, rendering it non-functional.
- **Task Reallocation**: The task of the faulty bot is reassigned to a healthy and operational bot.
- **Maintenance Handling**: If the faulty bot has a bin on it, the new bot will move to the maintenance pick point to retrieve the bin before resuming the task.

This operation ensures uninterrupted task execution while addressing the issue with the faulty bot.

NOTE:

- The non-recovery operation is highly critical and cannot be used arbitrarily. It should only be executed following specific key guidelines to ensure proper usage and avoid potential issues.
- The non-recovery button will only be enabled once both the Reset and Home feedback are true and only when a non-recovery alarm is generated. It must be **clicked only after these conditions are met, following the sequential steps precisely.**

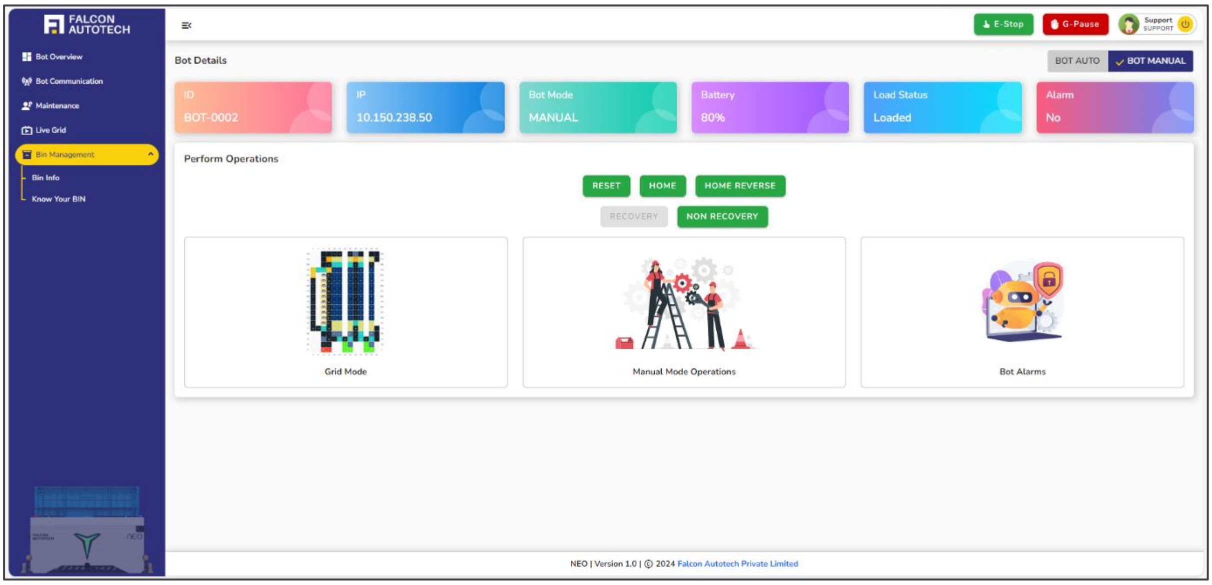


Figure 6.10 Non-Recovery Button Enabled

6.3.1.5 Grid Mode:

As mentioned above, The **Bot Manual** comprises three key sections: **Grid Mode**, **Manual Mode Operations**, and **Bot Alarms**.

The **Grid Mode** is a vital feature represented by an illustrated layout of the site. By selecting **Grid Mode**, users gain access to the **Grid Mode Screen**, where they can create manual tasks for the robots.

Before initiating task creation, it is essential to understand the intricacies of the layout and its coordinate system to ensure precise task assignment. The layout is designed with the following components:

- **Towers**: Structures housing bins at multiple levels.
- **Aisles**: Pathways between towers, enabling bot movement.
- **Levels**: Each tower contains bins arranged at different levels, with access points on the left and right for efficient robot operations.
- **Stations**: Designated zones where pick and put operations for articles are carried out.

On clicking grid mode, the user can access grid mode screen which will allow user to create manual tasks for the robots. The robots navigate these areas using a network of tracks marked with barcodes to facilitate accurate tracking and positioning. Users must familiarize themselves with the layout's **waypoints**, which are clearly identified in the grid, as these play a crucial role in guiding the robots to their designated tasks. Understanding these elements ensures the successful creation of manual tasks and seamless operation of the robots within the system.

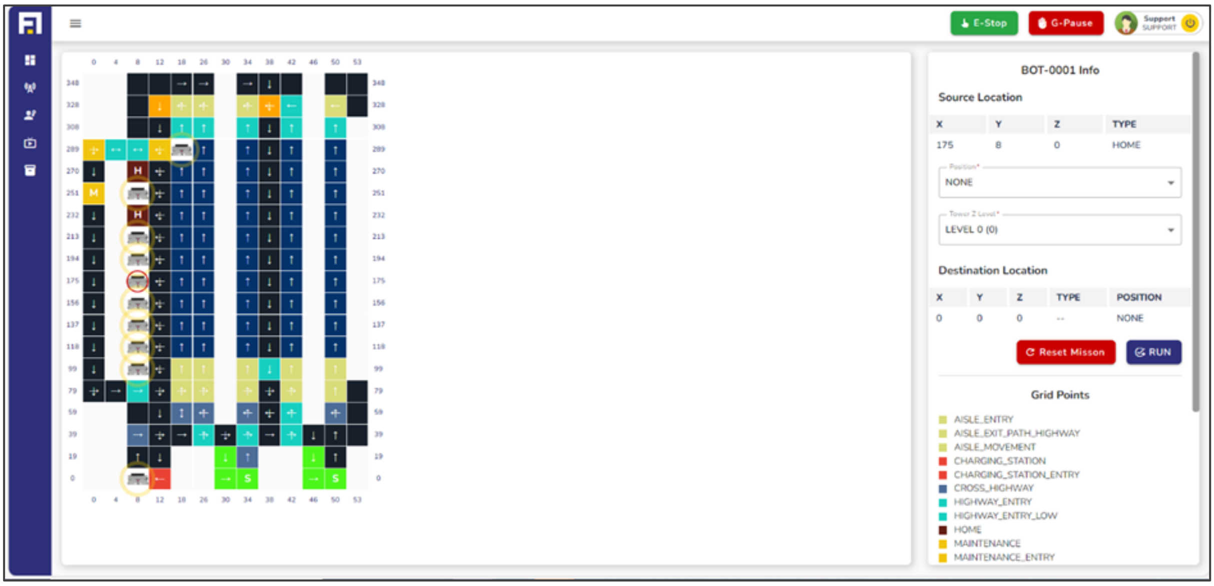


Figure 6.11 Grid Mode Main Page

The **Grid Representation** (Figure 5.3.1.5) provides a dynamic visual layout of the site, offering a comprehensive view of the grid's structure and the current positioning of all bots within it. Key features include:

- **Waypoints and Bot Locations**: The grid highlights various waypoints critical to task execution. It displays the current location of all bots, with the bot in use emphasized by a **red circle** for quick identification, while other bots are marked with **yellow circles** for easy distinguishment.
- **Interactive Hover Details**: By hovering over any bot, users can view essential information such as the bot's **current location**, the **type of location**, and its **coordinates**, ensuring a precise understanding of the bot's whereabouts.

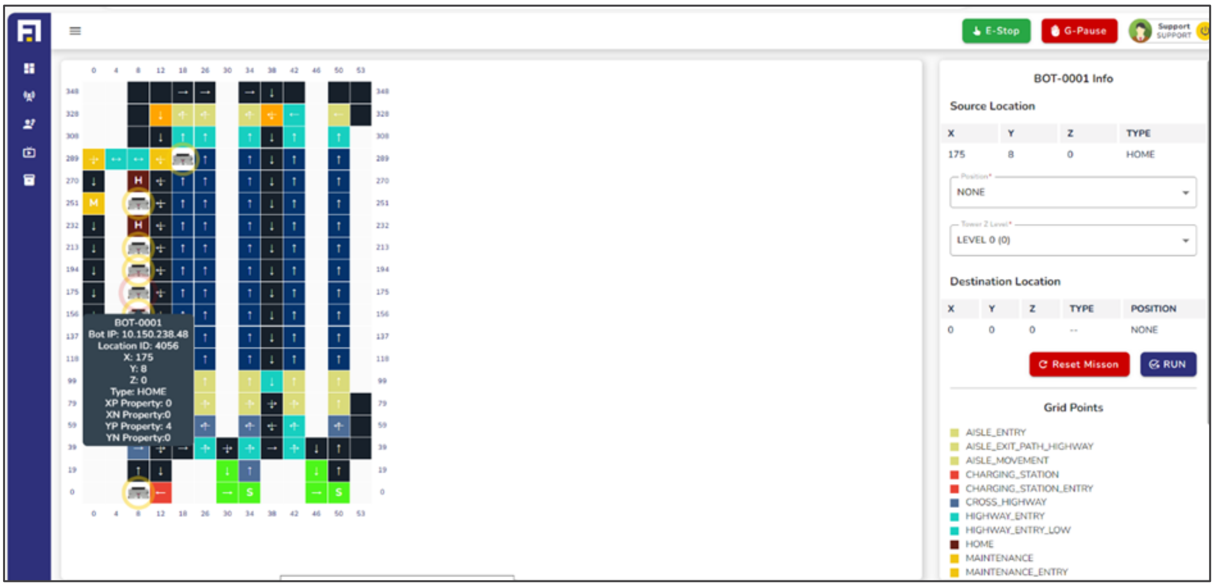


Figure 6.12 Hover on robot in grid mode

- **Coordinate System**: The grid's sides are labelled with coordinate values derived from precise measurements, converted into a coordinate system for intuitive navigation and ease of use.
- **Direction Indicators**: Arrows and directional markers guide users on the permissible paths the bots can traverse within the grid.

The grid is fully interactive, with each cell being **clickable**. This feature allows users to effortlessly create tasks by selecting a **source location** (the bot's current position) and a **destination**. The color-coded and uniquely named waypoints further streamline the task assignment process, ensuring clarity and accuracy. To the right of the grid, the **Bot Info Panel** provides detailed insights about each bot, including waypoint details, current tasks, and status. This serves as a valuable cross-reference tool, enhancing confidence and precision in task creation. By combining visual clarity, interactivity, and detailed information, the grid representation empowers users with a robust platform for seamless robot's manual task creation.

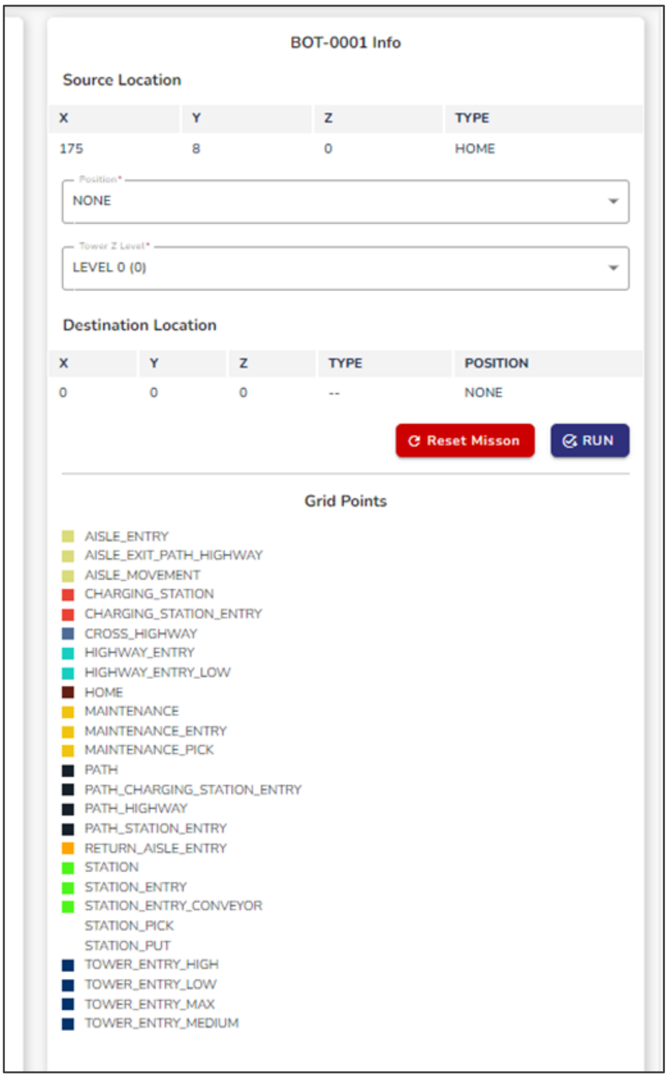


Figure 6.13 Bot Info Screen in grid mode

The **Bot Info Panel**, prominently displayed on the Grid Mode screen, is an essential tool for seamless task creation and precise execution. This intuitive panel allows users to define and verify task details with accuracy before dispatching them to a robot.

Manual Task Creation Workflow

- **Selecting the Destination**
    - To initiate a task, the user simply clicks on a cell in the grid.
    - The selected cell's coordinates are automatically reflected in the **Destination Info** section of the Bot Info Panel, enabling the user to cross-verify the chosen location.
- **Destination-Specific Options**
    - Tower as Destination:
        - If the destination is a tower, the user must specify additional operations.
        - Position Dropdown: The user can select the desired operation—Pick Left, Pick Right, Put Left, Put Right, or choose None if the bot needs to simply climb the tower without interacting with a bin.
        - Level Dropdown: The level within the tower must be selected using the dropdown menu. This option is exclusively enabled when a tower is chosen as the destination.

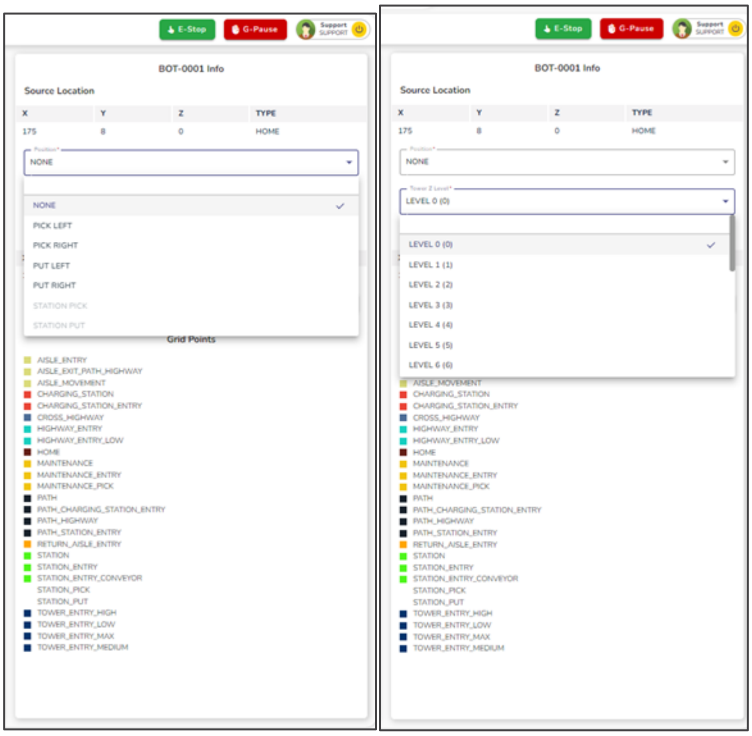


Figure 6.14 Tower Selection in grid mode

- Station as Destination:
    - If the destination is a station, the Position Dropdown limits the options to Station Pick or Station Put, and the Level Dropdown remains disabled, as levels are not applicable to stations.

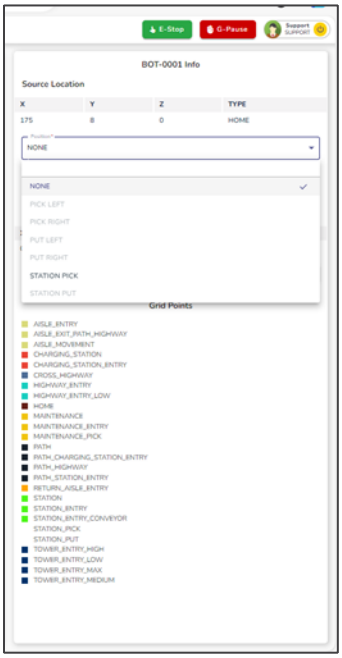


Figure 6.15 Station as Destination in grid mode

- **Other Grid Points:**
  - For all other locations on the grid, no additional inputs for position or level are required, ensuring a straightforward task creation process.

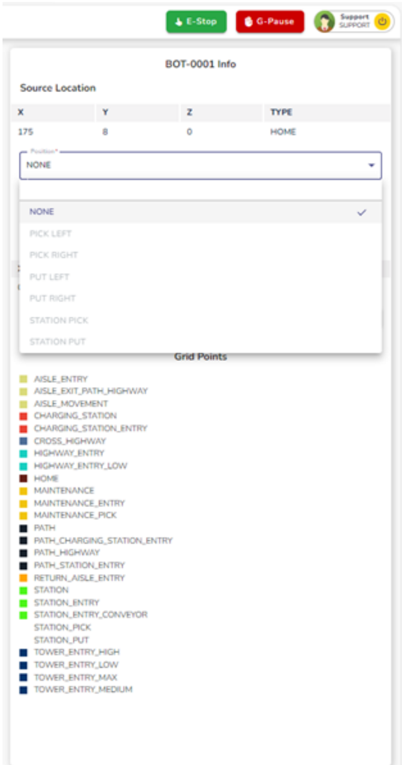


Figure 6.16 Other grid points selected as destination in grid mode

- **Running the Task**
    - Once the destination and relevant options are selected, the user can click the **Run** button to finalize the task.
    - Upon execution, the bot immediately begins performing the assigned task with precision.

- **Reset Mission**

  This option allows users to reset tasks previously assigned to a robot, ensuring smooth operations when tasks remain incomplete due to unforeseen issues or interruptions. This feature specifically clears only the missions assigned through **Grid Mode**, leaving tasks from other modes unaffected. By using this option, users can free the robot from its current mission queue, enabling it to accept new tasks without residual conflicts. This ensures efficiency and flexibility in managing robotic operations, especially in time-sensitive scenarios.

  The Bot Info Panel's design ensures clarity and precision, empowering users to effortlessly create tasks tailored to the bot's environment. By dynamically enabling or disabling dropdown options based on the selected destination, the panel minimizes errors and streamlines the task creation process. With the Bot Info Panel, users have a robust interface that simplifies task assignment while maintaining flexibility for complex operations.

6.3.2 Bot HMI Mode:

The **Manual Mode Operations**, also referred to as the **Bot HMI Mode**, should be accessed only when the situation is necessary, not as a routine action. It is critical to note that when a wave or process is actively running, the bot should not be switched to manual packet mode unless absolutely required.

As a subset of the **Bot Manual Mode**, the **HMI Mode** acts as the primary interface for manual control, enabling precise command packet exchanges between the robot and the server. Upon selecting the **Manual Mode Operations** section, the user is redirected to the HMI screen. This action is visually signaled by a purple light on the robot, visible physically, indicating that the bot is now actively transmitting manual command packets to the server. This mode offers a robust solution for scenarios requiring immediate manual intervention, ensuring seamless communication and control.

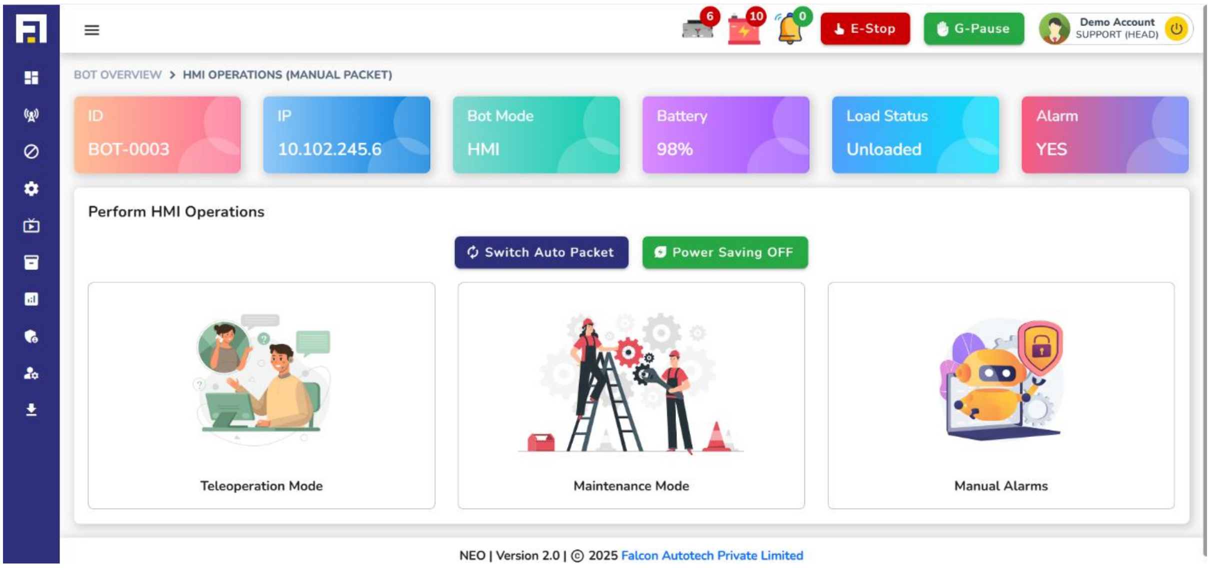


Figure 6.17 Bot HMI Mode

The **Bot HMI Mode** is divided into three distinct sections: **Teleoperation Manual Mode**, **Maintenance Mode**, and **Manual Alarms**.

- **Teleoperation Manual Mode (HMI Mode):** This section serves as the Human-Machine Interface (HMI) and functions as the primary control hub for manually operating the robot. It facilitates precise, real-time interaction between the user and the robot, allowing for effective teleoperation.
- **Maintenance Mode:** This section provides critical feedback and diagnostic information about the robot's maintenance status. Users can access details regarding the bot's health and performance, ensuring that any maintenance requirements are promptly addressed.
- **Manual Alarms:** When the bot is operating in manual mode, all active alarms are displayed in this section. This enables users to monitor and address any issues affecting the bot during manual operations.

Located at the top of the Bot HMI Mode interface, two prominent buttons offer additional operational control:

- **Switch Auto Packet**: Used to **transition the bot from manual to automatic (packet) mode** after maintenance or testing. Ensures a smooth handover back to autonomous system control.
- **Power Saving ON/OFF:**

6.3.2.1 Teleoperation Manual Mode:

This mode, as outlined above, functions as the Human-Machine Interface (HMI) for robots, offering an intuitive platform for manual control. It proves invaluable in scenarios involving alarms, error recovery requiring manual intervention, or during manual testing phases.

Designed for straightforward operation and efficient troubleshooting, the interface empowers users to seamlessly navigate and control the robot. It includes comprehensive options to maneuver the robot along the X, Y, and Z axes while also enabling precise actions such as lifting, sliding, and actuator operations, ensuring a robust and user-friendly experience.

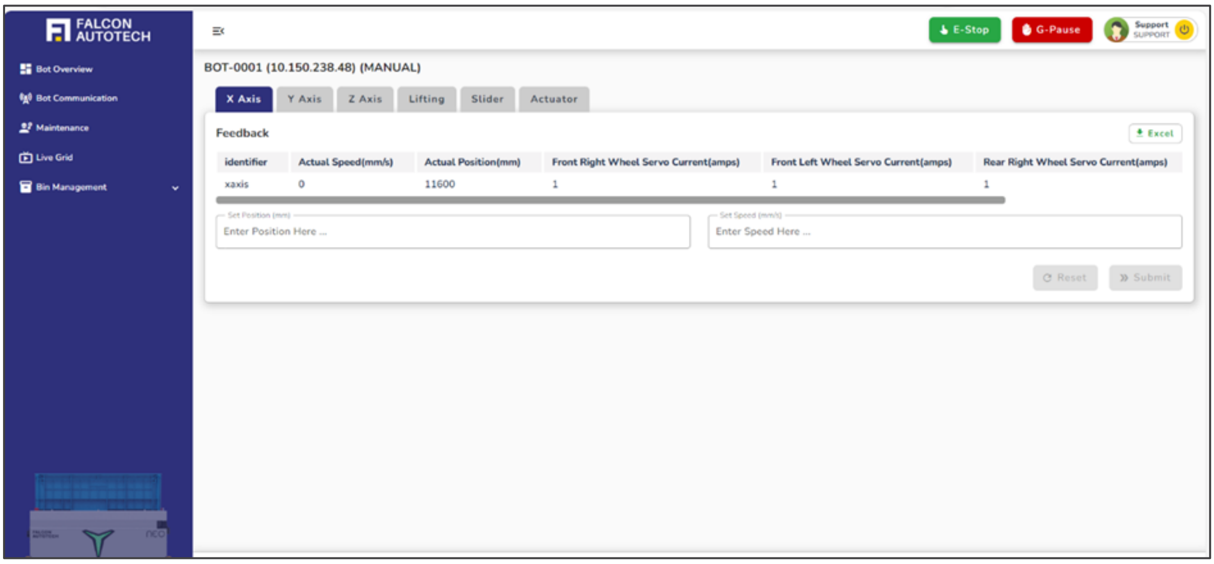


Figure 6.18 Bot HMI Main Screen

6.3.2.1.1 X Axis:

The X Axis section allows manual control of the robot along the X axis.

- Users can define **position** and **velocity** by entering values in the designated input fields.

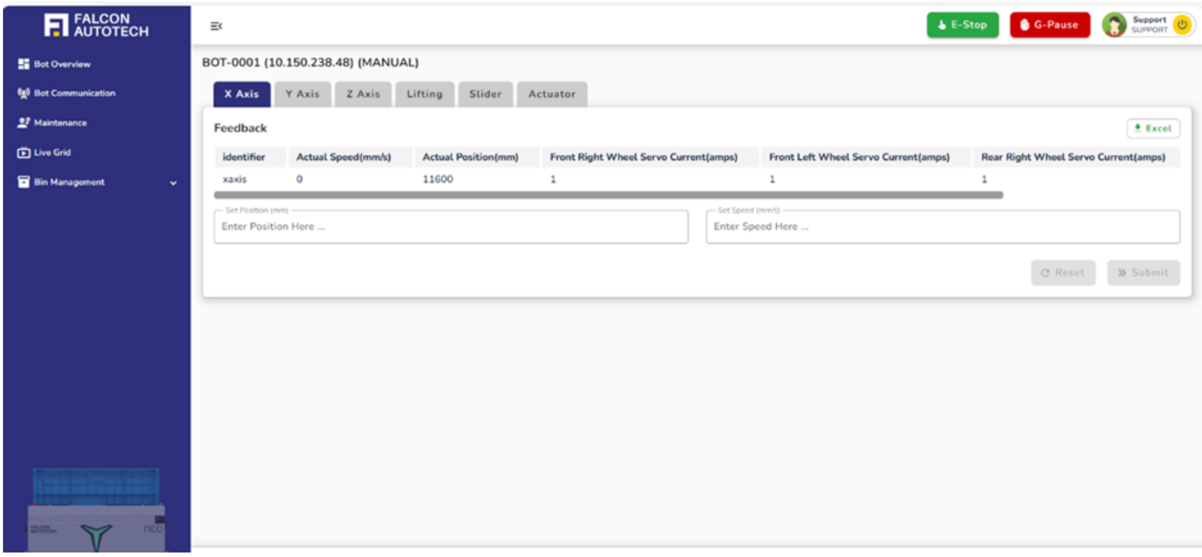


Figure 6.19 Bot X-Axis HMI Mode

- Real-time feedback for the robot's actual position and velocity is displayed, providing clear monitoring of its status.
- Once the desired values are set, clicking **Submit** applies the changes, while the **Reset** button allows users to modify or reset the inputs if needed.
- A summary table of the configured values is displayed below for easy reference.

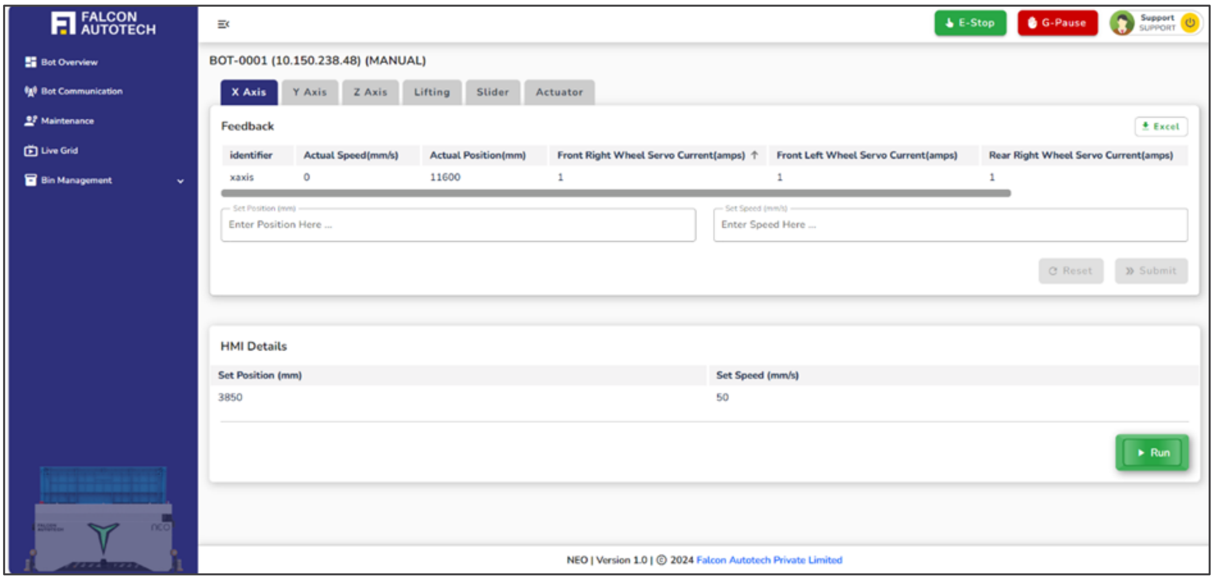


Figure 6.20 Bot X axis HMI Mode After Submitting Values

- To initiate the robot's motion, the user can press and hold the **Run** button, prompting the robot to move according to the preset parameters.
- The speed range for this operation is constrained to a safe limit of **50 to 300 mm/s**, ensuring both precision and safety during manual control.
- If the speed or position values exceed the defined thresholds, an **error message** is displayed below the input fields on the dashboard. This message specifies the permissible limits, guiding the user to input valid values within the allowed range. This ensures operational safety and prevents unintended robot behavior.

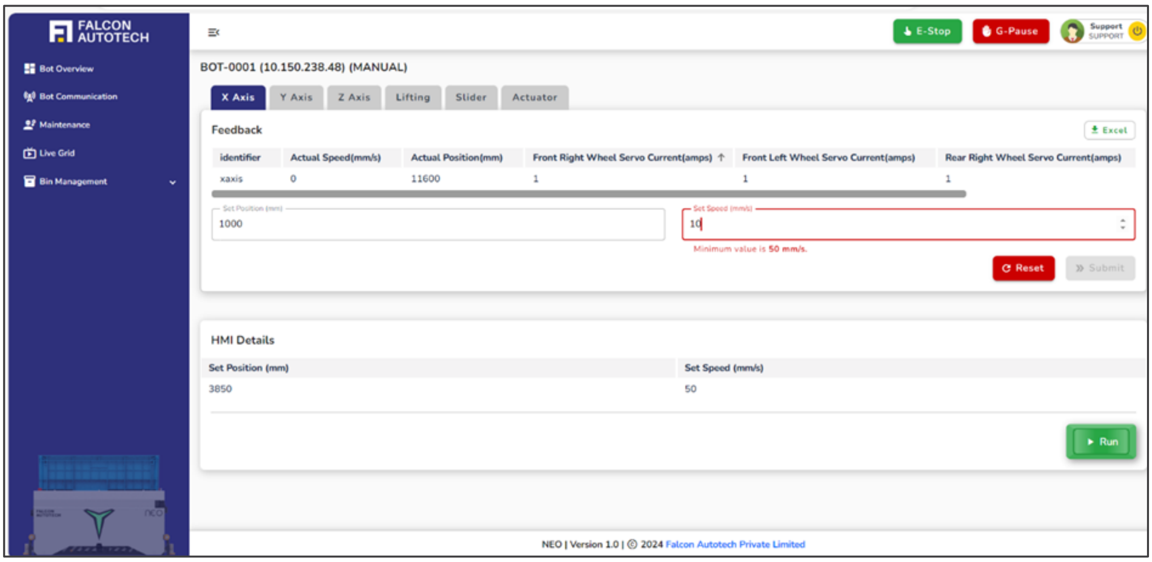


Figure 6.21 Bot X axis Error Message

6.3.2.1.2 Y Axis:

The **Y Axis** section enables manual control of the robot along the Y axis.

- Users can enter the desired position and velocity in the input fields provided.

- Real-time feedback for actual position and servo current is displayed, allowing users to track the robot's status.

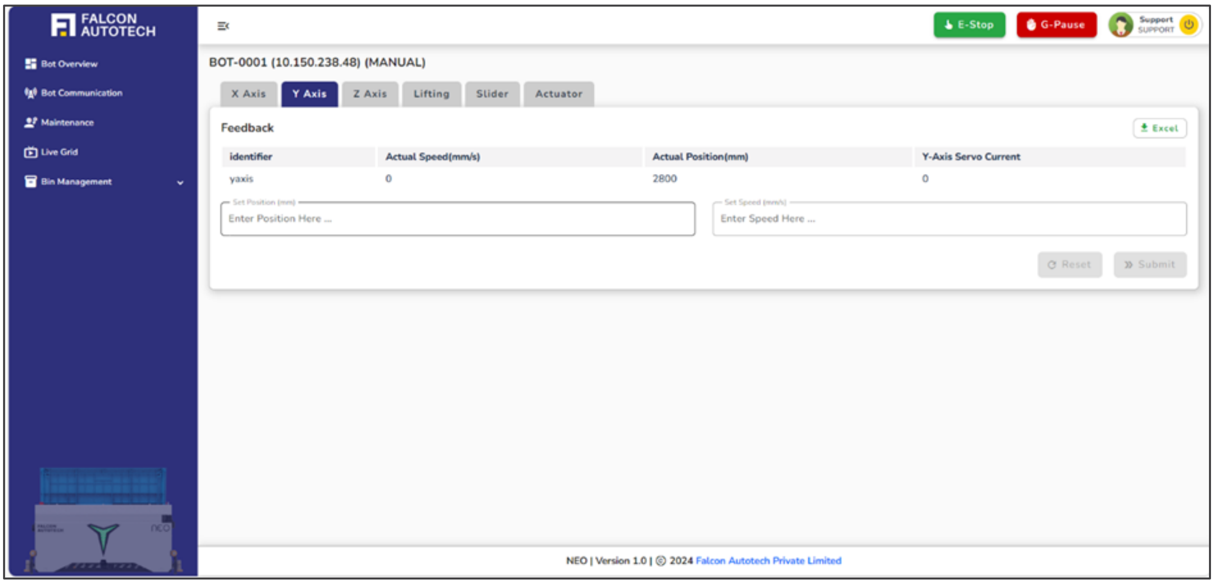


Figure 6.22 Y-Axis HMI Screen

- After setting the values, clicking Submit applies them. If adjustments are needed, users can click Reset to clear the inputs.
- A table below the section shows the configured values for verification.

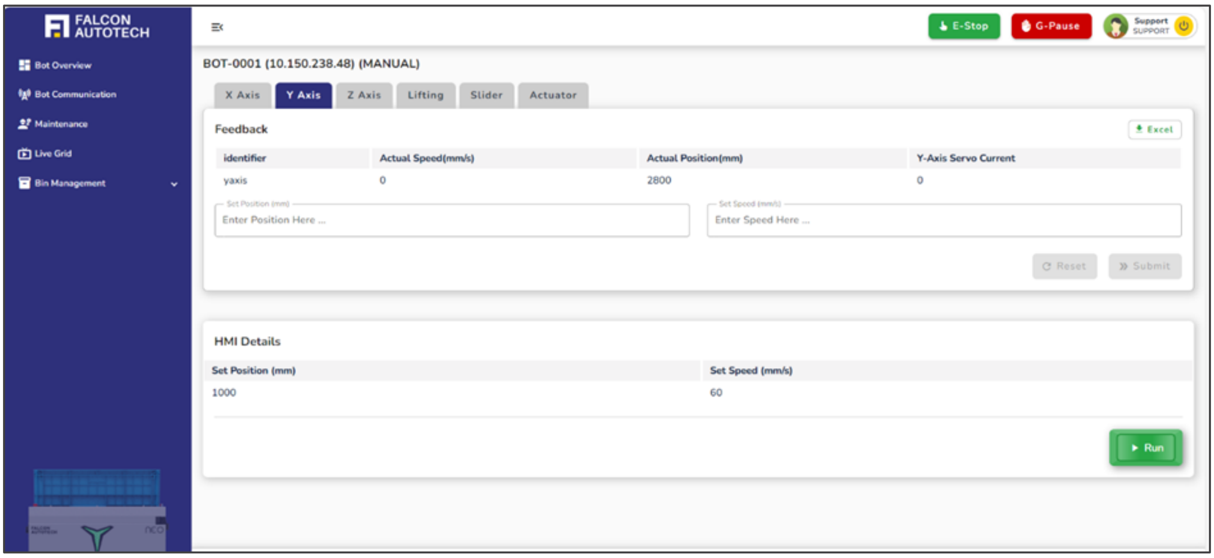


Figure 6.23 Y axis After Submitting Values

- To execute the motion, users must press and hold the Run button, initiating the robot's movement as per the preset values.
- The speed is limited to a safe range of 50 to 300 mm/s, ensuring precise and controlled operation.

- If the speed or position values exceed the defined thresholds, an error message is displayed below the input fields on the dashboard. This message specifies the permissible limits, guiding the user to input valid values within the allowed range. This ensures operational safety and prevents unintended robot behavior.

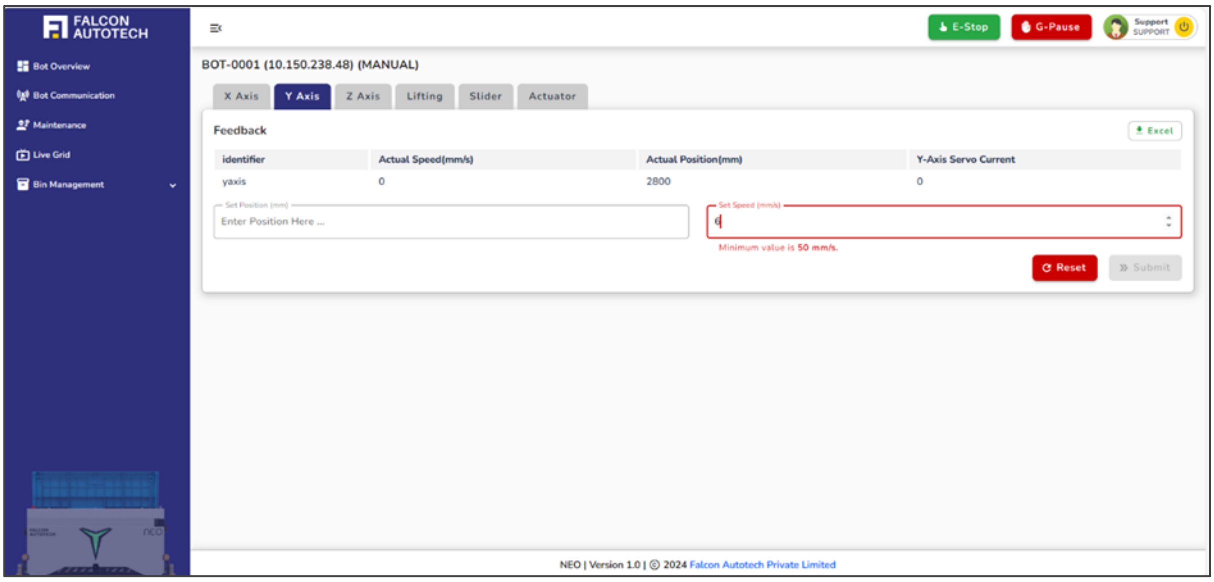


Figure 6.24 Y axis Error Message

6.3.2.1.3 Z Axis:

The Z Axis section empowers users with precise manual control over the robot's vertical movement along the Z axis, ensuring seamless operation in critical scenarios.

Users can define the minimum position, maximum position, and jog velocity within the input fields, adhering to the grid's predefined Z-direction limits for safety and accuracy. The minimum and maximum positions serve as safety constraints, ensuring the robot remains within the defined boundaries of the layout, preventing unintended overreach or collision.

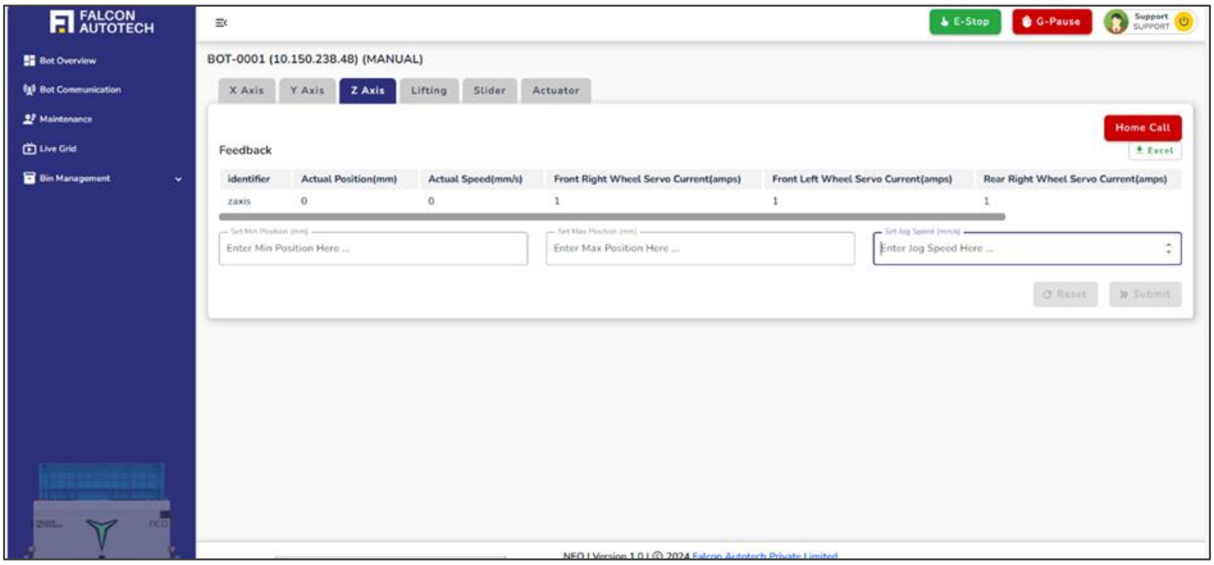


Figure 6.25 Z-axis HMI Screen

- The system provides real-time feedback on the robot's current position and velocity, allowing users to monitor the robot's exact state and make informed adjustments.
- Once the desired values are set, clicking Submit applies them, while the Reset button offers the flexibility to modify inputs.
- A detailed table below displays the configured values for easy verification and reference.

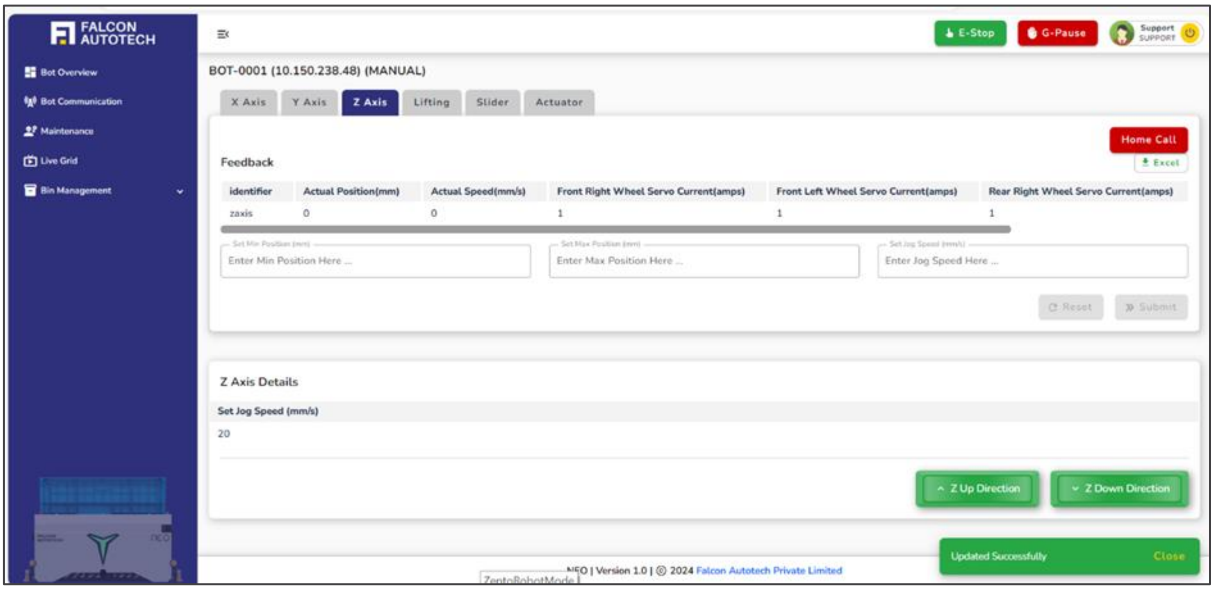


Figure 6.26 Z axis After Submitting Values

- To initiate movement, users can press and hold the Z Up Direction or Z Down Direction buttons, directing the robot to move vertically based on the preset parameters with unparalleled precision.
- The speed range is thoughtfully capped between 10 to 50 mm/s, ensuring smooth and controlled vertical motion.
- Additionally, the Home Call button adds an extra layer of functionality. When activated, it localizes the robot to a barcode position, mirroring the Home feature but tailored specifically for manual mode. This ensures effortless localization and accurate positioning during manual operations.
- If the speed or position values exceed the defined thresholds, an error message is displayed below the input fields on the dashboard. This message specifies the permissible limits, guiding the user to input valid values within the allowed range. This ensures operational safety and prevents unintended robot behavior.

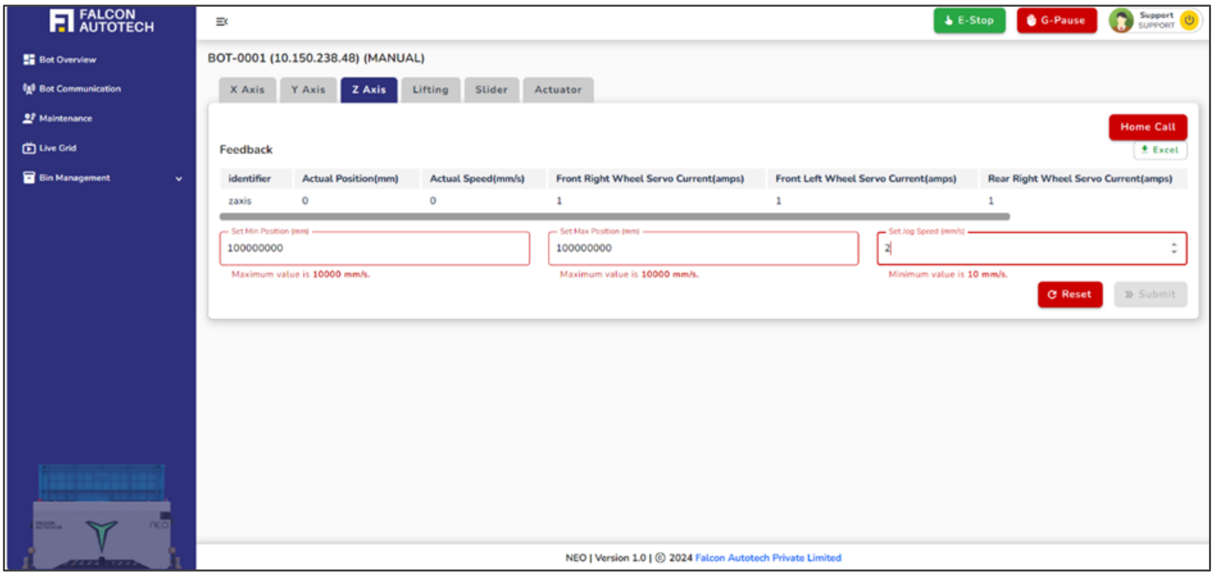


Figure 6.27 Z axis Error Message

6.3.2.1.4 Lifting:

The Lifting Section provides users with precise control to lift the robot along specific axes, ensuring accurate adjustments during manual operations.

The interface features three dedicated buttons for lifting, each corresponding to movement along one of the three predefined axes: X, Y, or Z. These buttons are configured with precise, predefined values to enable seamless and controlled movement.

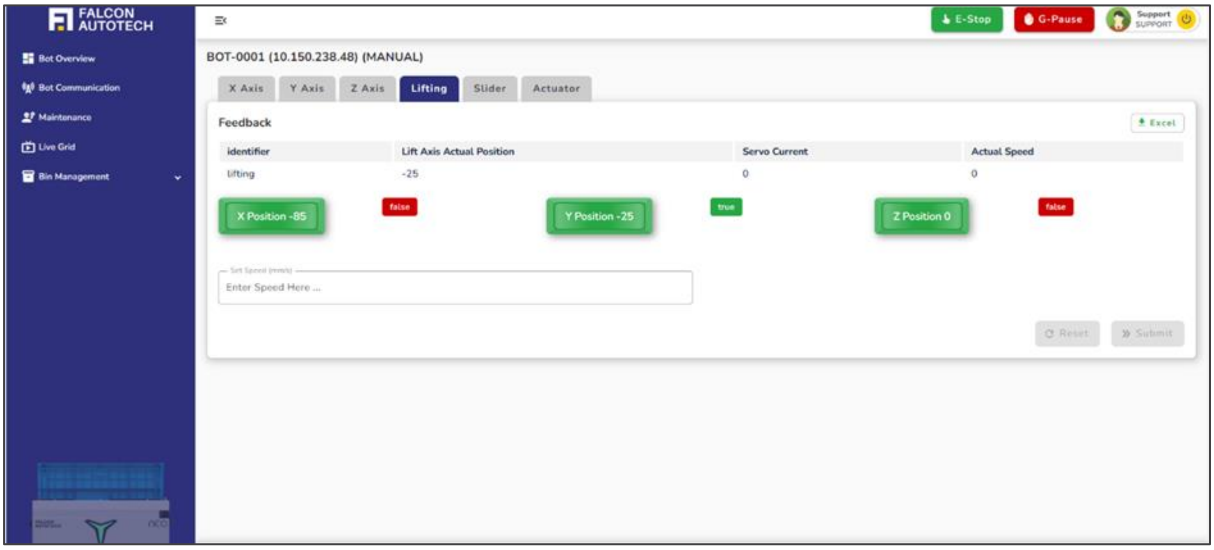


Figure 6.28 Bot Lifting HMI Mode

- A feedback panel dynamically updates in real-time, displaying critical parameters such as the actual velocity, current position, and servo current. This ensures users have continuous visibility into the robot's operational status.
- Users can configure the desired velocity to execute lifting operations, tailoring the movement to suit specific requirements.

- Once the desired values are set, clicking Submit applies them, while the Reset button offers the flexibility to modify inputs.

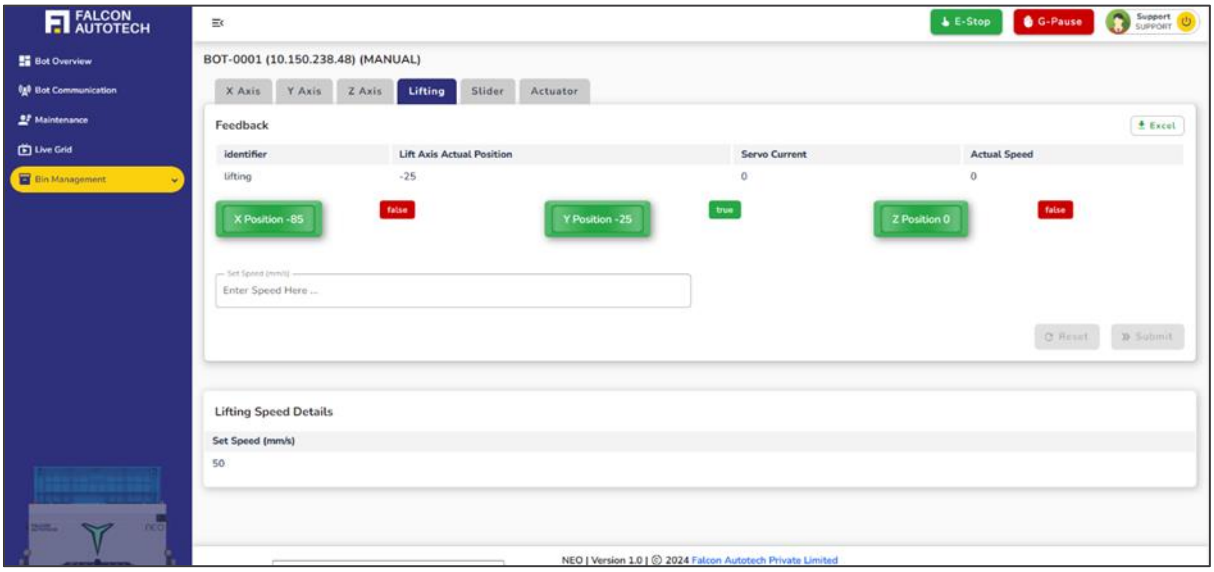


Figure 6.29 Bot Lifting Screen after submitting values

- A detailed table below displays the configured values for easy verification and reference.
- Once the parameters are set, pressing the corresponding **lifting position buttons (X, Y, Z)**—designed as momentary controls—enables the robot to lift in the specified direction. The robot's lift position is adjusted with precision, and feedback values are instantly reflected on the interface for user confirmation.
- Upon successfully completing the lift, a **true/false feedback indicator** next to the button confirms whether the operation was executed as intended.
- The **speed limit** for lifting operations is carefully regulated between **10 to 50 mm/s**, ensuring safe, smooth, and efficient movements. This section combines user-friendly controls with robust feedback mechanisms, making it indispensable for meticulous robot handling.

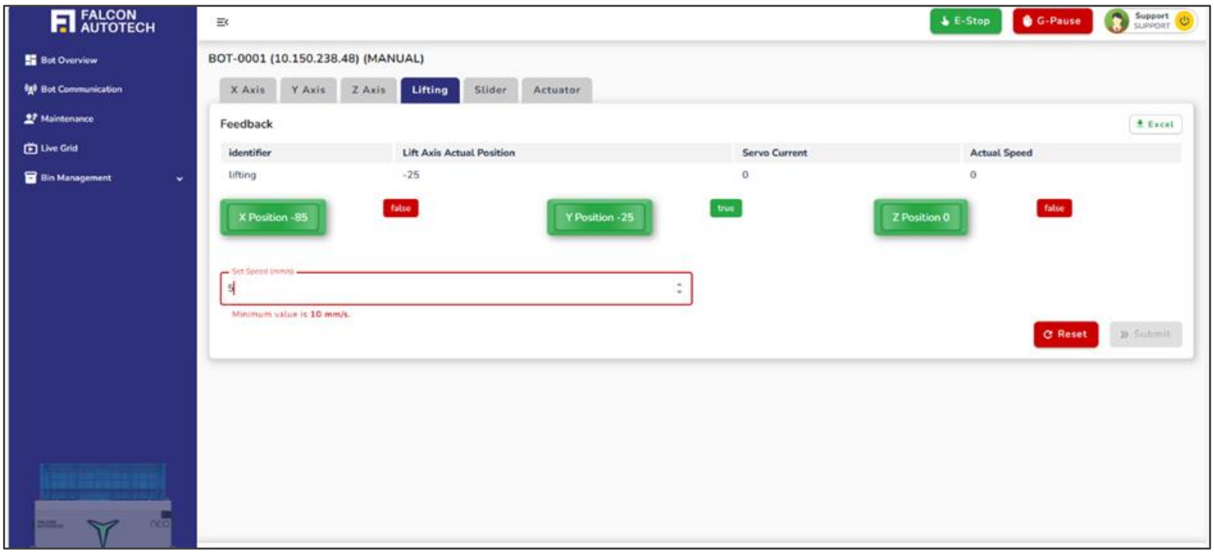


Figure 6.30 Bot Lifting Error Message

6.3.2.1.5 Slider:

The slider is a critical component of the NEO system, functioning as the robot's arm to facilitate precise pick-and-place operations

The interface offers three dedicated buttons for slider positions: Left, Right, and Centre. Each button corresponds to a predefined direction, configured with precise values to ensure accurate and controlled movements.

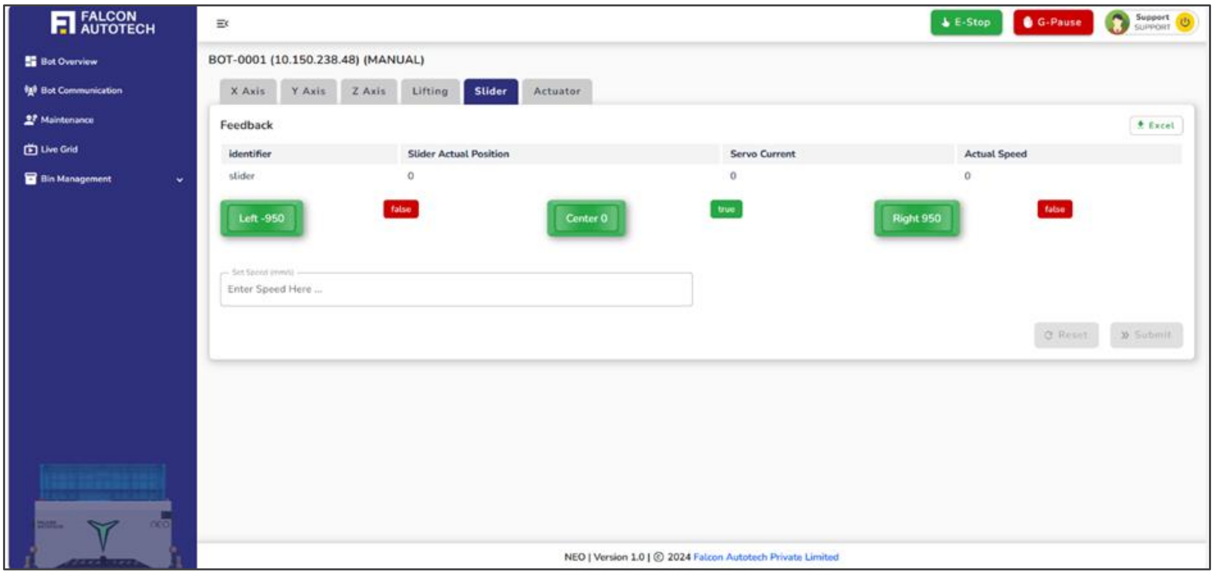


Figure 6.31 Bot Slider HMI Screen

A real-time feedback panel displays essential metrics, including actual velocity, current position, and servo actual position, providing users with continuous insights into the slider's operational status.

Users can set the desired velocity to move the robot's slider in the selected direction, tailoring operations to specific requirements.

Once the desired values are set, clicking Submit applies them, while the Reset button offers the flexibility to modify inputs.

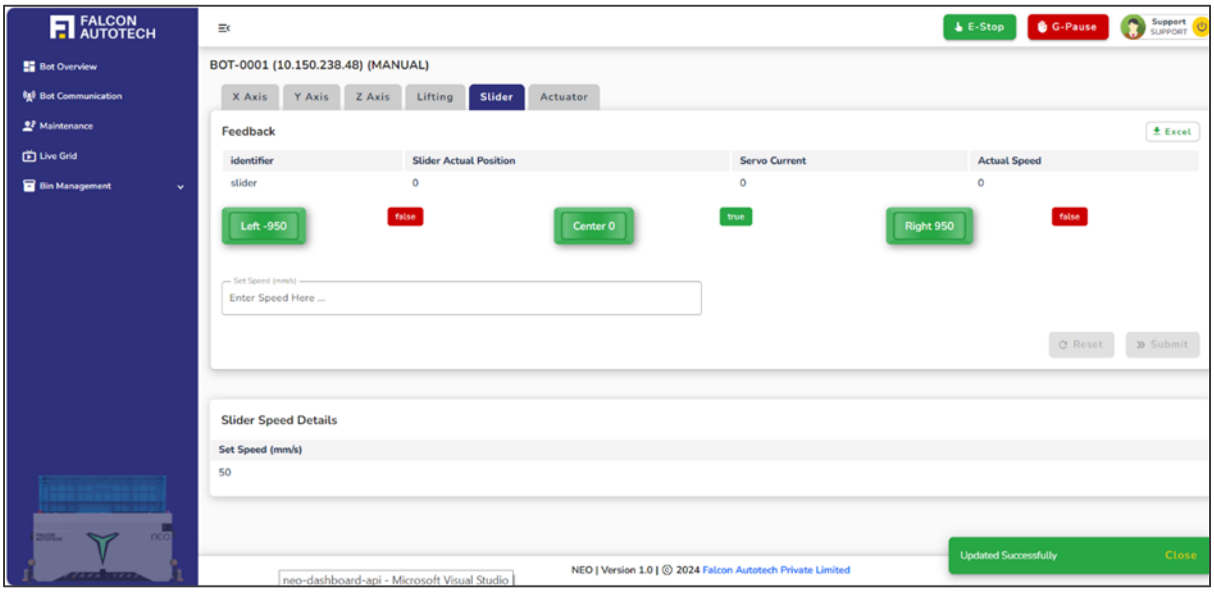


Figure 6.32 Bot Slider Axis after submitting values

A detailed table below displays the configured values for easy verification and reference.

Once parameters are configured, the user can activate the slider position buttons, which function as momentary controls to guide the slider's movement. The slider adjusts to the specified position seamlessly, with the feedback panel updating values dynamically for real-time monitoring.

Upon reaching the intended position, a true/false feedback indicator next to the button confirms the success of the operation.

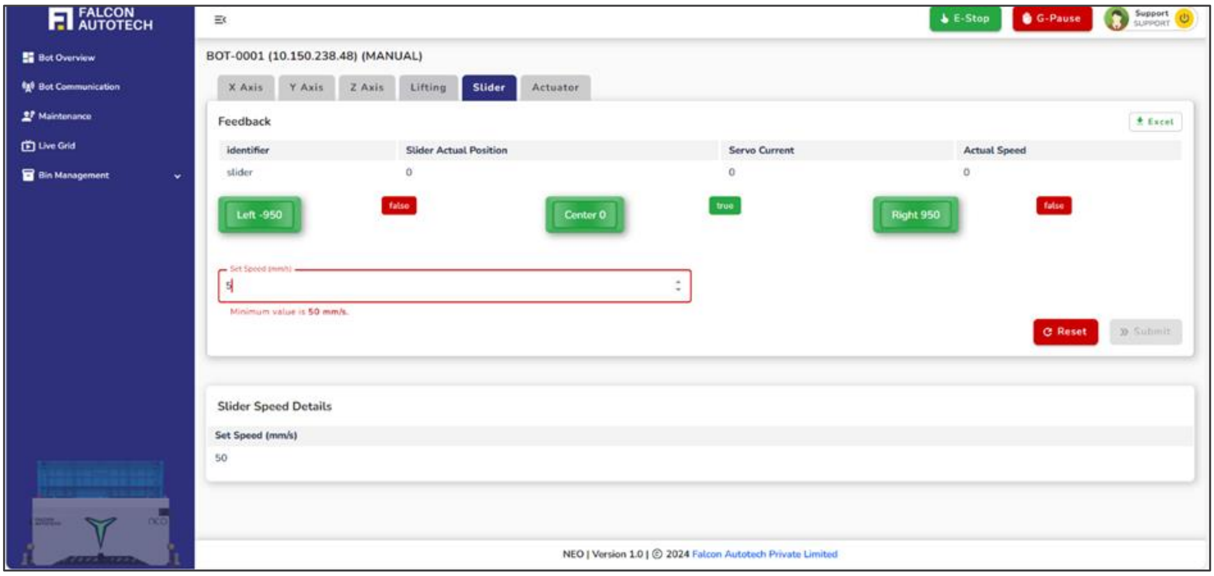


Figure 6.33 Bot Slider Axis Error Message

The slider's speed is regulated within a safe range of 50 to 300 mm/s, ensuring smooth and efficient functionality. This precise control mechanism ensures seamless integration with the NEO system, optimizing the robot's performance for handling bins.

6.3.2.1.6 Actuator:

The robot is equipped with **four actuators**, also referred to as **fingers**, which play a vital role in the precise handling of bins. These actuators are named: **Front Left**, **Front Right**, **Rear Left**, and **Rear Right**.

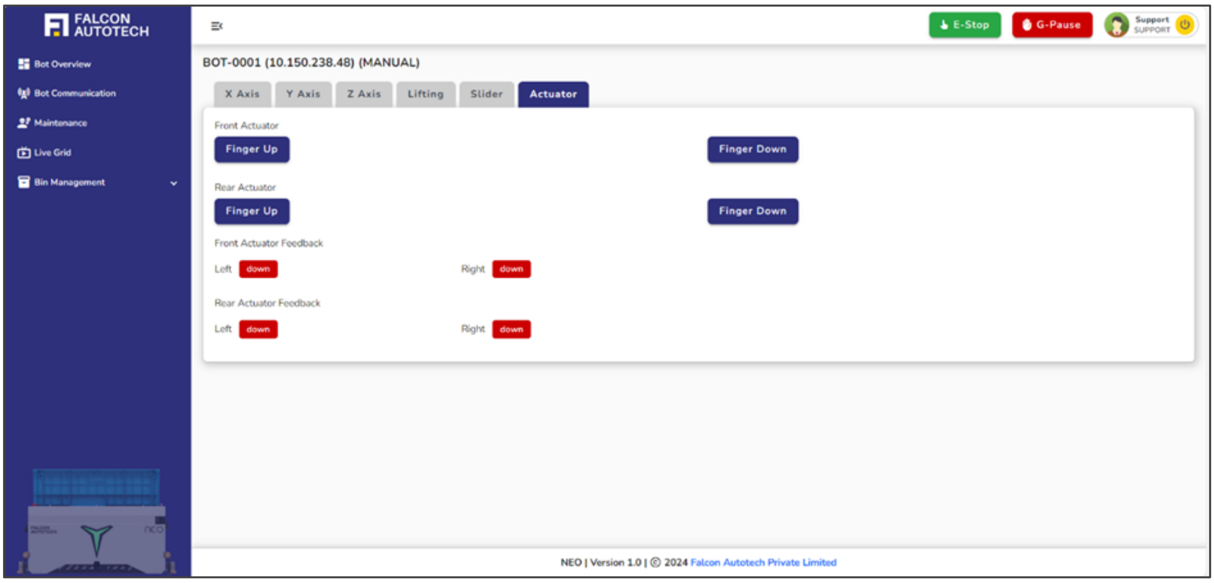


Figure 6.34 Bot Actuator HMI Screen

- Users can activate or control each actuator individually based on operational requirements by simply clicking the corresponding button for the desired finger.
- A real-time feedback panel provides instant updates on the status of each actuator, allowing users to monitor their functionality and ensure smooth operation.
- This feedback ensures confidence in the system's performance, providing critical insights into whether the actuators are operating correctly or require attention.
- This intuitive control system, paired with detailed feedback, ensures that the robot's actuators function with precision, reliability, and efficiency in every task.

6.3.2.2 Maintenance Mode:

When the **Maintenance Mode** section is selected on the Bot HMI Mode page, the user is seamlessly redirected to the **Maintenance Screen**—a dedicated interface designed to provide essential feedback and diagnostic information about the robot's overall health and performance.

- This section serves as a critical resource, offering detailed insights into the robot's **maintenance status**. Users can monitor key aspects, such as battery **health** and in-depth data on vital components like **sensors, lidars**, and more.
- By consolidating these diagnostics into an intuitive and accessible interface, the Maintenance Screen empowers users to proactively manage the robot's upkeep, ensuring smooth operations and minimizing downtime. It is an indispensable tool for maintaining the system's reliability and efficiency.

6.3.2.2.1 Barcode Scanner Data:

The **Barcode Scanner Data** section provides essential details about barcodes previously read during the automatic cycle. Each barcode's coordinates and control information are meticulously recorded to ensure precision and traceability in operations.

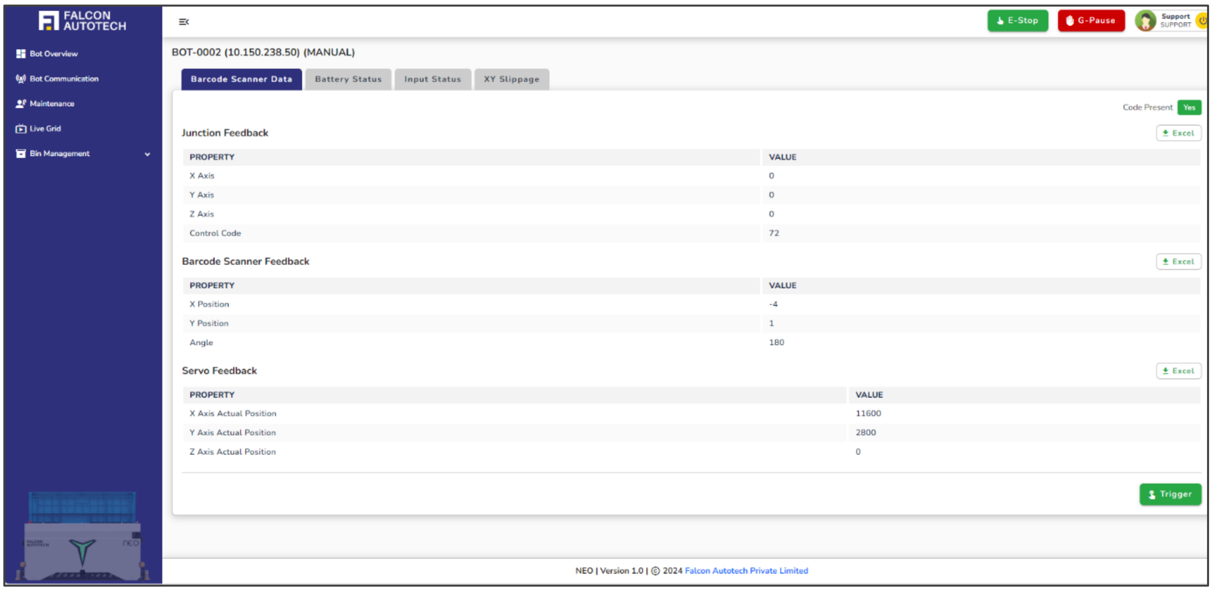


Figure 6.35 Barcode Scanner Data

- Junction Coordinate Feedback:
- **X Axis**: Displays the X-coordinate value of the barcode as read in the last auto cycle, offering spatial clarity for its horizontal position.
- **Y Axis**: Captures the Y-coordinate value, representing the vertical alignment of the barcode during the previous cycle.
- **Z Axis**: Reflects the Z-coordinate value, denoting the depth or height at which the barcode was read.
- **Control Code**: Represents the unique tag number associated with the barcode, enabling precise identification and tracking within the system.
- Barcode Scanner Feedback:
- **X Position**: Shows the current horizontal position of the barcode, updated in real time.
- **Y Position**: Indicates the vertical alignment of the barcode as it is actively scanned.
- **Angle**: Displays the orientation of the barcode, ensuring accurate readings irrespective of its alignment.
- Servo Feedback:
- **X Axis Actual Position**: Displays the real-time horizontal position of the servo.
- **Y Axis Actual Position**: Monitors the vertical alignment of the servo during operations.
- **Z Axis Actual Position**: Tracks the depth or height position, ensuring precision in all axes.

6.3.2.2.2 Battery Status:

The **Battery Status Feedback** section provides a detailed overview of the robot's battery health, offering critical metrics to ensure optimal performance and reliability. Users can access the following vital details:

- **Serial Number**: Uniquely identifies the battery for tracking and maintenance purposes.
- **Voltage**: Displays the current voltage level, indicating the battery's operational state.
- **Current**: Monitors the real-time flow of current for effective power management.
- **Temperature**: Tracks the battery's temperature, ensuring it remains within safe operating limits.
- **Remaining Ah**: Shows the available ampere-hours, providing an accurate measure of the remaining battery capacity.
- **Full Ah**: Indicates the total ampere-hours of the battery when fully charged, enabling users to compare and assess performance.

This feedback ensures users are equipped with the necessary data to monitor, maintain, and optimize the robot's power system efficiently.

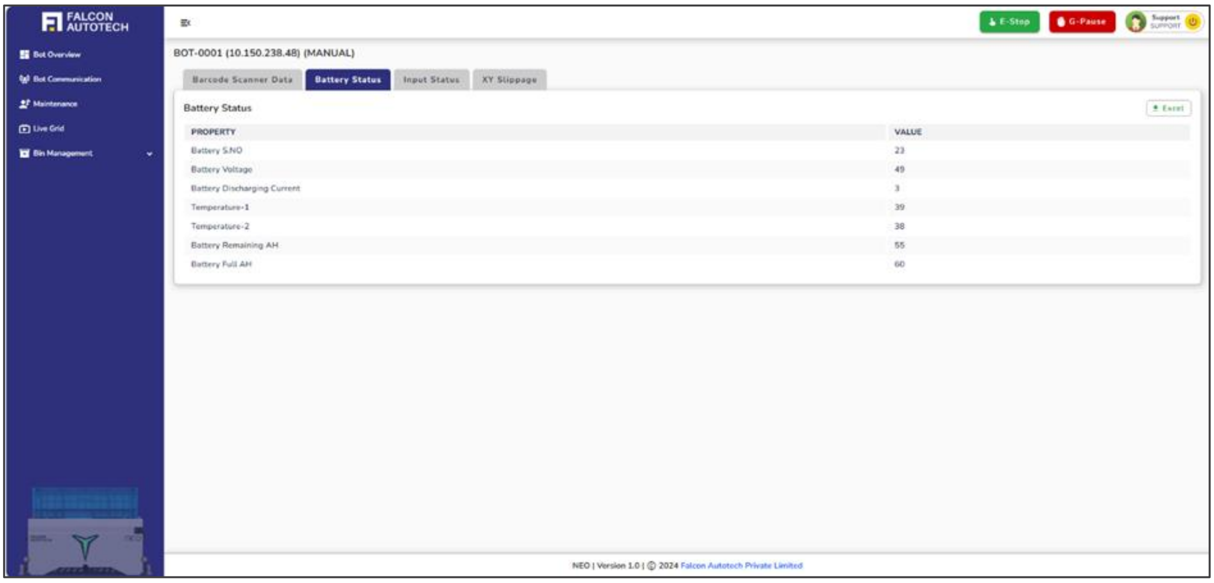


Figure 6.36 Battery Status Display

6.3.2.2.3 Input Status:

The **Input Status Feedback** section provides detailed insights into the operational state of the robot's sensors and lidars. This feature enables the user to assess their functionality and ensure the robot's systems are performing optimally.

- **Health Status**: The condition of each sensor and lidar is categorized as *Healthy*, *Alert*, or *Danger*, based on their performance.
- **Feedback Indicators**: The status is represented by **True** or **False** values:
  - A **False** status signifies the sensor or lidar is functioning correctly and within the optimal range.
  - A **True** status indicates an issue requiring immediate attention, signaling that the component is either in an alert or danger state.

This reverse feedback mechanism ensures that users focus on addressing the critical components flagged as **True**, enabling effective diagnostics and prompt corrective actions for maintaining peak robot performance.

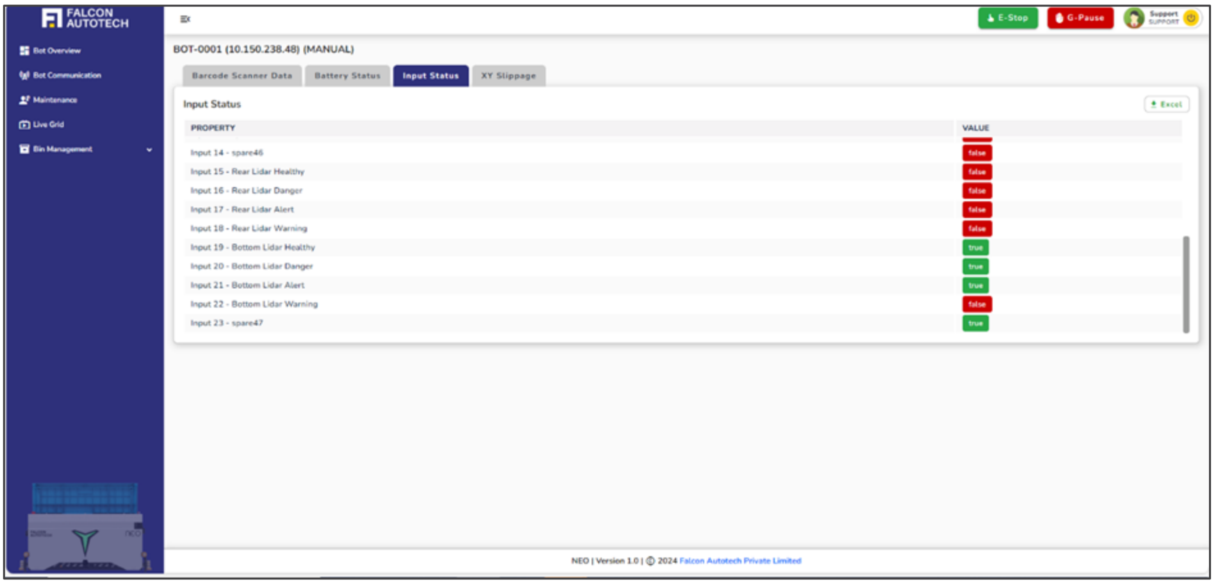


Figure 6.37 Input Status Display

6.3.2.2.4 X Y Slippage:

This section provides detailed insights into critical operational parameters, offering users a clear understanding of the system's performance and behavior. It highlights key metrics and factors, enabling precise diagnostics and adjustments. By presenting accurate, real-time data, this section plays a vital role in maintaining operational efficiency, ensuring system reliability, and facilitating proactive decision-making.

- XY Before Slippage:
- This parameter represents the slippage factor, expressed as a percentage, based on the robot's total travel distance. It applies to movements with a minimum travel distance of 20mm, ensuring precise measurement of minor positional shifts.
- XY After Slippage:
- This parameter signifies the slippage factor, also expressed as a percentage, calculated for the robot's total travel distance after significant movement. It applies to travel distances with a minimum threshold of 150mm, capturing larger-scale positional deviations for accurate diagnostics and adjustments.

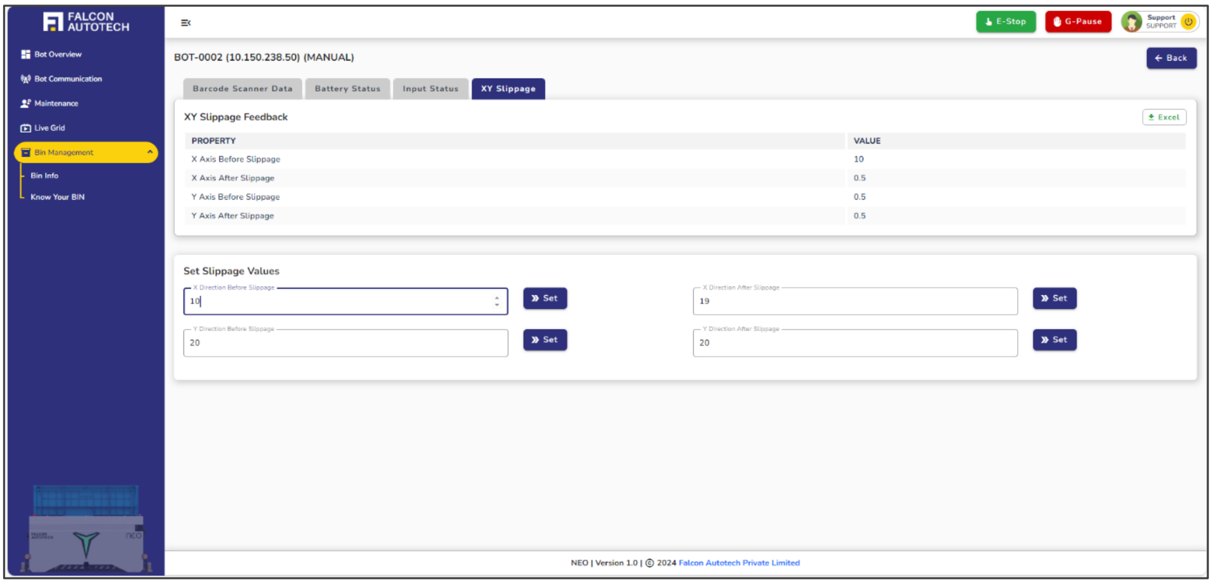


Figure 6.38 XY Slippage Screen

6.3.2.3 Manual Alarms:

The **Manual Alarms** section is an integral part of the Manual Mode page, designed to provide users with critical insights into the bot's operation during manual mode. On selecting this section, users are redirected to a dedicated **Alarms** page featuring two tabs:

- **Manual Alarms**
- **Bypass Alarms**

- **Manual Alarms Tab:**

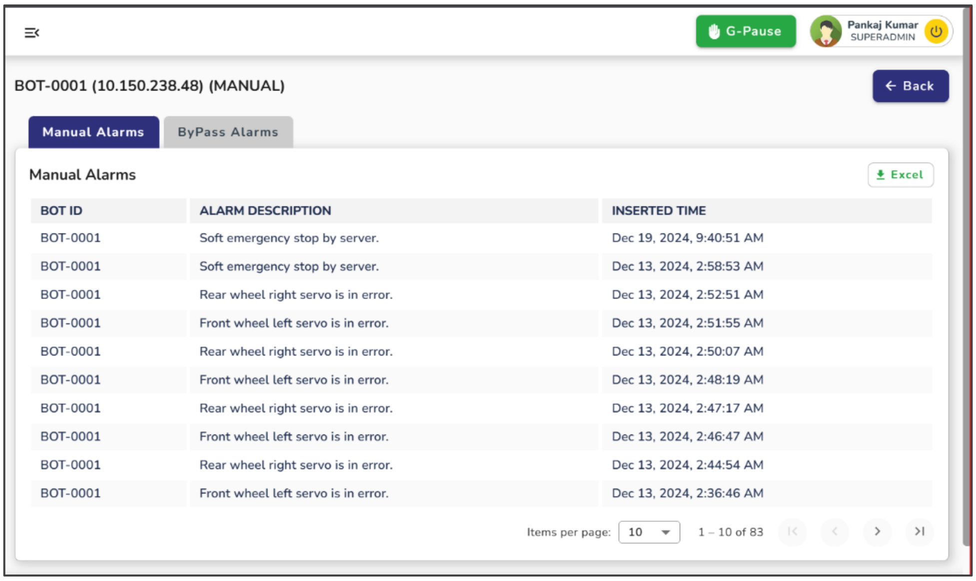


Figure 6.39 Manual Alarms Screen

This tab focuses on alarms generated during manual packet exchanges between the bot and the server—specifically when the bot operates in HMI manual mode. These alarms primarily highlight issues related to the bot's performance in this mode.

The interactive table under this tab provides the following key details:

- **Bot ID**: Identifies the bot encountering the alarm, enabling targeted troubleshooting.
- **Alarm Description**: Offers a concise explanation of the issue's cause for quick understanding.
- **Timestamp**: Records the precise time the alarm was triggered, ensuring accurate tracking and diagnostics.

To enhance user efficiency, the table supports advanced functionalities, including:

- **Sorting**: Organize data for better readability.
- **Searching**: Locate specific alarms quickly using keywords.
- **Excel Export**: Download the table as an Excel file for report generation and further analysis.

This combination of features ensures users can address errors effectively and maintain operational continuity.

- **Bypass Alarms Tab:**

The **Bypass Alarms** tab provides a solution for critical scenarios where certain alarms can be bypassed to restore partial functionality. This operation, however, is highly sensitive and requires strict authentication.

When a user selects this tab, they are prompted to enter a password for authorization. Upon successful authentication, the following table is displayed:

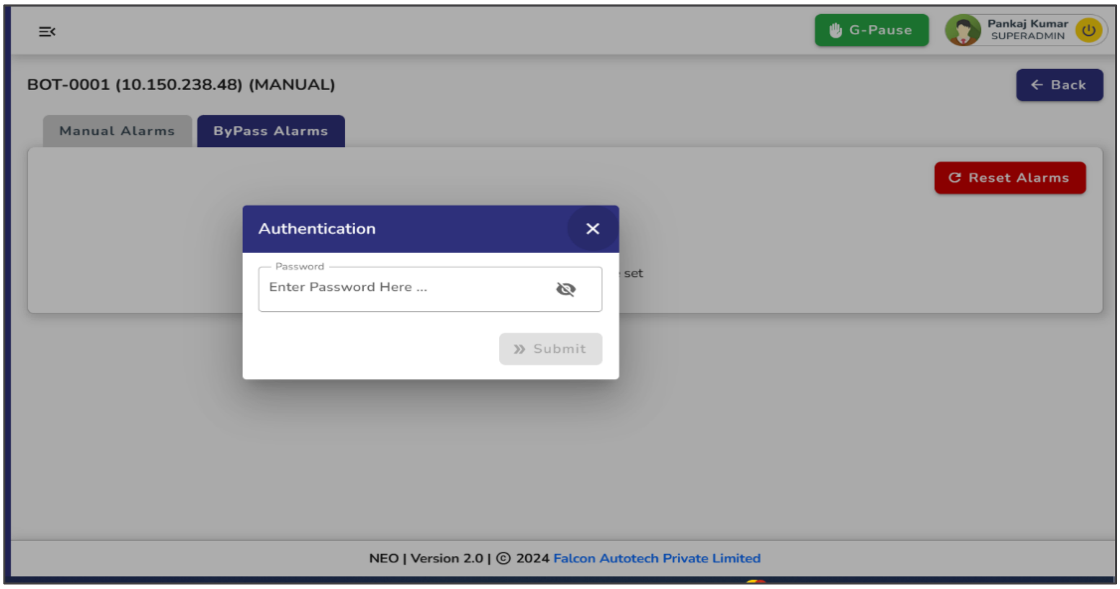


Figure 6.40 Authentication for Bypass Alarms

- **Bot ID**: Identifies the bot with the alarm.

- **Alarm Description**: Describes the specific alarm to be addressed.
- **Timestamp**: Indicates when the alarm was triggered.
- **Bypass Status**: Shows whether the alarm has already been bypassed (True/False).

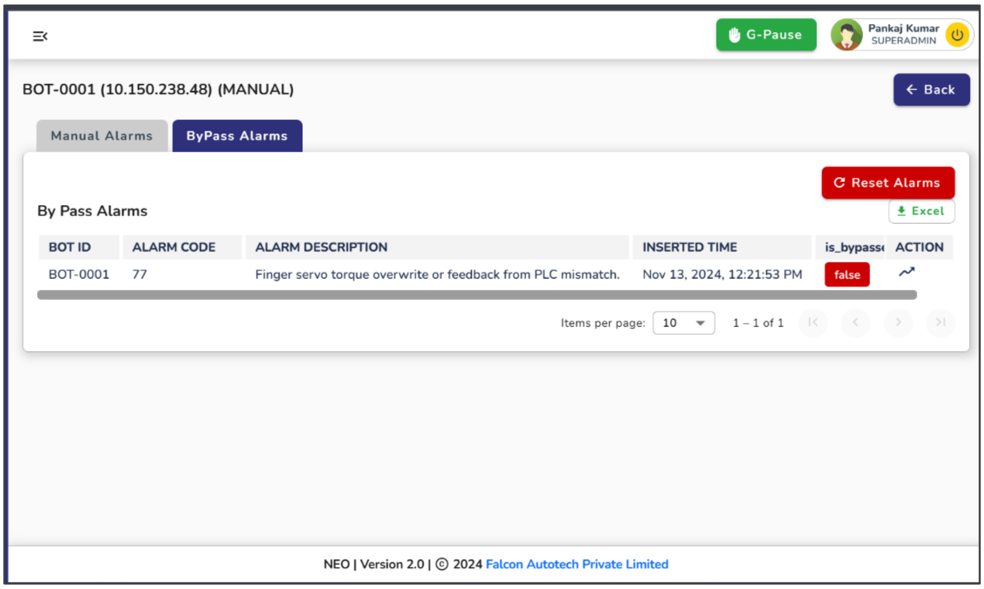


Figure 6.41 Bypass Alarms Screen

Each alarm entry includes an **Action Button**, allowing the user to bypass it. Once the bypass action is performed, the status is updated in real time, and the change is communicated to the PLC.

This meticulously designed interface ensures users can navigate alarms, authenticate critical operations, and maintain the bots' efficiency with precision and control.

6.3.3 Bot Alarms:

The Bot Alarms section serves as the final and vital part of the Bot Overview page. Upon clicking this button, the user is redirected to the Bot Alarms page, which provides a comprehensive table displaying the alarm history for the selected bot.

Alarms play a crucial role in diagnosing and understanding the issues affecting the robot's performance. They detail the reasons behind the bot's errors, offering clear insights into any malfunctions or operational anomalies. This information is instrumental in identifying and resolving problems efficiently, ensuring seamless operation and minimal downtime.

The **Bot Alarms** page is divided into two primary tabs, each catering to specific alarm scenarios:

- **Normal Errors:** This tab displays alarms encountered during the bot's normal operations in auto mode, highlighting any issues that may disrupt routine tasks.
- **Slider Recovery Issues:** This tab focuses on alarms related to slider recovery, providing targeted insights into malfunctions occurring during bin pick-and-place operations.

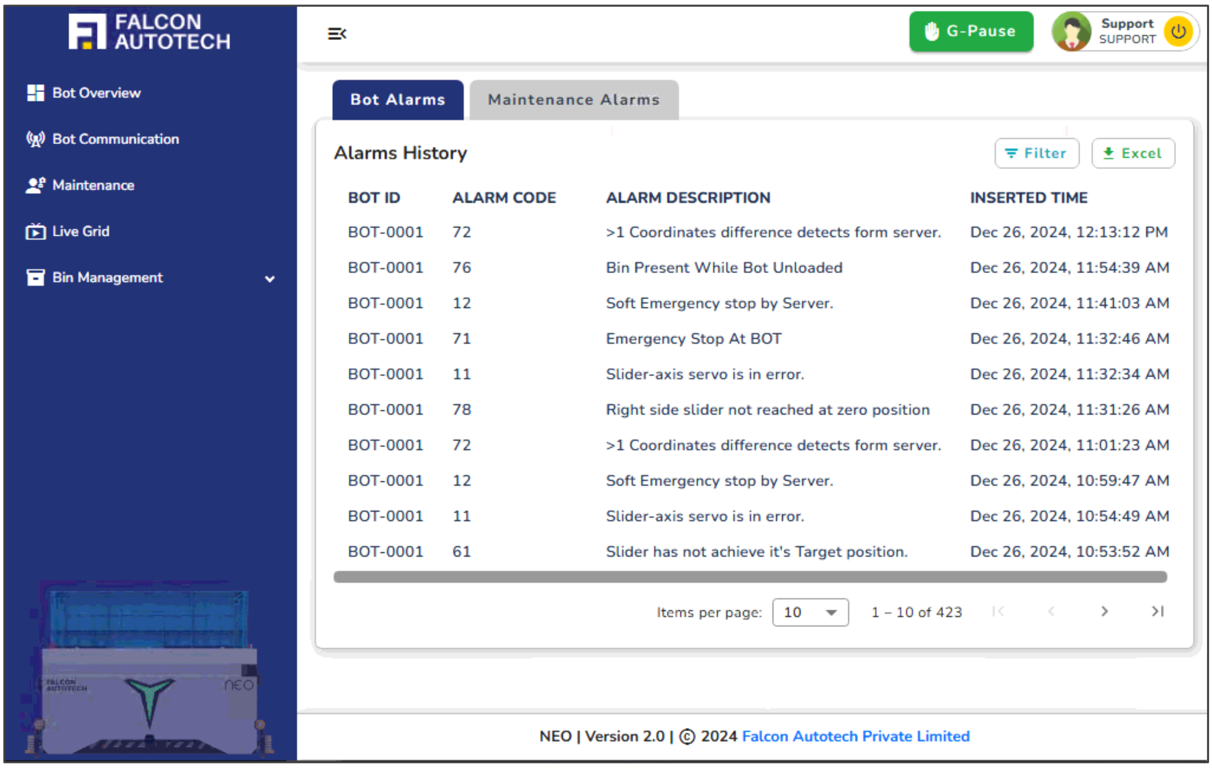


Figure 6.42 Bot Alarms screen

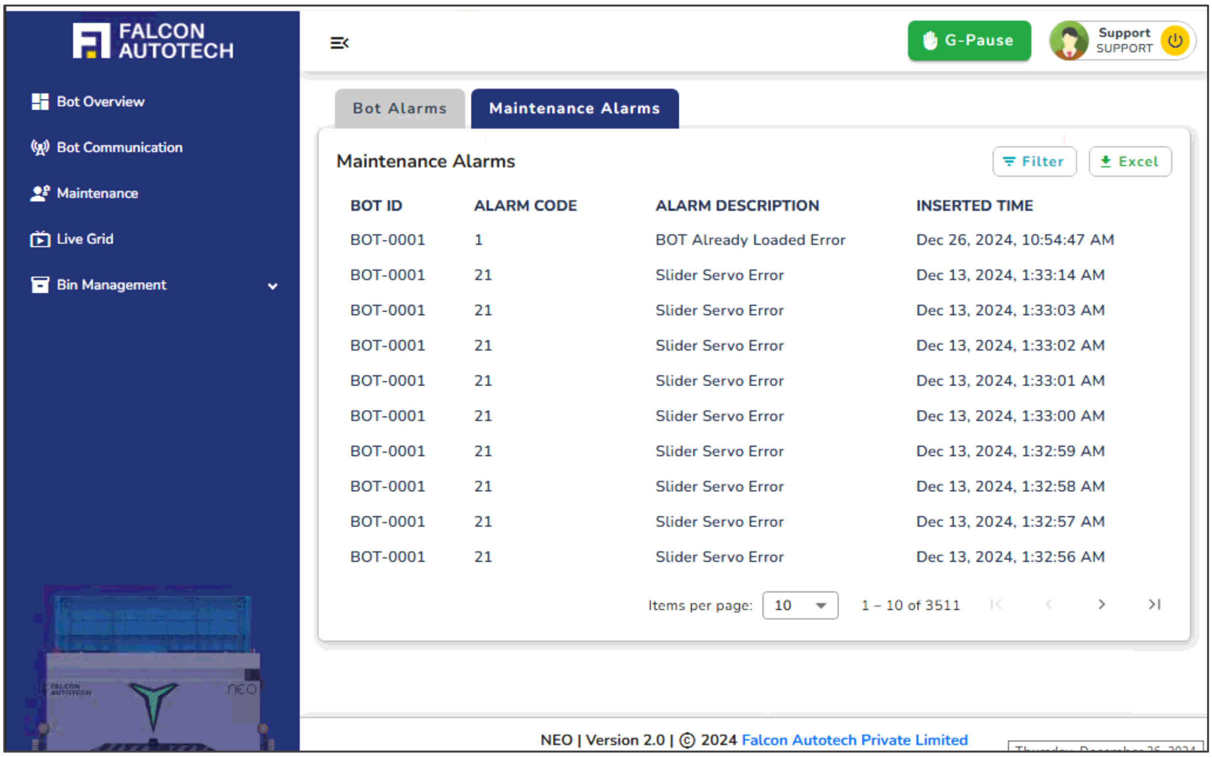


Figure 6.43 Maintenance alarms screen

Both tabs present a detailed and interactive table with the following information:

- **Bot ID:** Identifies the bot experiencing the issue.
- **Alarm Code:** Provides a unique code for the alarm, aiding quick diagnostics.
- **Alarm Description:** A brief explanation of the alarm's cause.
- **Timestamp:** The exact time the alarm was triggered for precise tracking.
- The table is designed for user convenience, offering features such as sorting, searching, and downloading data as an Excel file. This functionality supports efficient report generation and streamlined analysis, ensuring users can address errors effectively and maintain smooth operations.

6.3.4 Bot Auto:

Bot Auto mode represents that the bot is ready for auto tasks and can be utilized in the current process or wave. In the bot auto section, the user will find operation buttons such as Auto Start, Auto Stop and Auto calibration. Below the buttons there is a section that shows **Bot Alarms**. Each button is clickable and includes a feedback mechanism. The color coding, along with the enabling and disabling of the buttons, depends on the feedback status, which is also displayed on the dashboard.

6.3.4.1 Auto Start:

The **Auto Start** button prepares the bot for any upcoming tasks in auto mode. Its functionality ensures that the bot is in optimal condition to resume operations. If the feedback indicator is **red**, the user must ensure that certain manual parameters need to be checked before proceeding. Users must verify the following prerequisites:

- Ensure the **Reset** and **Home** statuses are set to **true**.

Once these conditions are met, clicking the **Auto Start** button transitions the bot to a healthy state, ready to undertake tasks. The bot can only be deployed for operations when the **Auto Start feedback** is **green**, indicating it is primed for smooth and uninterrupted functionality in auto mode.

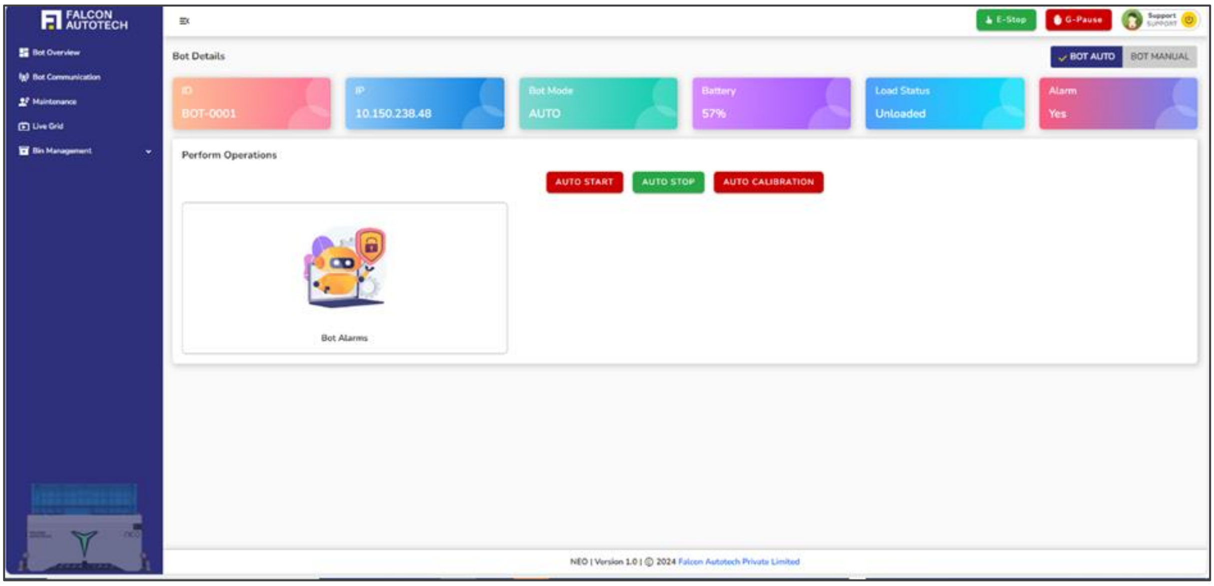


Figure 6.44 Auto Stop Enabled

Once when user clicks on the button, the screen will display a successful message, as shown in the image below.

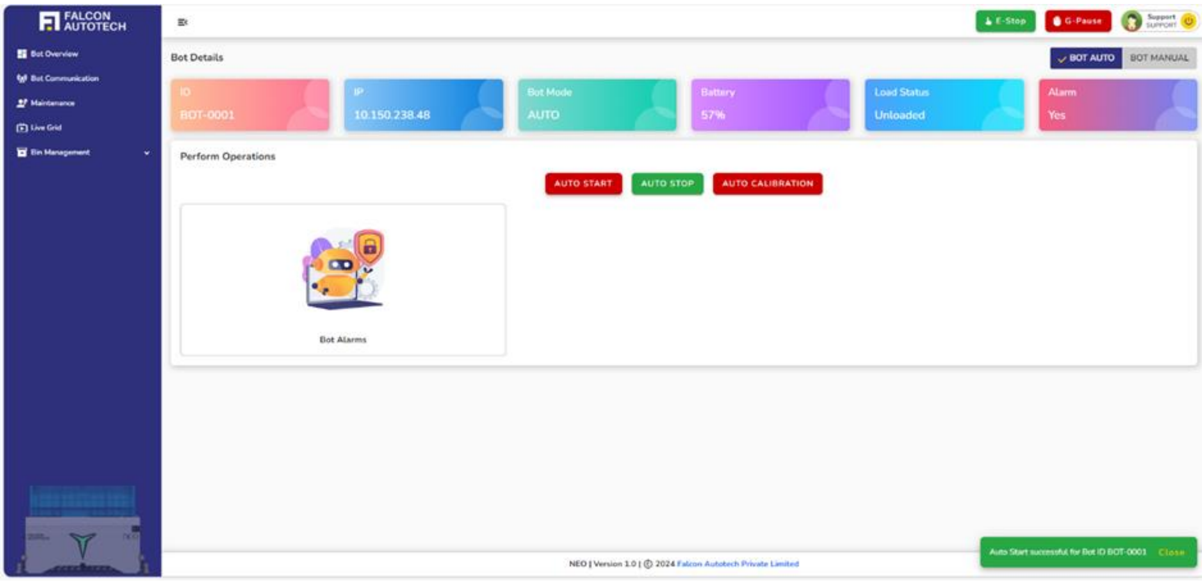


Figure 6.45 Auto Stop Successful Message

Once the feedback is true from PLC, the auto-start button turns green, and the screen will look like the one in image below.

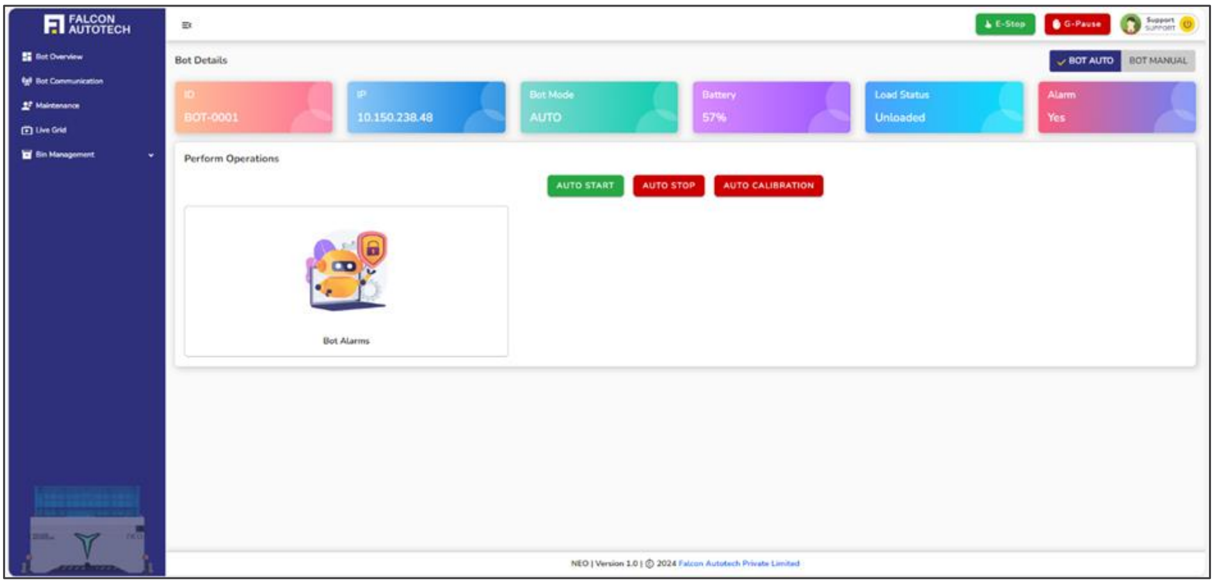


Figure 6.46 Auto Stop Feedback true from PLC

6.3.4.2 Auto Stop:

The **Auto Stop** button allows users to halt the robot mid-operation if necessary. This feature is particularly useful for pausing tasks temporarily due to unforeseen circumstances, with the option to resume operations later. To maintain seamless control, the **Auto Start** and **Auto Stop** buttons operate in a mutually exclusive manner: If **Auto Start** is active, the **Auto Stop**

feedback will display as **red**, indicating the robot is in operation. Conversely, if **Auto Stop** is engaged, the **Auto Start feedback** will show **red**, signaling that the robot is paused. This inverse functionality ensures clear, reliable feedback for the robot's operational state, allowing for safe and efficient management of tasks.

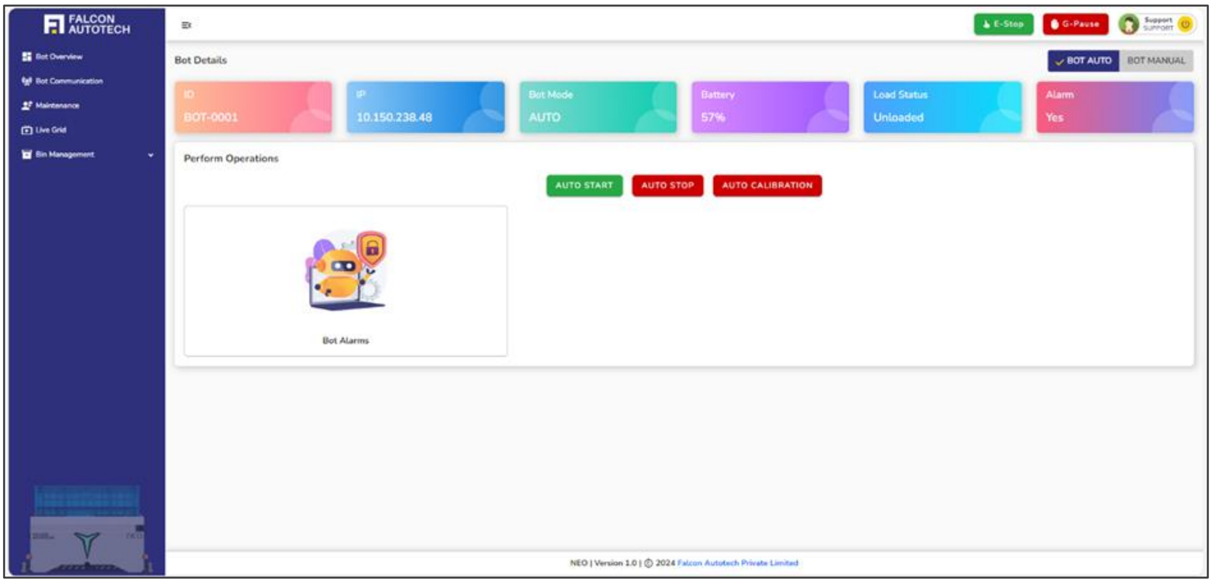


Figure 6.47 Auto Start Enabled

Once the user clicks the Auto Stop, the screen will display the message that auto stop is successfully activated.

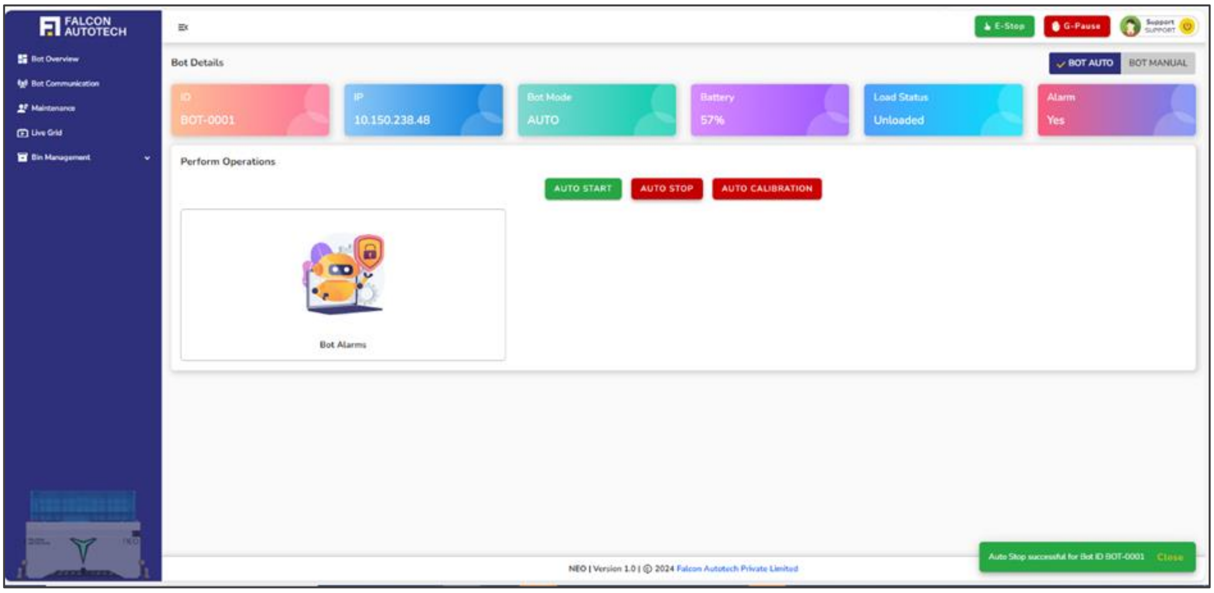


Figure 6.48 Auto Start Successful message

Once the feedback from PLC is received, then the **Auto Stop** button turns green, and the action has been successfully performed on robot.

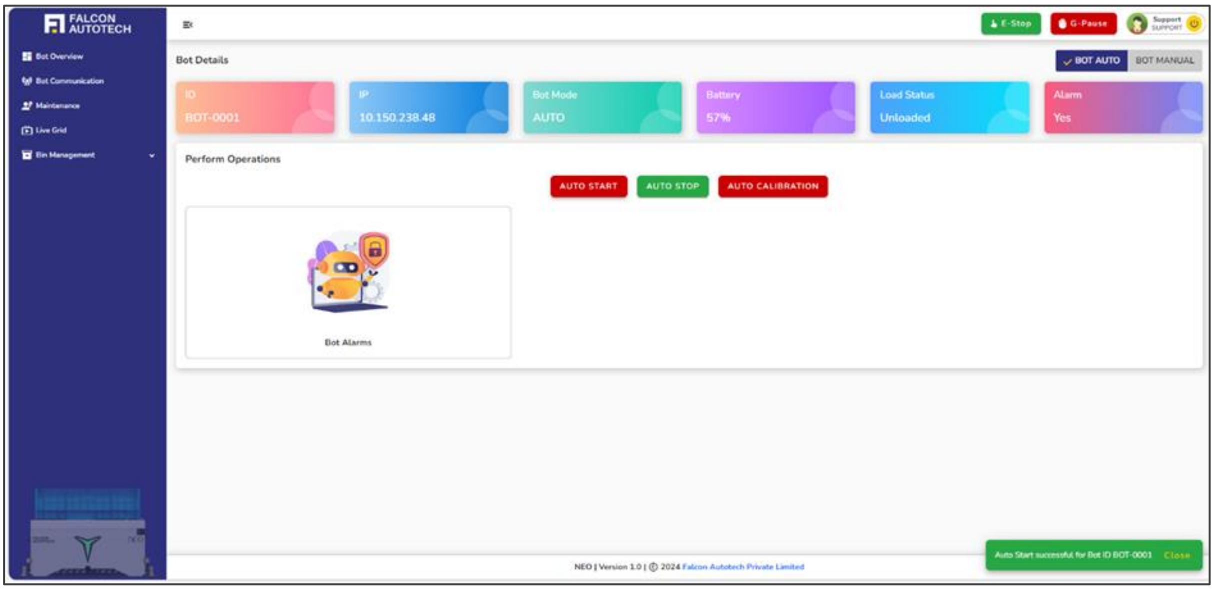


Figure 6.49 Auto Start Feedback true from PLC

7 Bot Communication

The **Bot Communication** section, accessible via the navigation bar, provides a dynamic graphical representation of the communication between the server and the bots. This visualization is crucial for monitoring real-time interactions and ensuring system health. The graph illustrates the **heartbeat signals** for all bots, plotting **time** on the X-axis (continuously updated) and **counter numbers (1 to 9)** on the Y-axis. Each bot is **color-coded** for easy identification, with corresponding labels displayed on the graph. The graph updates frequently, reflecting every communication packet exchanged between the **PLC (Programmable Logic Controller)** and the bots. Users can **hover over the lines** to view detailed information about specific communication frequencies or counter values at various time intervals. If communication ceases, the graph stabilizes, displaying a straight line, making it immediately apparent when a bot is inactive. This tool ensures comprehensive visibility into the bot-server communication flow, enabling quick troubleshooting and performance analysis.

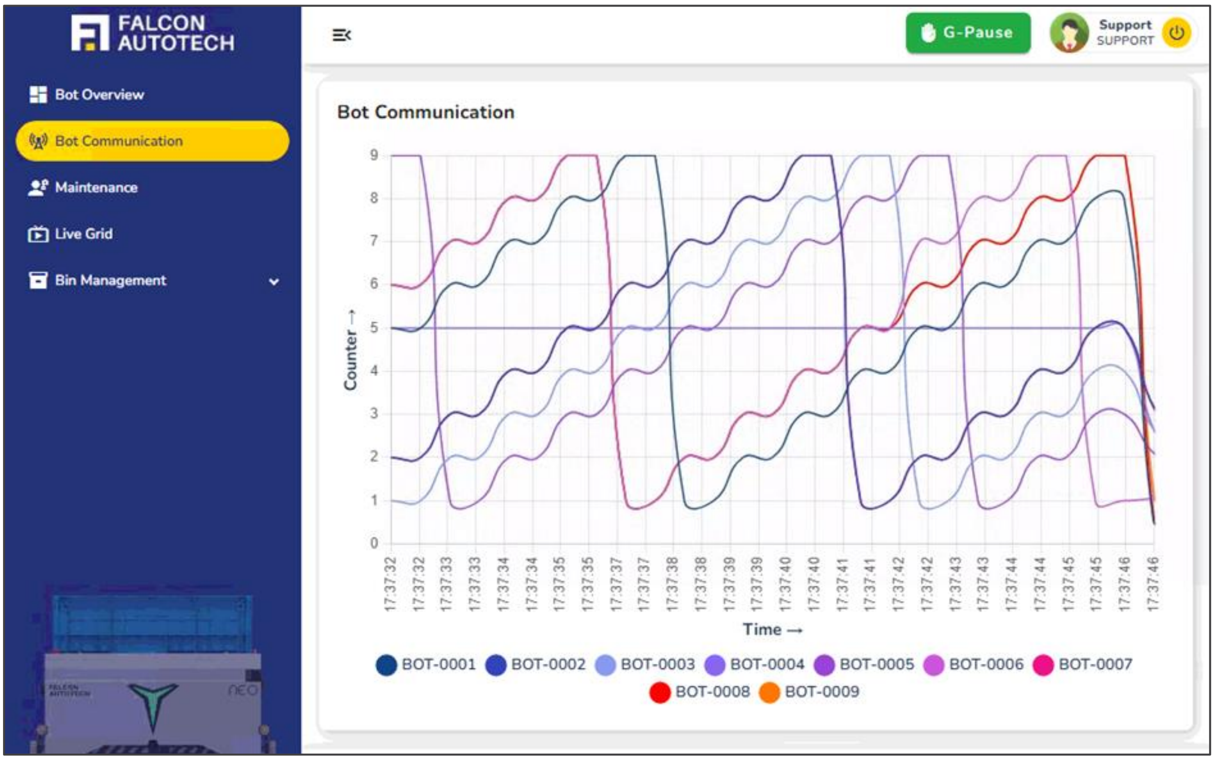


Figure 7.1 Robot's continuous communication

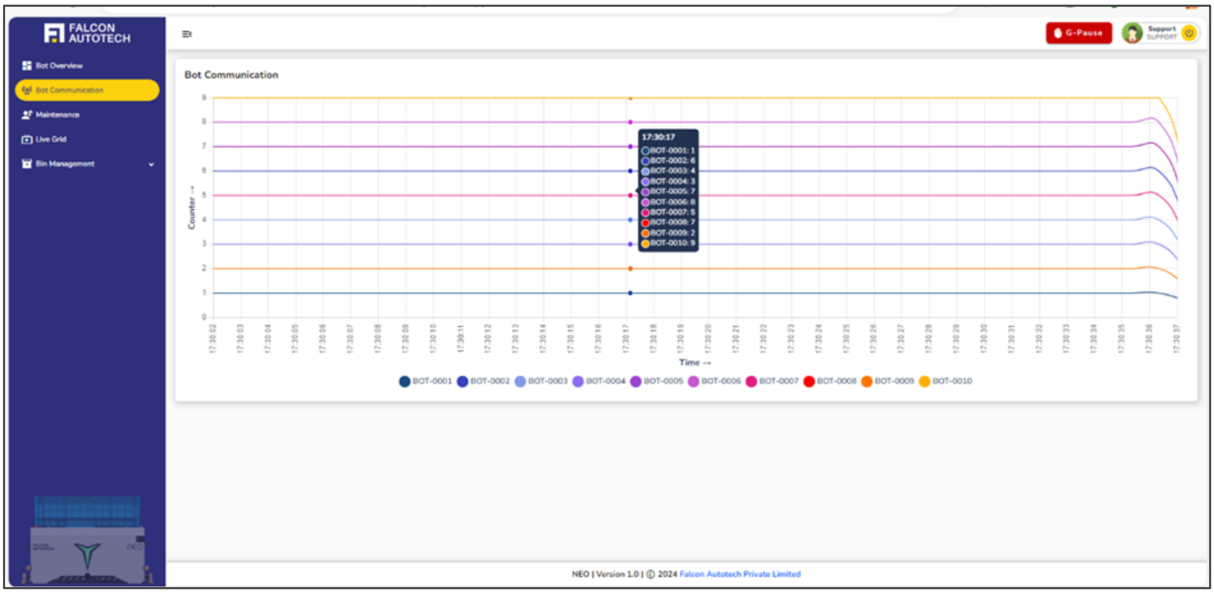


Figure 7.2 Inactive Bots

8 Block/Unblock Location

The Block/Unblock Location feature provides users with the ability to temporarily restrict access to specific storage locations within the live grid. This functionality is particularly useful during maintenance, cleaning, inspections, or operational adjustments, where certain bins or sections must be excluded from robot activity.

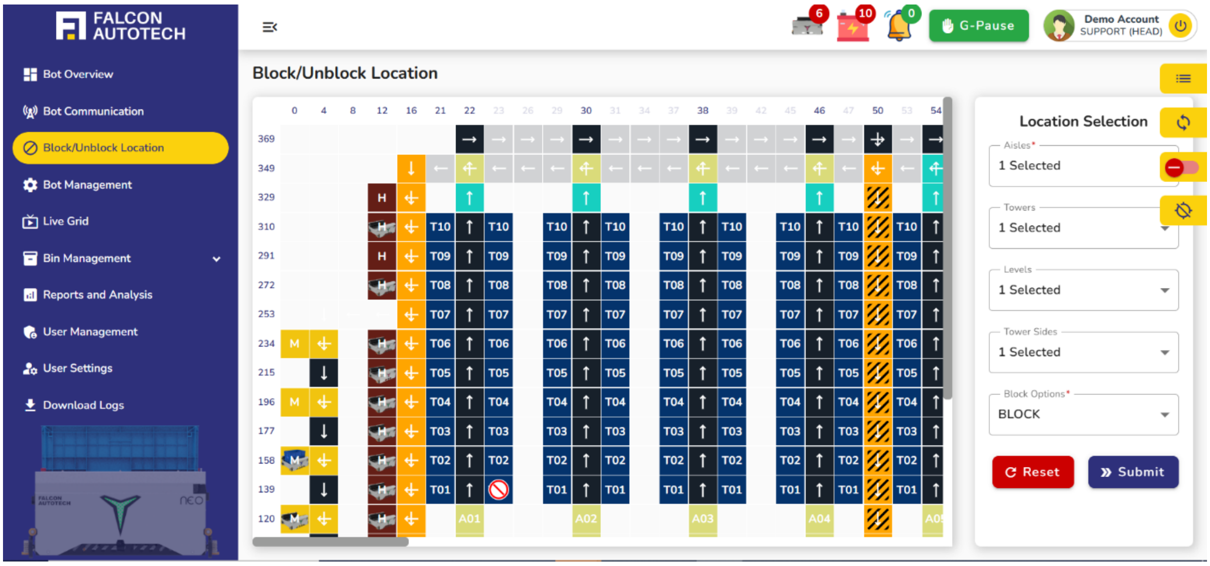
The primary goal of this module is to **prevent robots from performing any pick, put, or preparation tasks** in the specified blocked areas. It helps ensure:

- **Safety** for maintenance teams working in the area
- **System accuracy** by avoiding task assignments in zones under maintenance
- **Operational control** by allowing selective blocking at various granularities.

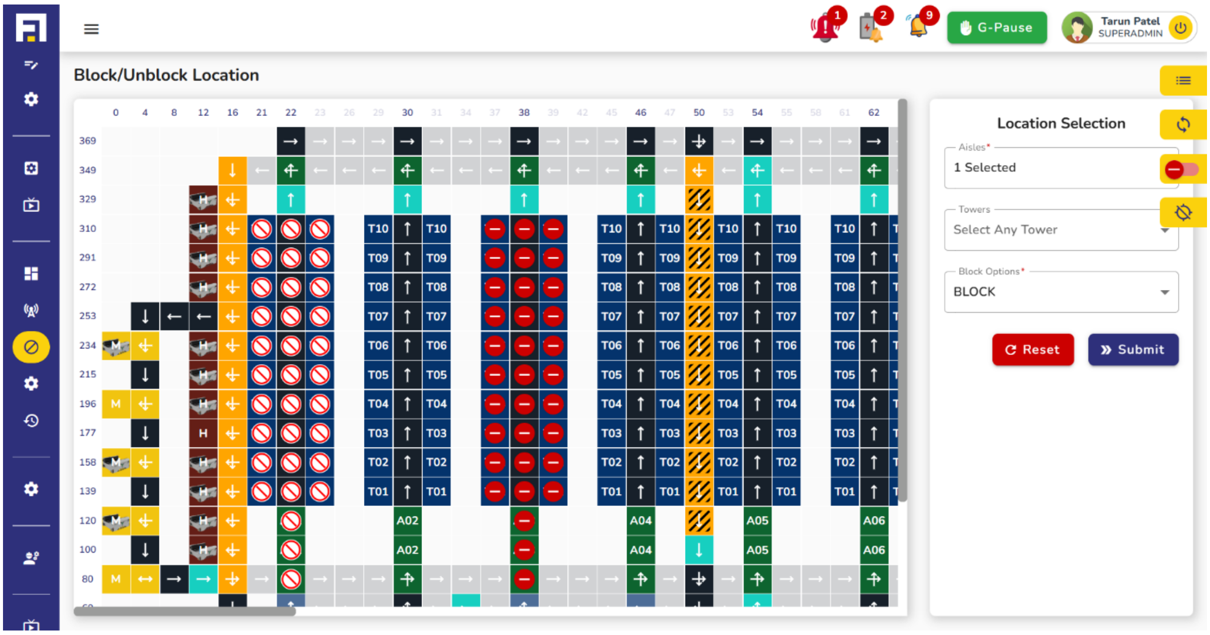
Users can block locations at different structural levels depending on the requirement:

- **Aisle Level** – Entire aisle across all towers and levels
- **Tower Level** – A specific vertical tower within an aisle
- **Level** – A particular horizontal level (e.g., shelf height)
- **Side of Level** – The physical side of the level (e.g., front/rear or left/right), useful when only part of a level needs blocking

This allows precise control and avoids unnecessary disruption in other active zones. Once a location is blocked. All bins within that location become inactive for pick, put, or preparation operations Robots are restricted from entering or interacting with those locations. Tasks involving these bins are rerouted to alternative available bins.



The system ensures that blocked areas are completely excluded from live task scheduling until explicitly unblocked. The same interface allows users to unblock locations once maintenance is complete. Upon unblocking, affected bins return to active status, and robots can resume operations as per normal logic.



To provide complete visibility into the areas that have been restricted from robot activity, the system offers a **"Click to show/hide blocked locations"** option. When a user clicks on this, a detailed table is displayed showing **all currently blocked locations** across the warehouse grid.

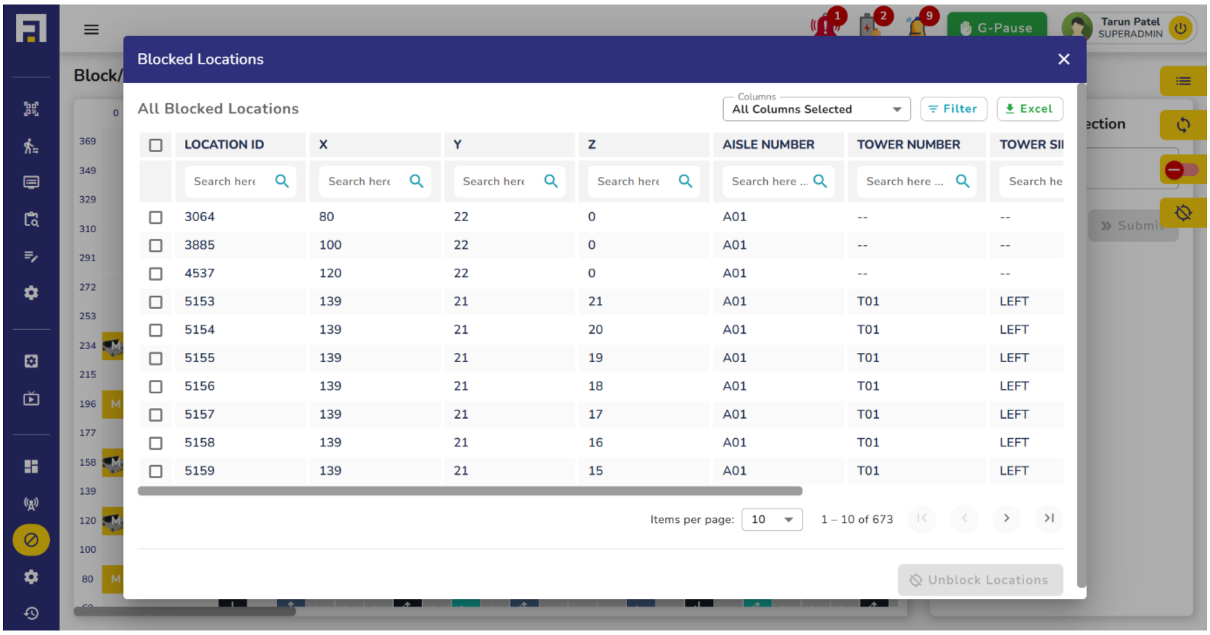


Figure 8.1 Robot's continuous communication

rrt

Maintenance

9 Maintenance

The **Maintenance Tab** on the navigation bar provides a critical interface for handling non-recovery scenarios where a bot cannot continue operations due to irreparable errors. This screen is designed to ensure smooth transitions for bots requiring maintenance and efficient management of any bins they carry.

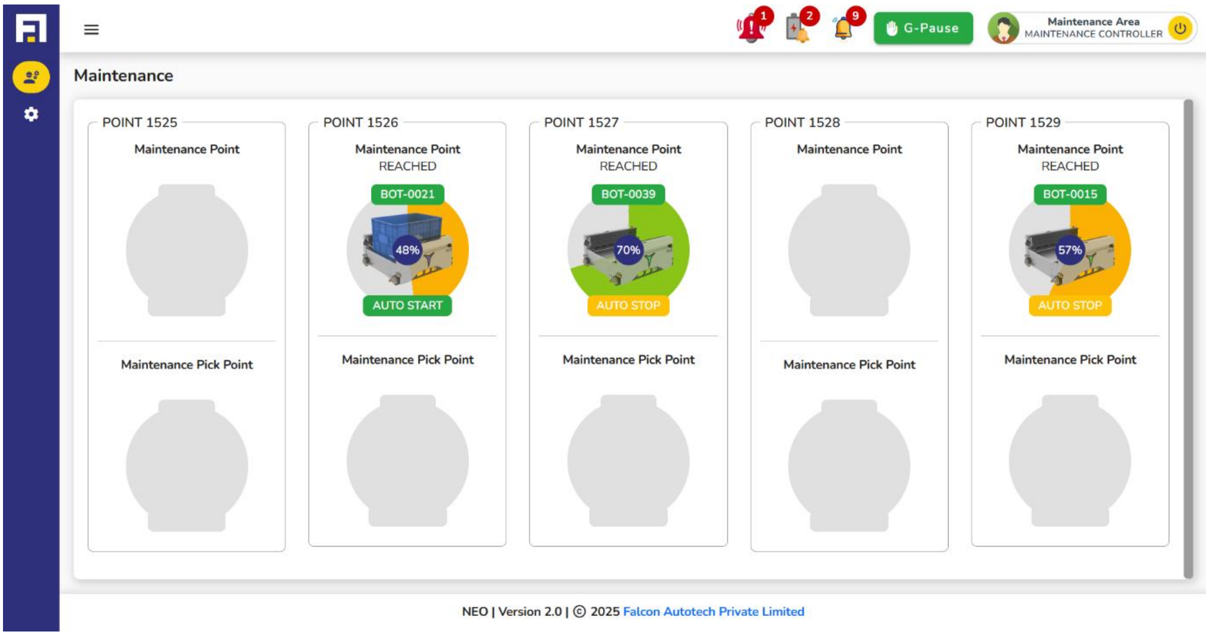


Figure 9.1 Maintenance Main Screen

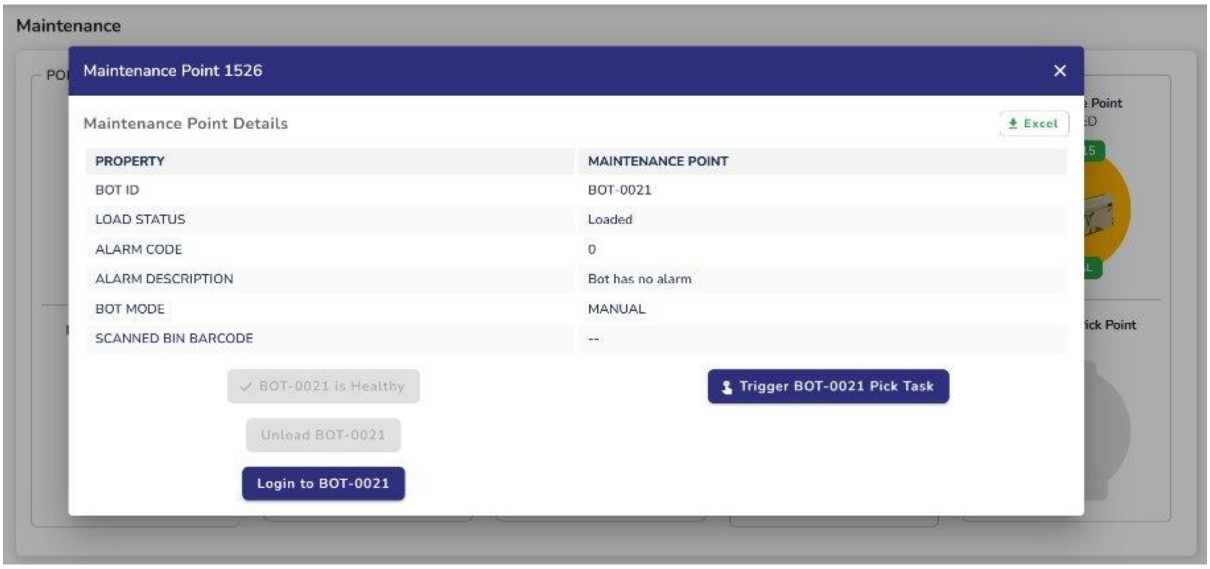
Users are redirected to the **Maintenance Screen** under two conditions:

9.1.1.1 Auto non-recovery:

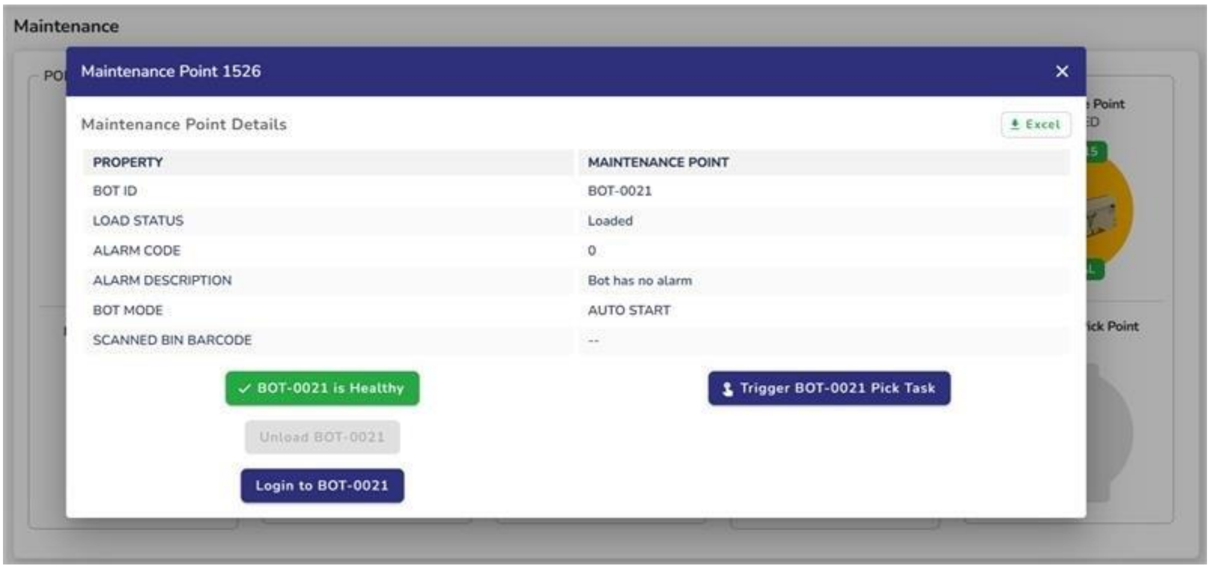
- During an active wave operation, the bot encounters an error and attempts retries but fails.
- If the bot is still functional enough to navigate, it automatically receives a task to move to the maintenance area.
- Any bin present on the robot must be transferred to another healthy bot.
- **Neo-FMS** handles this by automatically assigning a replacement bot to retrieve the bin.

9.1.1.2 Manual non-recovery:

- During an active wave operation, if a bot encounters an error requiring maintenance, it must be sent manually to the maintenance point via a grid-based task. The user initiates this by clicking the **Non-Recovery Button** on the **Bot Manual Page**, triggering the reassignment of tasks to a replacement bot.
- Actions performed:
- Once the bot reaches maintenance, its details appear on the screen as show in the figure above.
- Clicking on the bot opens a popup with two actions as shown in the figure below.



- The "Healthy Bot" button is **enabled only if the bot is in Auto Start mode**. If not, the operator must enable Auto Start first.



- Upon clicking "Bot is Healthy", a popup prompts for bin ID scan using the maintenance scanner. If the bot carries a bin, then Bin ID is validated via scanner, and bot resumes its original task linked to the bin. No reallocation or duplication required. If the bot is not loaded then bot rejoins the active fleet and is eligible for new tasks from the system scheduler.
- When Bot cannot be marked healthy, a popup appears prompts the operator to scan the bin barcode. System generates a Recovery Pick Task for the scanned bin and ensures the original task continues without loss.
- A new bot is auto assigned to the recovery task. It navigates to the Maintenance Pick Point appears in the Pick section on the maintenance screen. On selecting the new bot, the operator is prompted to unload the bin from the maintenance bot and load it onto the pick bot. Once loaded, the pick bot leaves the maintenance area and resumes task execution

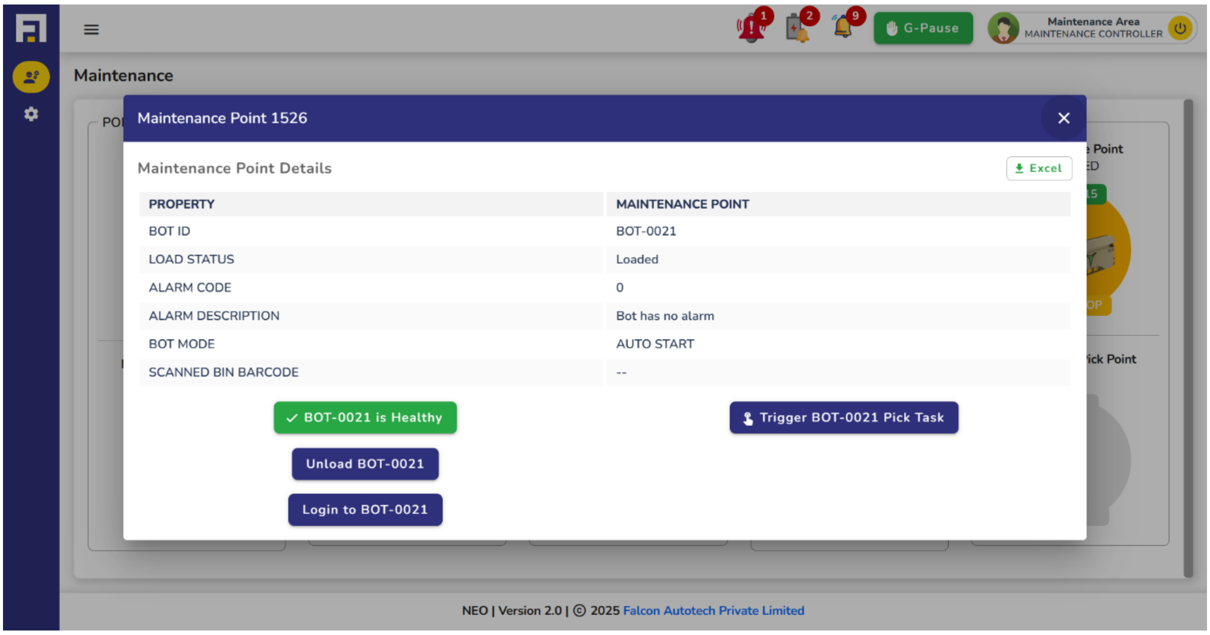


Figure 9.2 Maintenance Screen with both maintenance and maintenance pick bot

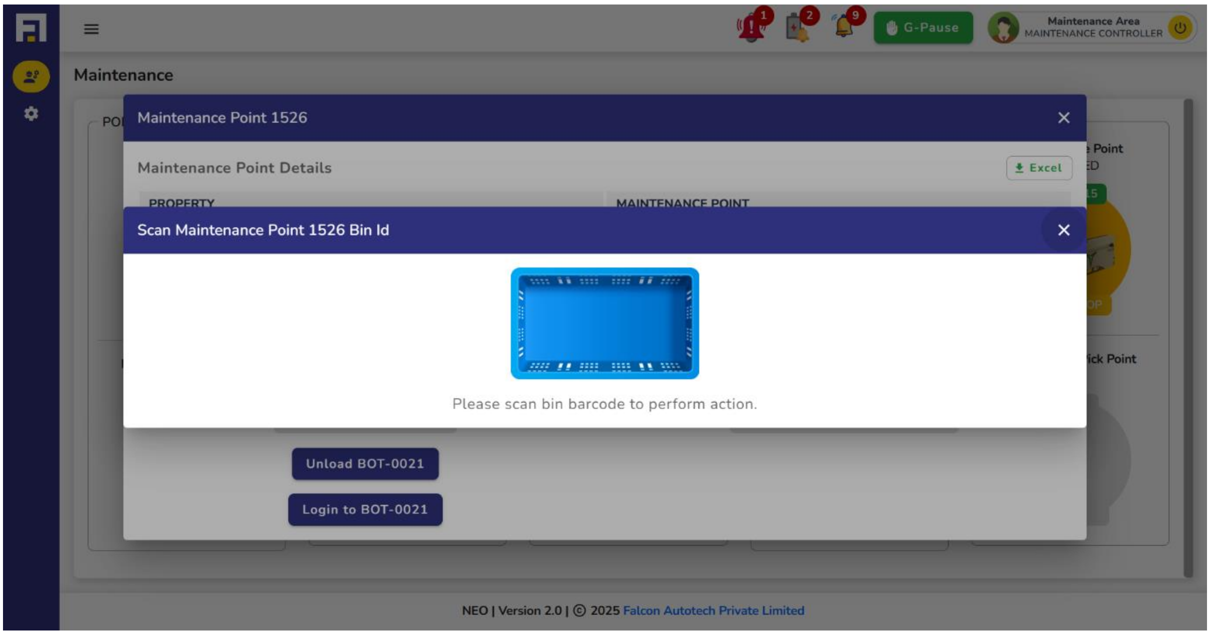


Figure 9.3 Maintenance Screen load/unload confirmation

10 Charging

The system dynamically manages robot charging by continuously monitoring battery levels and operating within defined system constraints to ensure optimal performance. Its primary objectives are to minimize operational downtime through proactive charging, prioritize robots with critically low battery levels, and efficiently utilize limited charging slots. Charging decisions are based on battery thresholds, where robots with a battery level at or below 18% are considered critical and are scheduled for immediate charging, while those with battery levels up to 39% are queued for regular charging. The system also manages concurrent charging sessions, allowing a maximum of two robots to charge simultaneously (a configurable limit), always giving priority to critically low robots to maintain uninterrupted operations.

The system refreshes the charging queue by identifying robots with battery levels at or below the batch threshold (≤39%) and prioritizing those with the lowest battery levels. Robots below the critical threshold (≤18%) are charged immediately. If all charging slots are occupied, the system will evict the robot with the highest battery level, as long as it's not in a critical state, to make room. As slots become available, the next robots in the queue begin charging. Charging is automatically stopped once a robot reaches the charging station and its battery level exceeds the batch threshold (>39%), at which point it is removed from the queue.

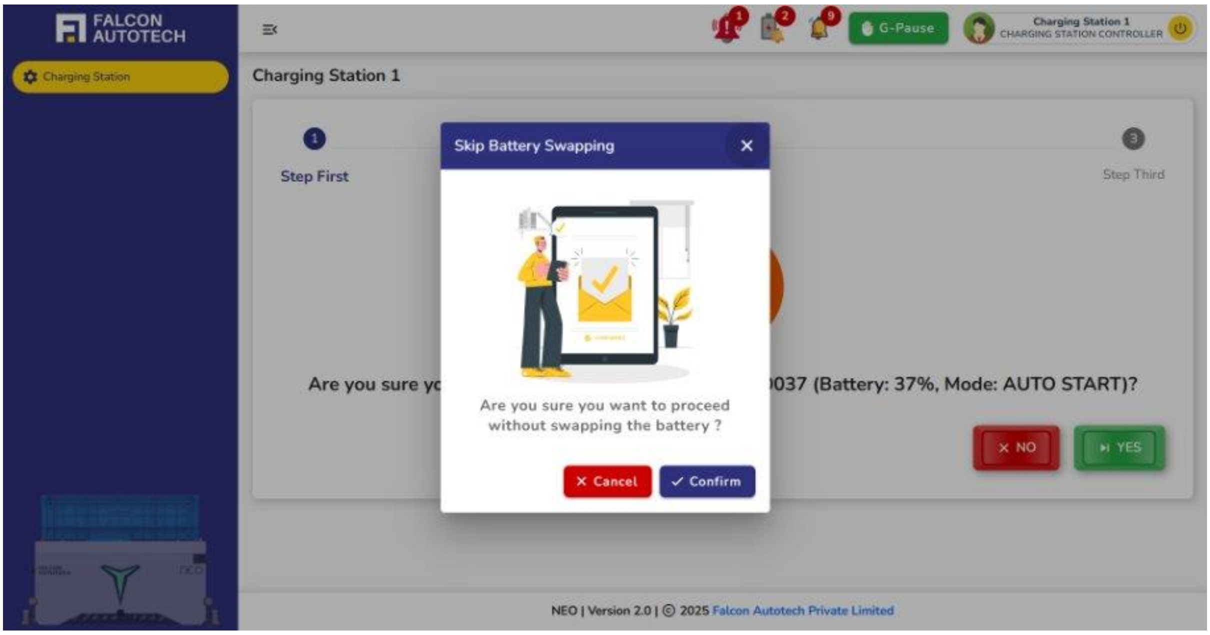
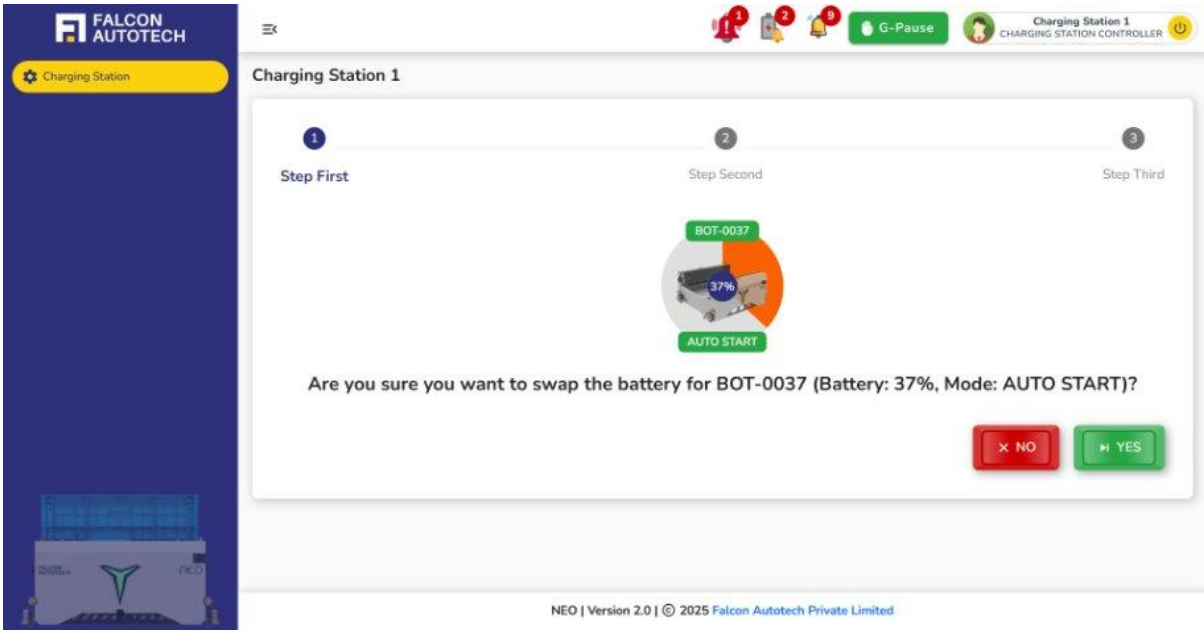
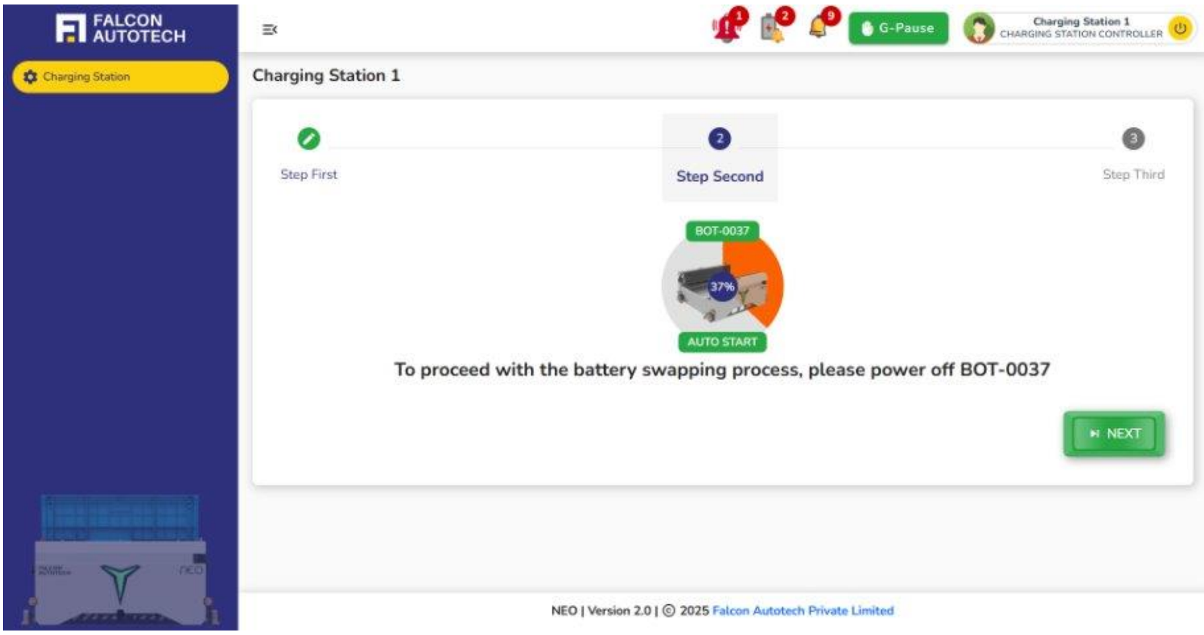
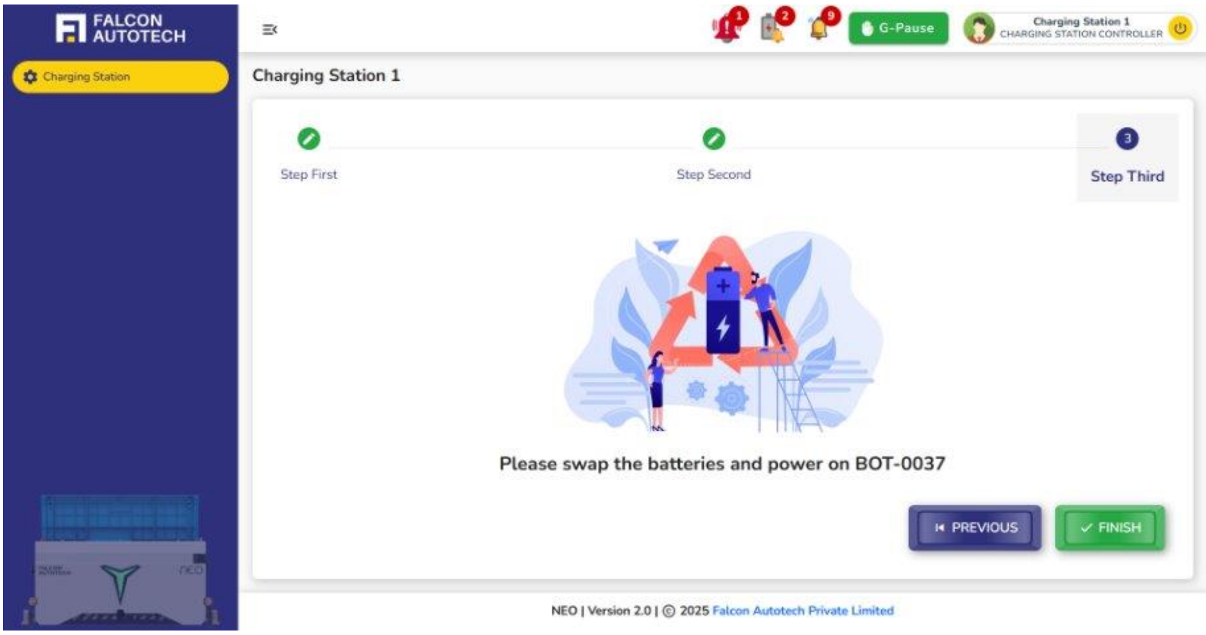
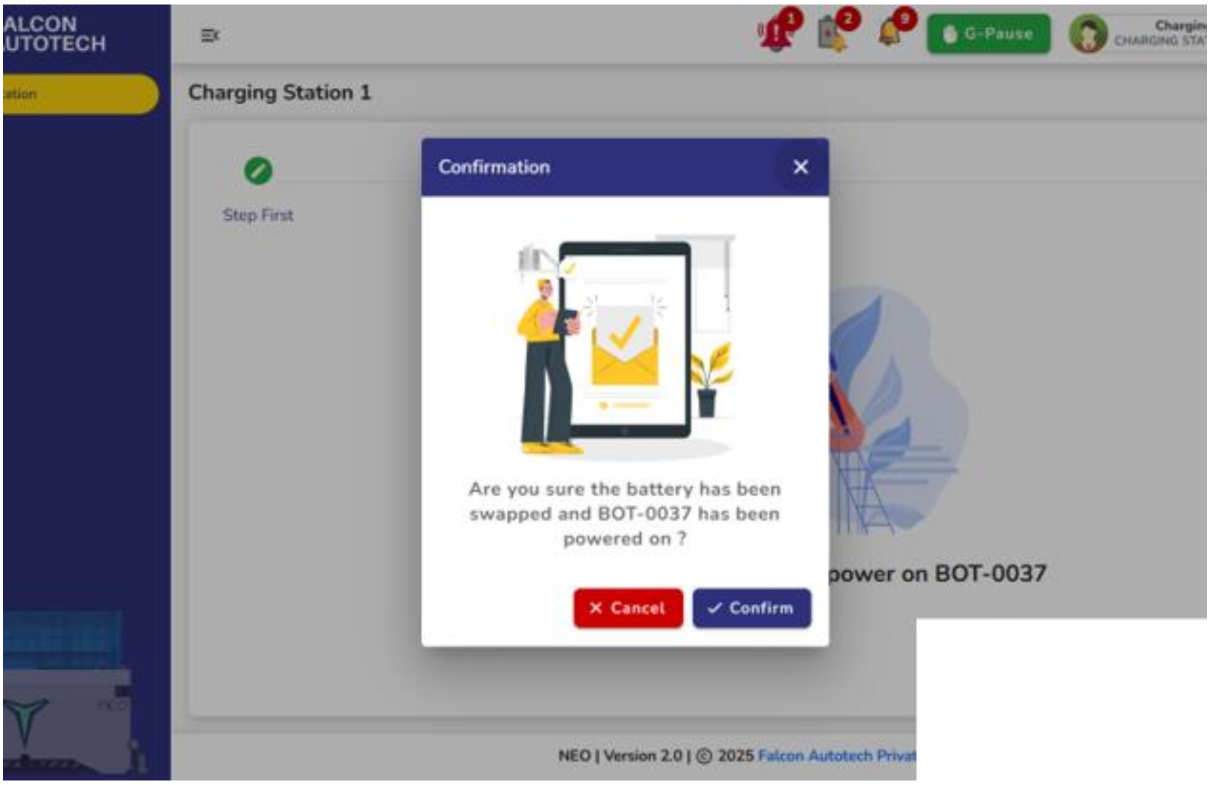


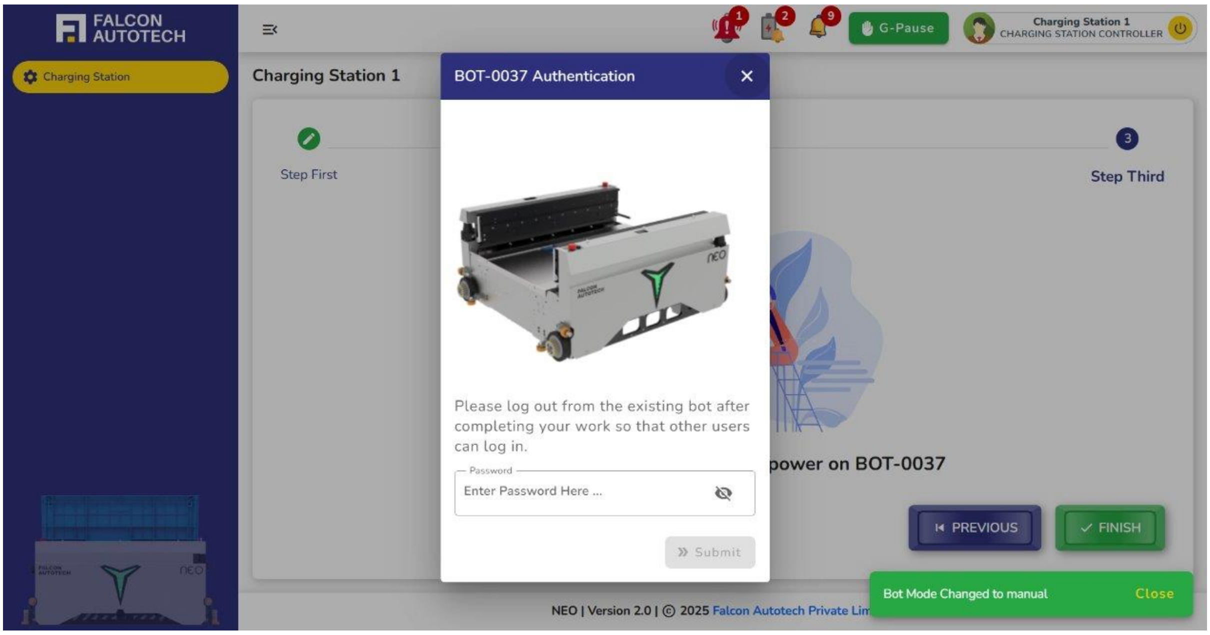
Figure 10.1 Charging Main Screen


FALCON AUTOTECH
Charging Station
G-Pause
Charging Station 1
CHARGING STATION CONTROLLER
Charging Station 1
Step First
Step Second
Step Third
BOT-0037
37%
AUTO START
Are you sure you want to swap the battery for BOT-0037 (Battery: 37%, Mode: AUTO START)?
NO
YES
NEO | Version 2.0 | © 2025 Falcon Autotech Private Limited


FALCON AUTOTECH
Charging Station
G-Pause
Charging Station 1
CHARGING STATION CONTROLLER
Charging Station 1
Step First
Step Second
Step Third
BOT-0037
37%
AUTO START
To proceed with the battery swapping process, please power off BOT-0037
NEXT
NEO | Version 2.0 | © 2025 Falcon Autotech Private Limited


FALCON AUTOTECH
Charging Station
G-Pause
Charging Station 1
CHARGING STATION CONTROLLER
Charging Station 1
Step First
Step Second
Step Third
Please swap the batteries and power on BOT-0037
PREVIOUS
FINISH
NEO | Version 2.0 | © 2025 Falcon Autotech Private Limited


Charging Station 1
Confirmation
Step First
Are you sure the battery has been swapped and BOT-0037 has been powered on ?
Cancel
Confirm



To perform charging or battery swapping, the robot first arrives at the designated charging station. Once it arrives, the system prompts the operator for confirmation to proceed with the battery swap. After receiving confirmation, the operator is instructed to power off the robot. Once the robot is powered off, the operator then swaps the battery and powers it back on and logs into the system to resume normal operations and continue the process.

11 Live Grid

The **Live Grid**, accessible from the navigation bar, offers a dynamic, read-only interface showcasing the real-time movements of all robots within the layout. This page is a visual representation of the operational flow, providing key insights into the bots' positions and movements without the need for any interaction or manual input.

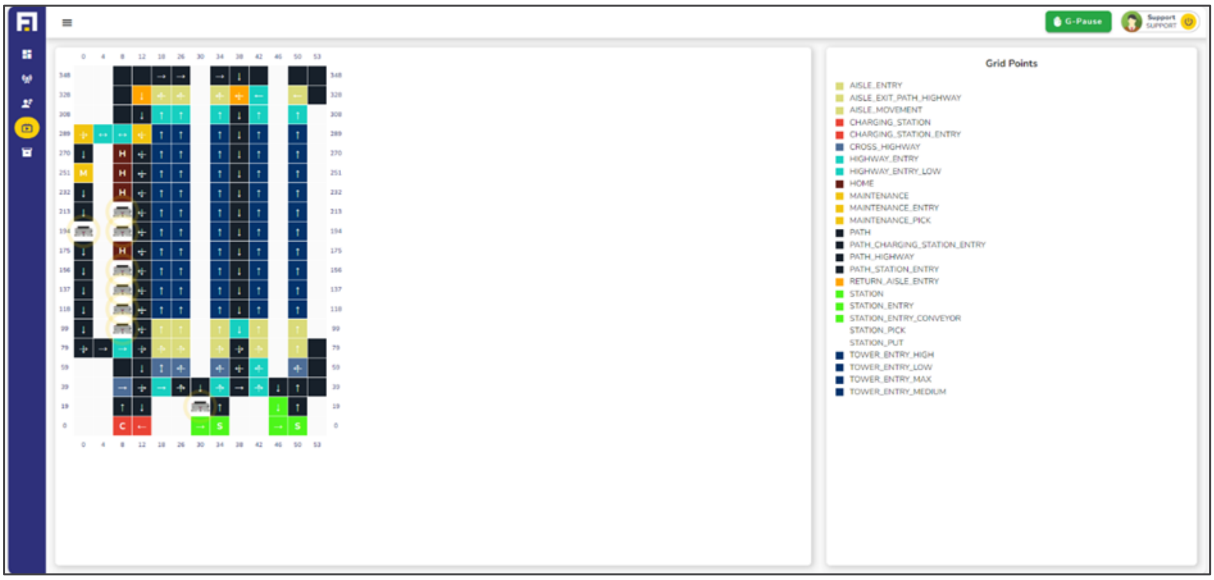


Figure 11.1 Live grid screen

The key features of live grid page:

- Real-Time Visualization: The grid continuously updates to display the current locations and paths of all bots, offering a live snapshot of their activities on the layout.
- Comprehensive Status Overview: Users can instantly identify where each bot is located and observe how they navigate through the grid, making it easier to track their tasks and ensure smooth operations.

As a read-only page, Live Grid removes any potential for accidental changes or disruptions, ensuring the focus remains solely on monitoring. This tool is invaluable for understanding the spatial distribution and movement patterns of the robots, helping users identify any congestion, inefficiencies, or delays in the system. With its clear, live updates and intuitive design, the **Live Grid** empowers users to stay informed about robot activity, offering a bird's-eye view of the layout in action.

12 Bin Management

The **Bin Management** page is an essential resource for users to understand and manage the inventory efficiently. It provides a comprehensive view of bin locations, contents, and associated details, making it an invaluable tool for inventory oversight and optimization. The page is divided into two intuitive submenus: **Bin Info** and **Know Your Bin**.

12.1 Bin Information

This section presents a dynamic table that lists all bins in the system, complete with their IDs, barcodes, locations within the tower, their placement (side and aisle), and other relevant attributes. Designed for flexibility, the table offers several user-friendly features:

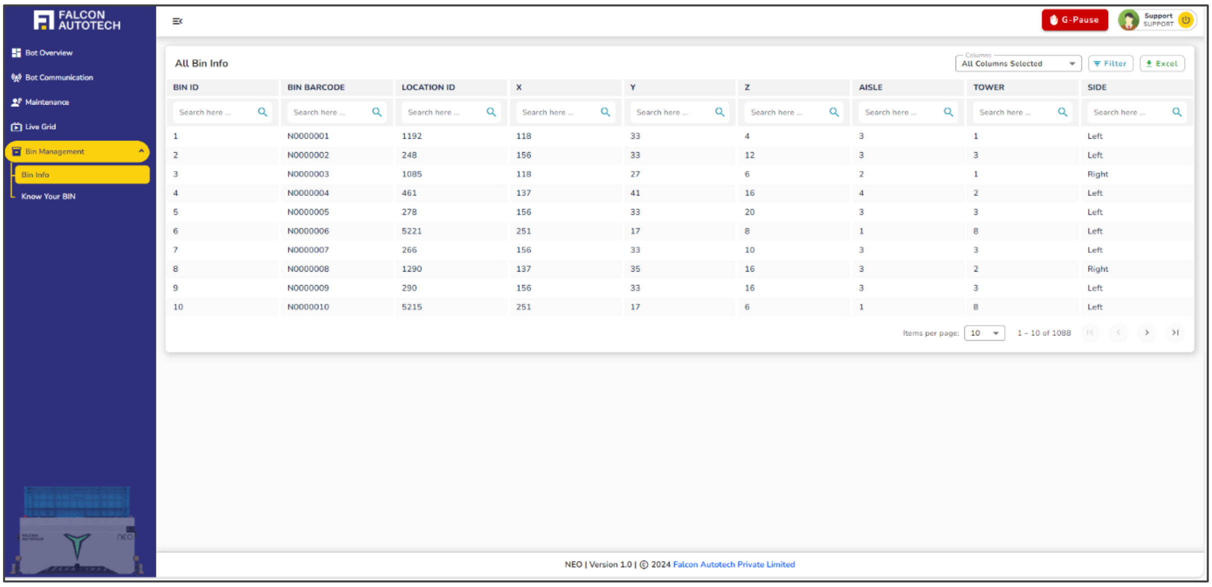


Figure 12.1 Bin Info Screen Bin Management

- **Customizable Columns**: Users can choose to display only specific columns by using the dropdown menu to select or deselect fields.

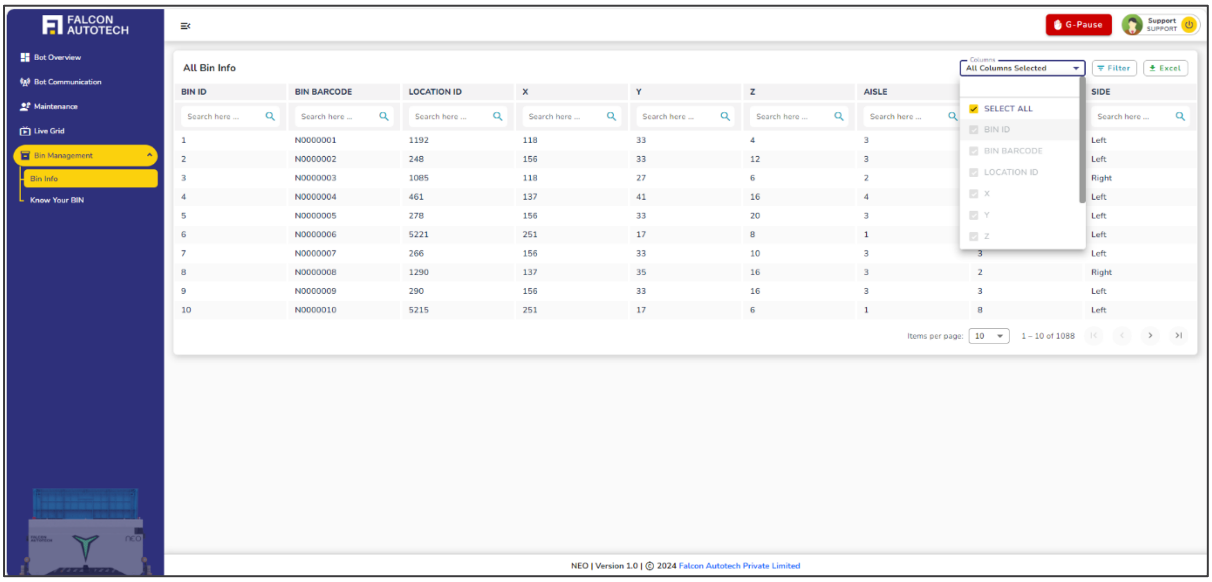


Figure 12.2 Bin Info Screen dropdown selection

- **Excel Export**: The data can be downloaded as an Excel file with a single click on the export button, enabling offline analysis and reporting.
- **Advanced Filtering**: Multiple conditions can be applied to filter the data, allowing users to search columns with comparison operators or exact values, streamlining the process of finding specific information.

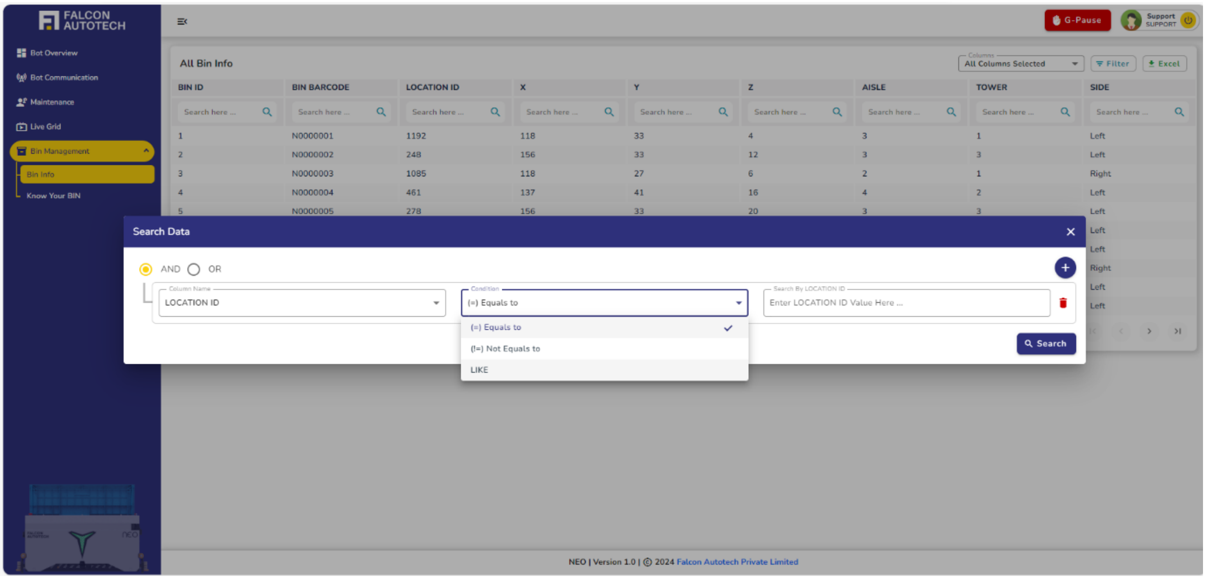


Figure 12.3 Bin info Advance Filtering

12.2 Know Your Bin

This subsection provides a powerful search feature to fetch detailed information about a specific bin. Users can select a search method (Bin ID or Barcode) from a dropdown menu and input the corresponding value. Upon searching, a detailed breakdown of the selected bin is displayed, organized into the following categories:

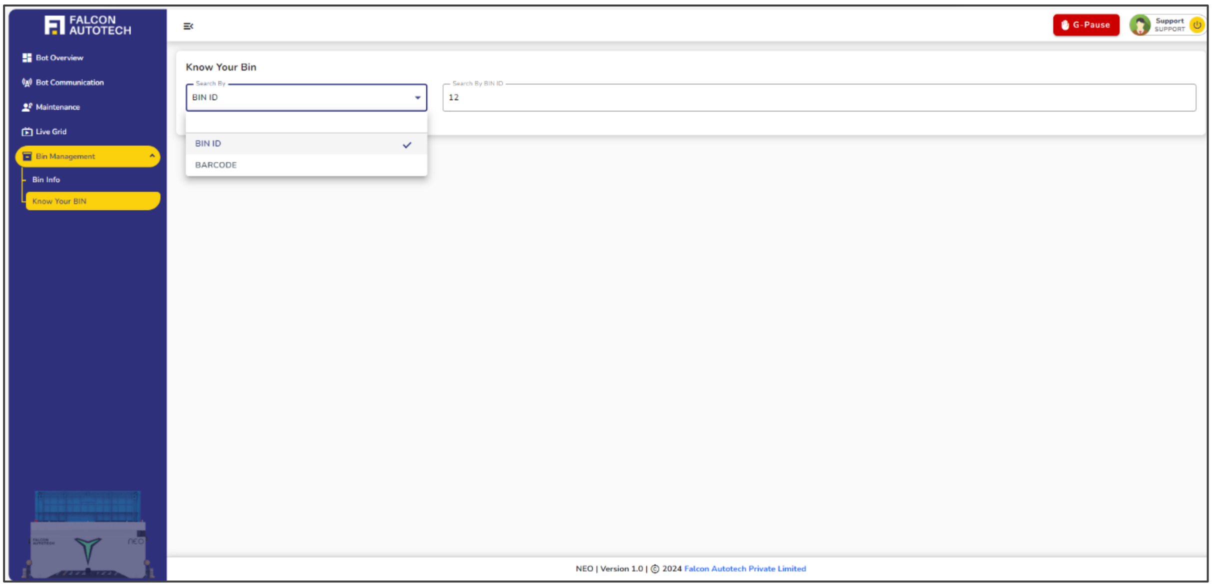


Figure 12.4 Know your Bin

- **Bin Info**: Highlights key details such as the bin's ID, barcode, and number of partitions/segments.

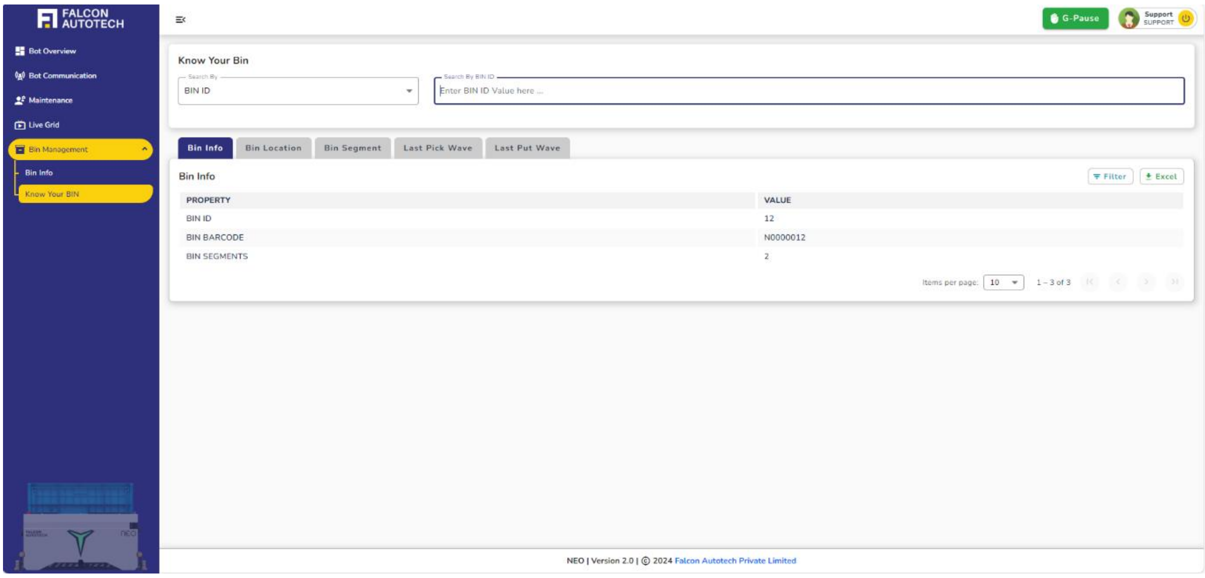


Figure 12.5 Bin General Information

- **Bin Location**: Specifies where the bin is situated within the inventory.

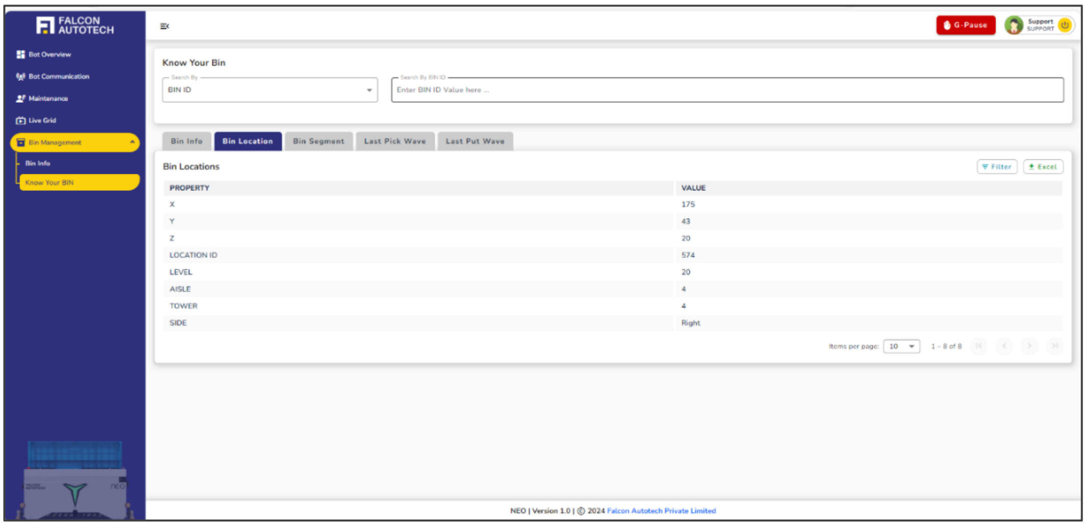


Figure 12.6 Know your bin location

- **Bin Segments**: Displays the contents of each segment within the bin, including SKUs and their respective quantities.

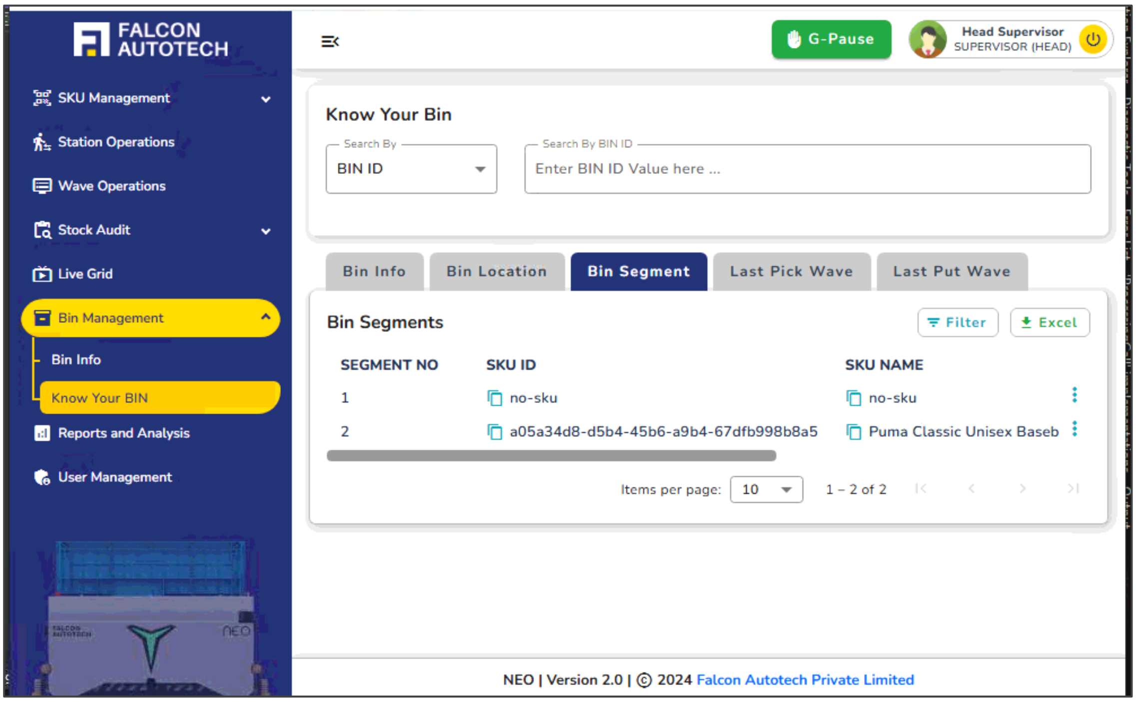


Figure 12.7 Know your bin segments

- **Last Pick Wave**: Provides insights into the bin's involvement in the most recent pick wave, including the frequency of its usage.

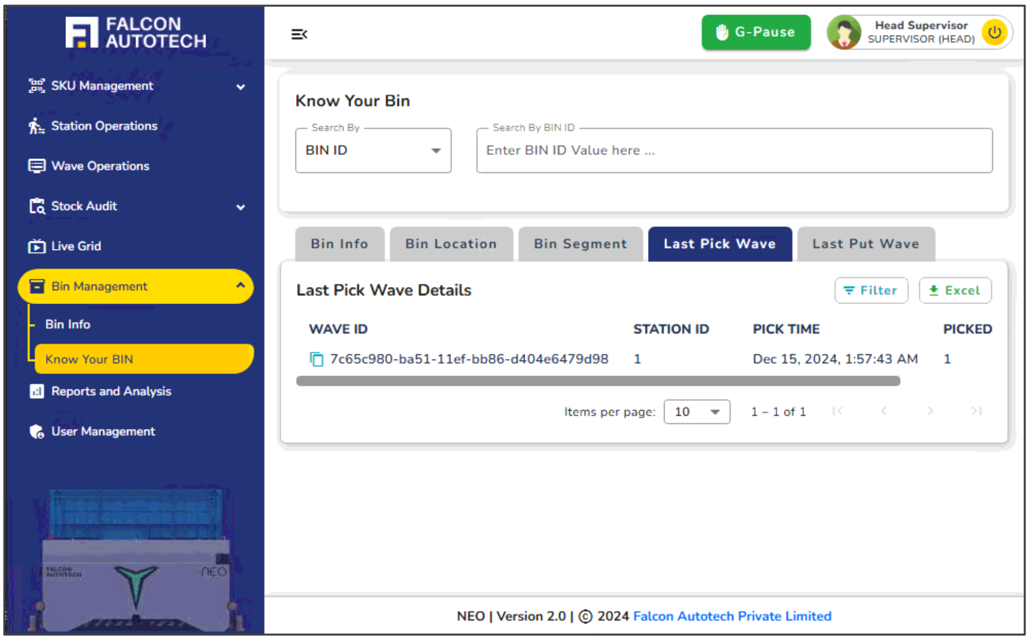


Figure 12.8 Last Pick wave

- **Last Put Wave**: Mirrors the above information for the most recent put wave, detailing its activity.

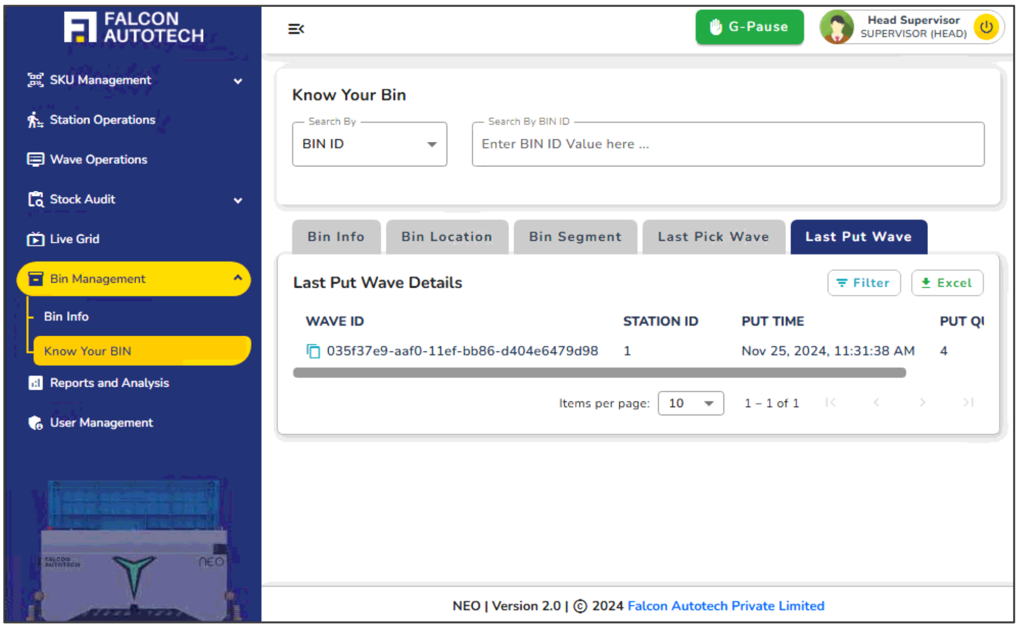


Figure 12.9 Last Put Wave

This thoughtfully designed page is a one-stop solution for tracking and managing bins, offering actionable insights to enhance operational accuracy and inventory control.

13 Summary

You have successfully completed the Dashboard Functionality Manual. We hope you have a better understanding of the overall functionality and requirements of the dashboard that Falcon team has offered.